

Cardiovascular System

August 2025!

Arteries of the Head and Neck

- Four pairs of arteries supply the head and neck:

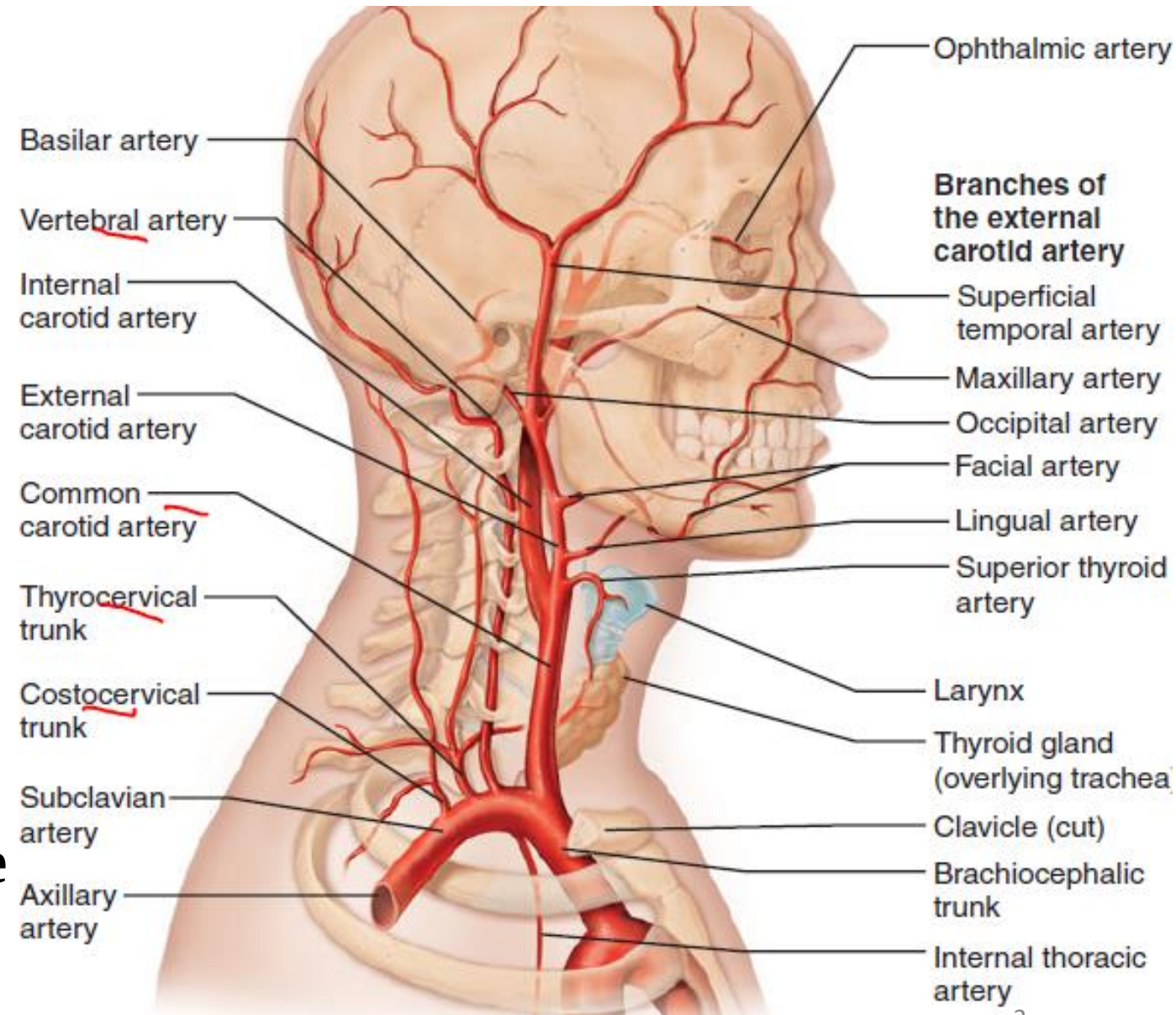
I. Common carotid arteries

II. Vertebral artery,

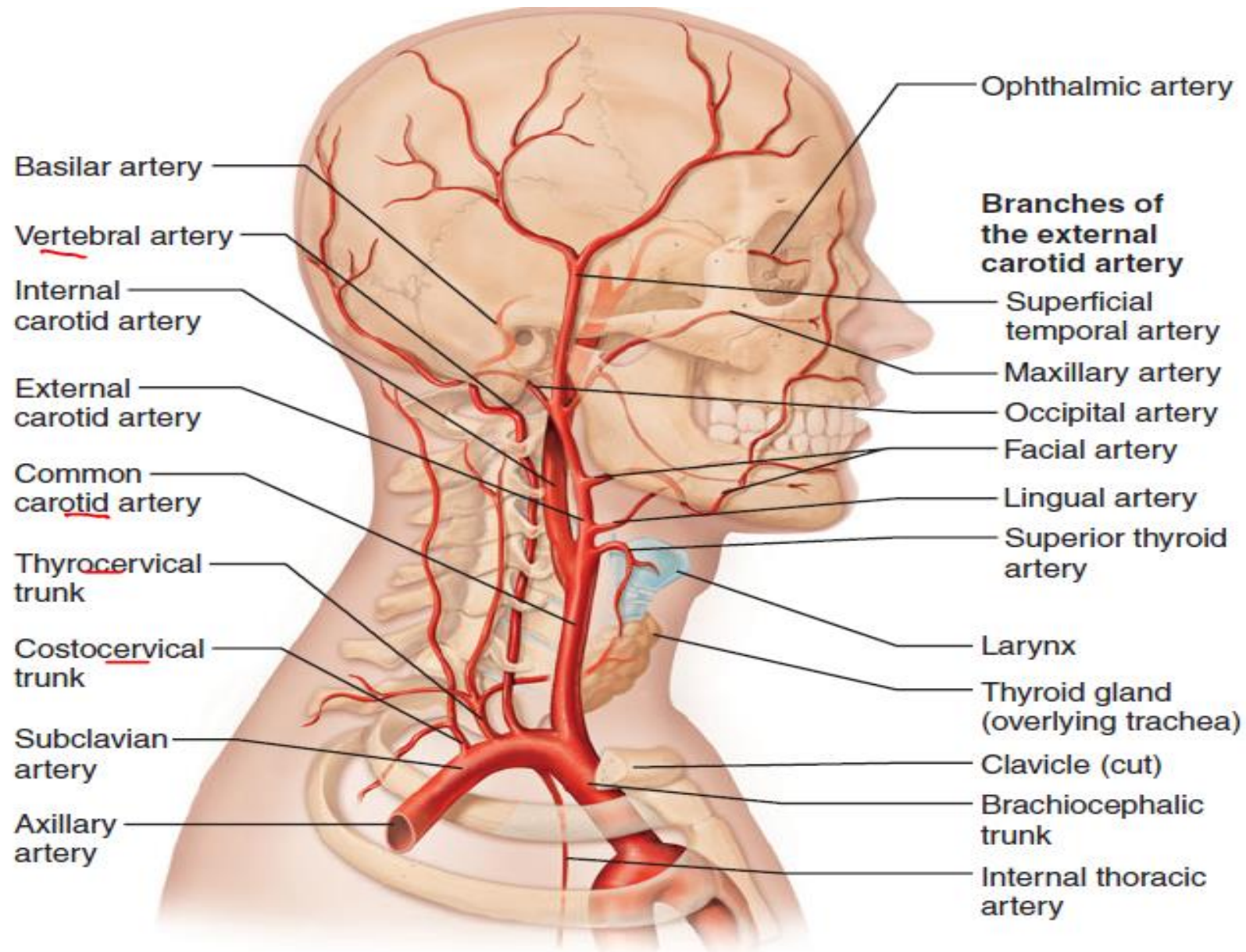
III. Thyrocervical trunk, and

IV. Costocervical trunk

- ❖ The 3 **arteries**(vertebral, thyrocervical& costocervical) **are** branches of subclavian artery.



Arteries of the Head and Neck,,



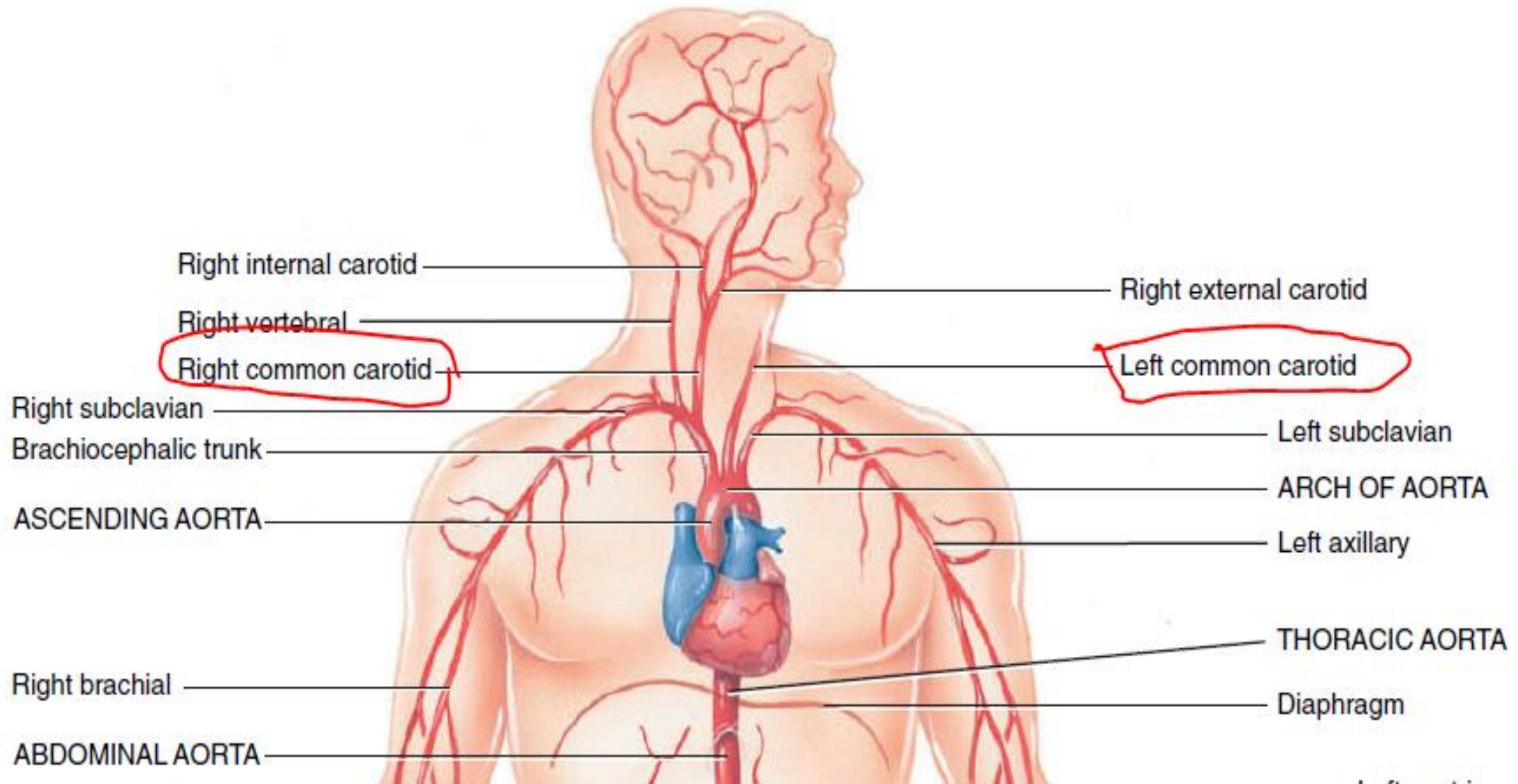
(a) Arteries of the head and neck, right aspect

Arteries of the Head and Neck,,

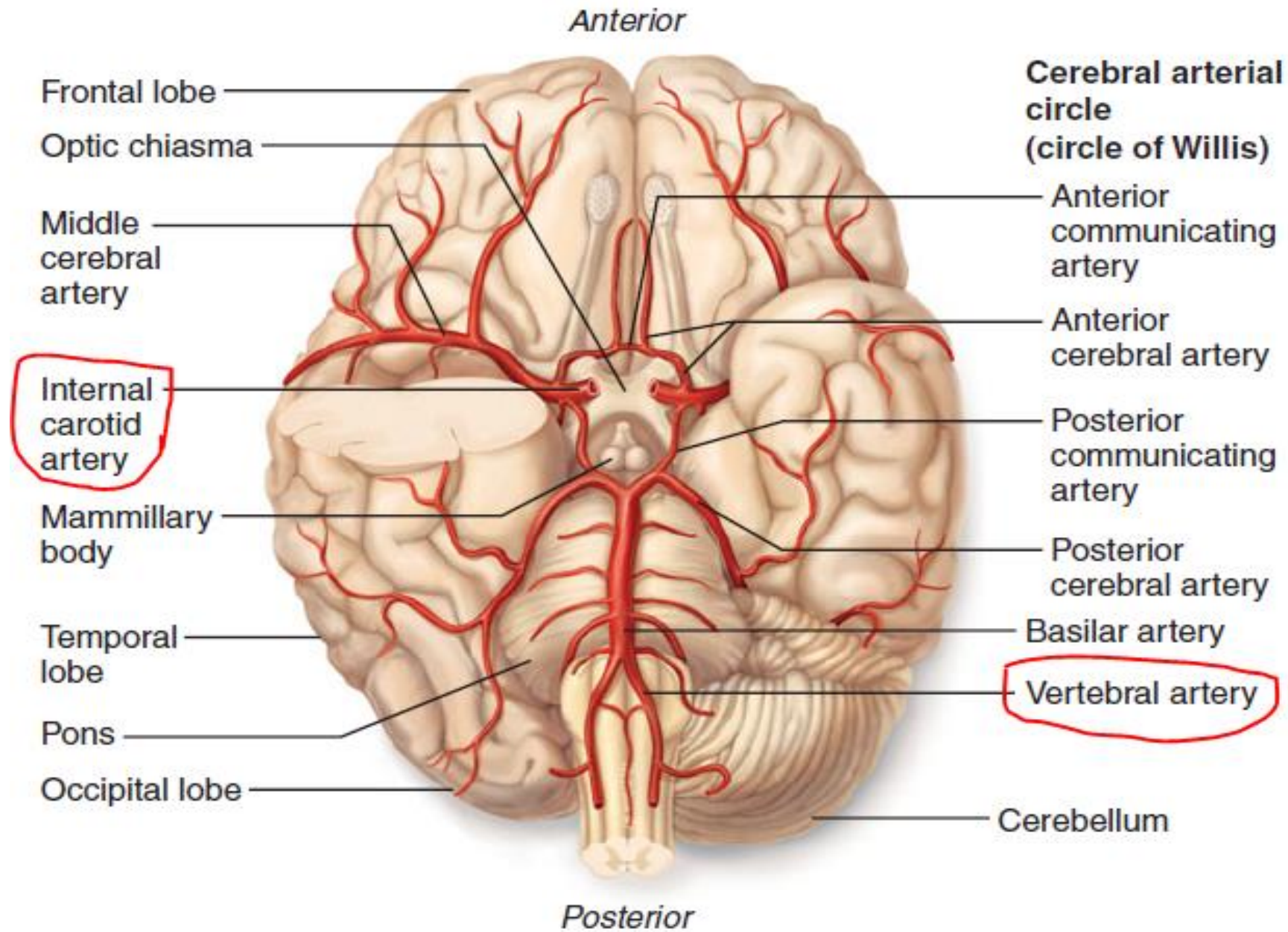
❑ **The common carotid arteries.** Most parts of the head and neck receive their blood from the **common carotid arteries**

- **The right common carotid** artery arises from the brachiocephalic artery and **the left common carotid artery** arises directly from the aortic arch.
- Each common carotid arteries branch into **internal & external carotid arteries.**
- **The external carotid arteries** give rise to a number of smaller arteries that carry blood to the neck, face, and scalp.
- **The internal carotid arteries along with the vertebral arteries** are the two primary suppliers of blood to the human brain.

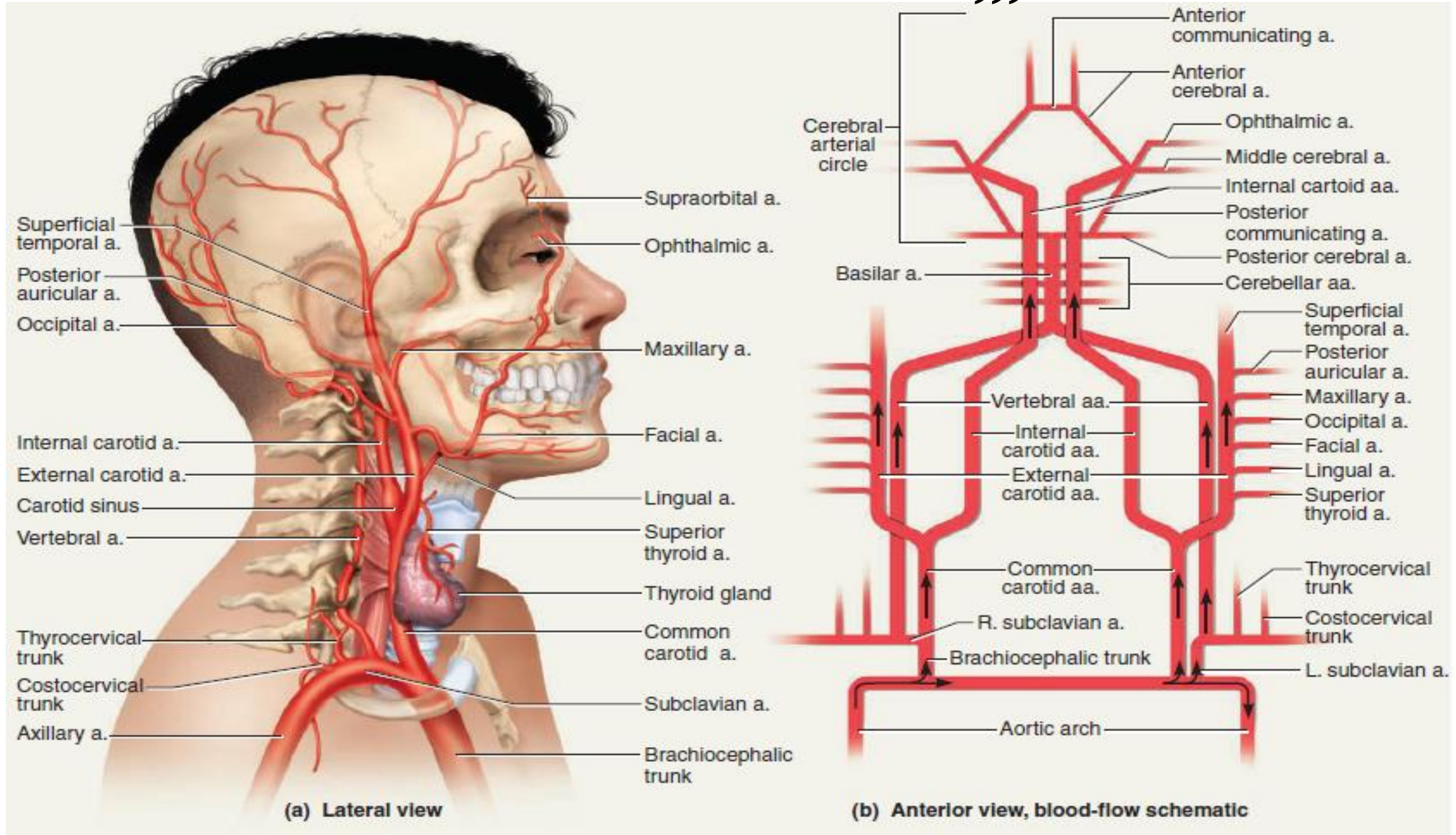
Arteries of the Head and Neck,,



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Arteries of the Head and Neck,,

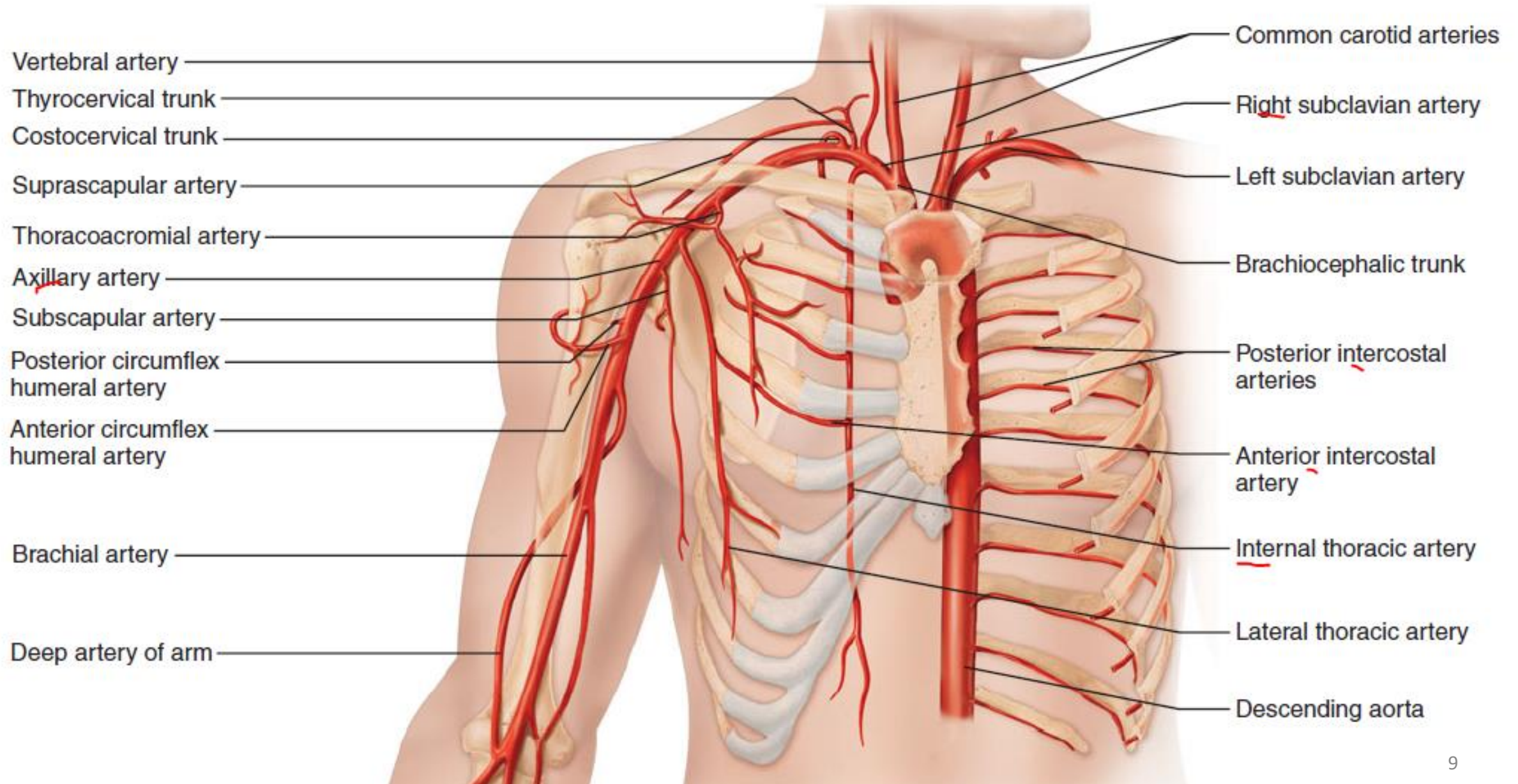


Arteries of the Thorax

❑ The arterial supply to the thoracic wall derives from the:

- ✓ *Thoracic aorta*, through the posterior intercostal and subcostal arteries.
- ✓ *Subclavian artery*, through the internal thoracic and supreme intercostal arteries.
- ✓ *Axillary artery*, through the superior and lateral thoracic arteries.
- The **intercostal arteries** course through the thoracic wall between the ribs.

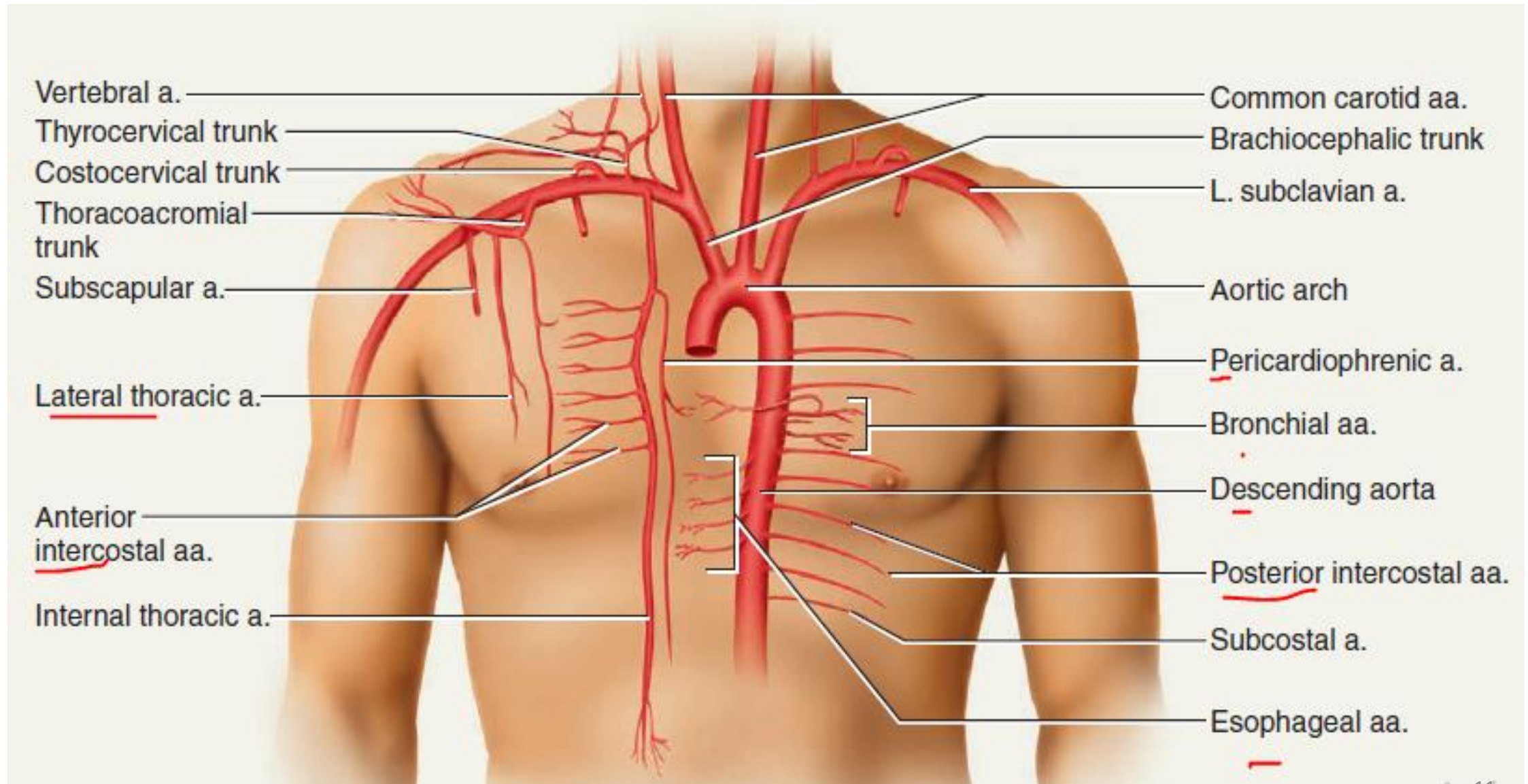
Arteries of the Thorax,,



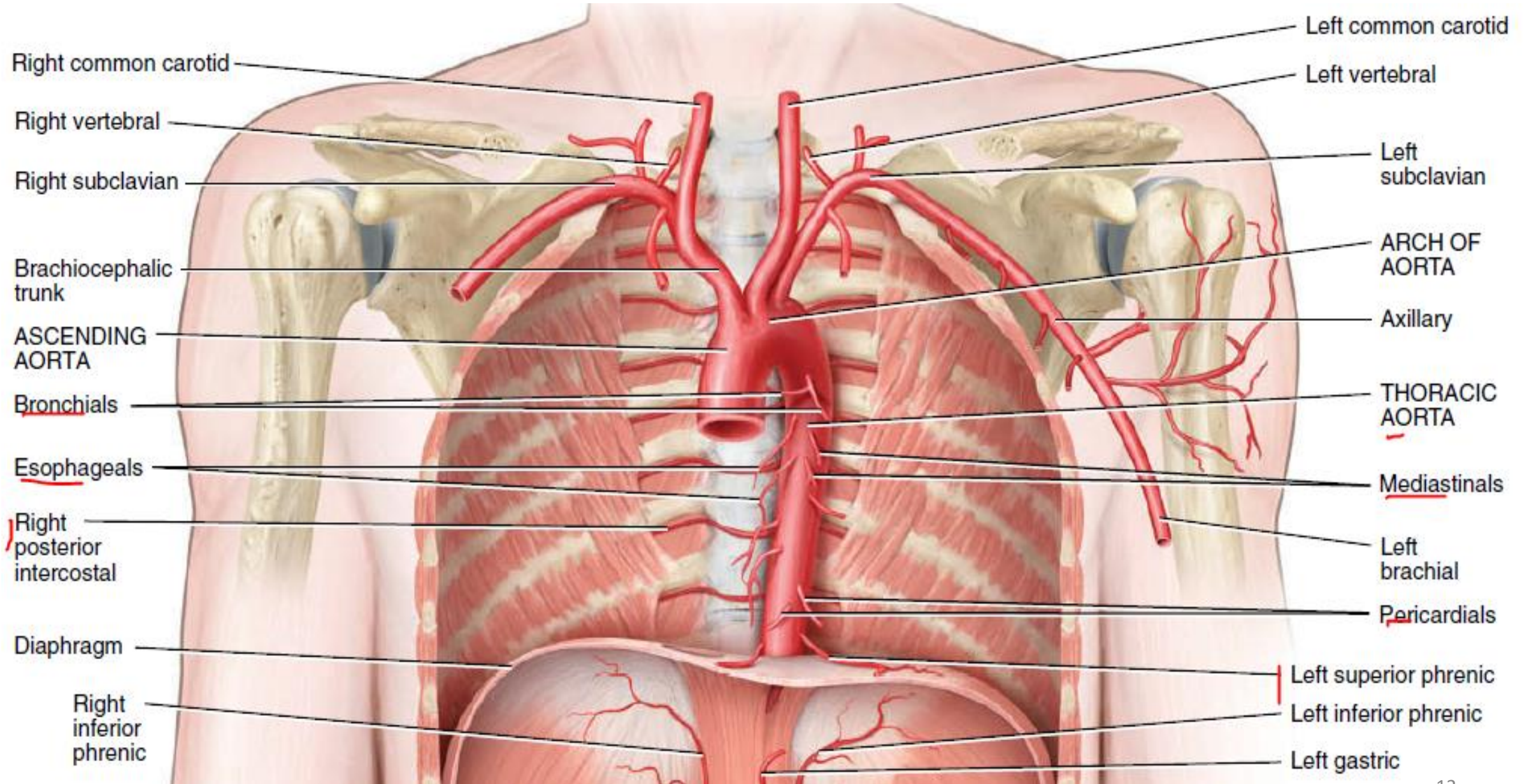
Arteries of the Thorax,,

- ❖ **The anterior thoracic wall receives** its blood from the **internal thoracic artery**(formerly the *internal mammary artery*).
- **Internal thoracic artery arises** from the subclavian artery.
- ❖ **Anterior intercostal arteries** branch off at regular intervals from **internal thoracic artery arises** and run horizontally to supply the ribs and the structures in the intercostal spaces.
- **The internal thoracic artery** also sends branches superficially to supply the skin and mammary gland.

Arteries of the Thorax,,,



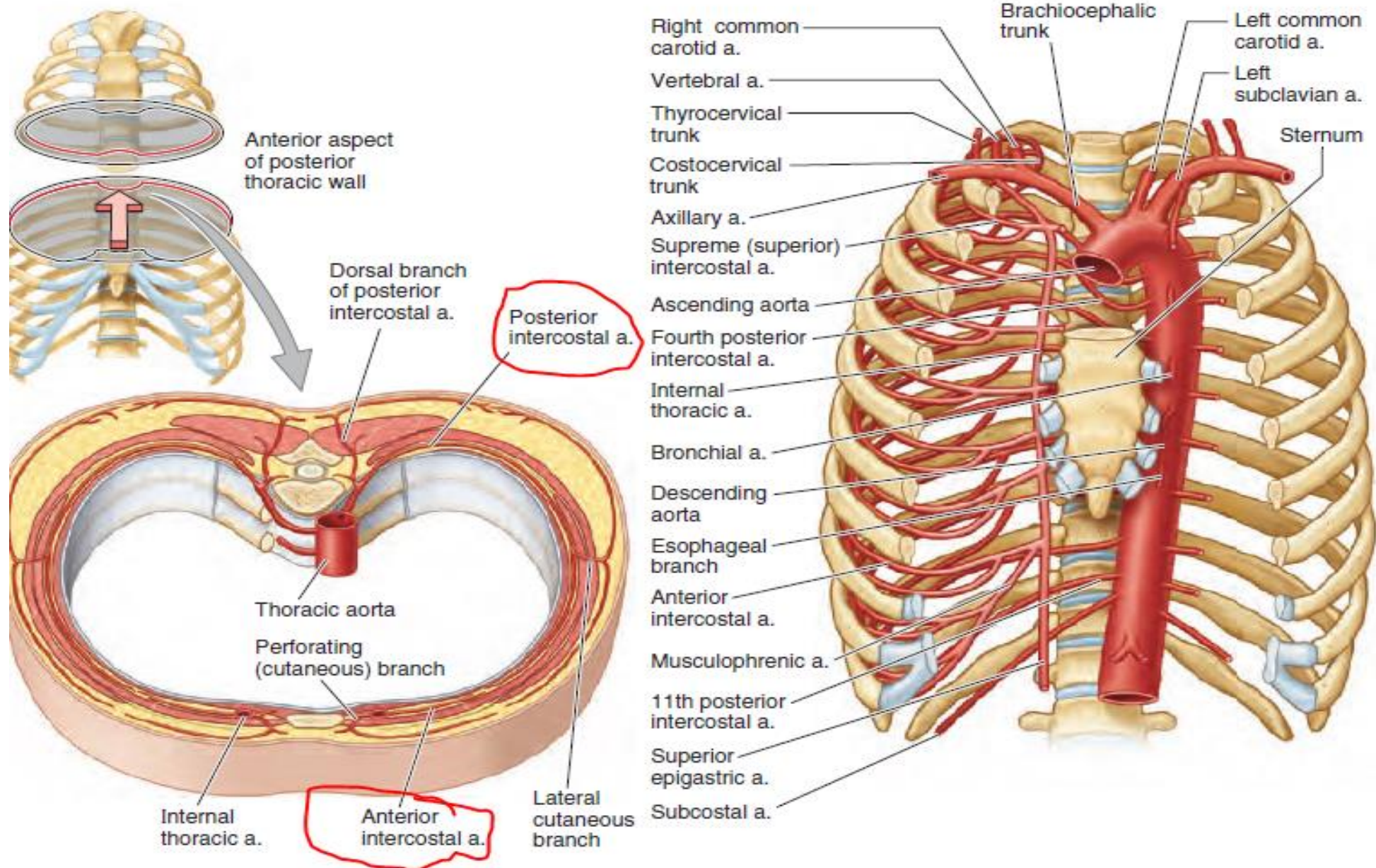
Arteries of the Thorax,,,



Arteries of the Thorax,,,

- ❖ **The posterior thoracic wall receives** its blood from the **posterior intercostal arteries which are branches of thoracic aorta.**
- **In the lateral thoracic wall,** they form anastomoses with the anterior intercostal arteries.
- Thoracic viscera receive **their functional blood** supply from small branches off the thoracic aorta.
- **Usually, two bronchial arteries** serve the left lung, and one serves the right lung; in some people they arise from posterior intercostal arteries instead of the aorta.

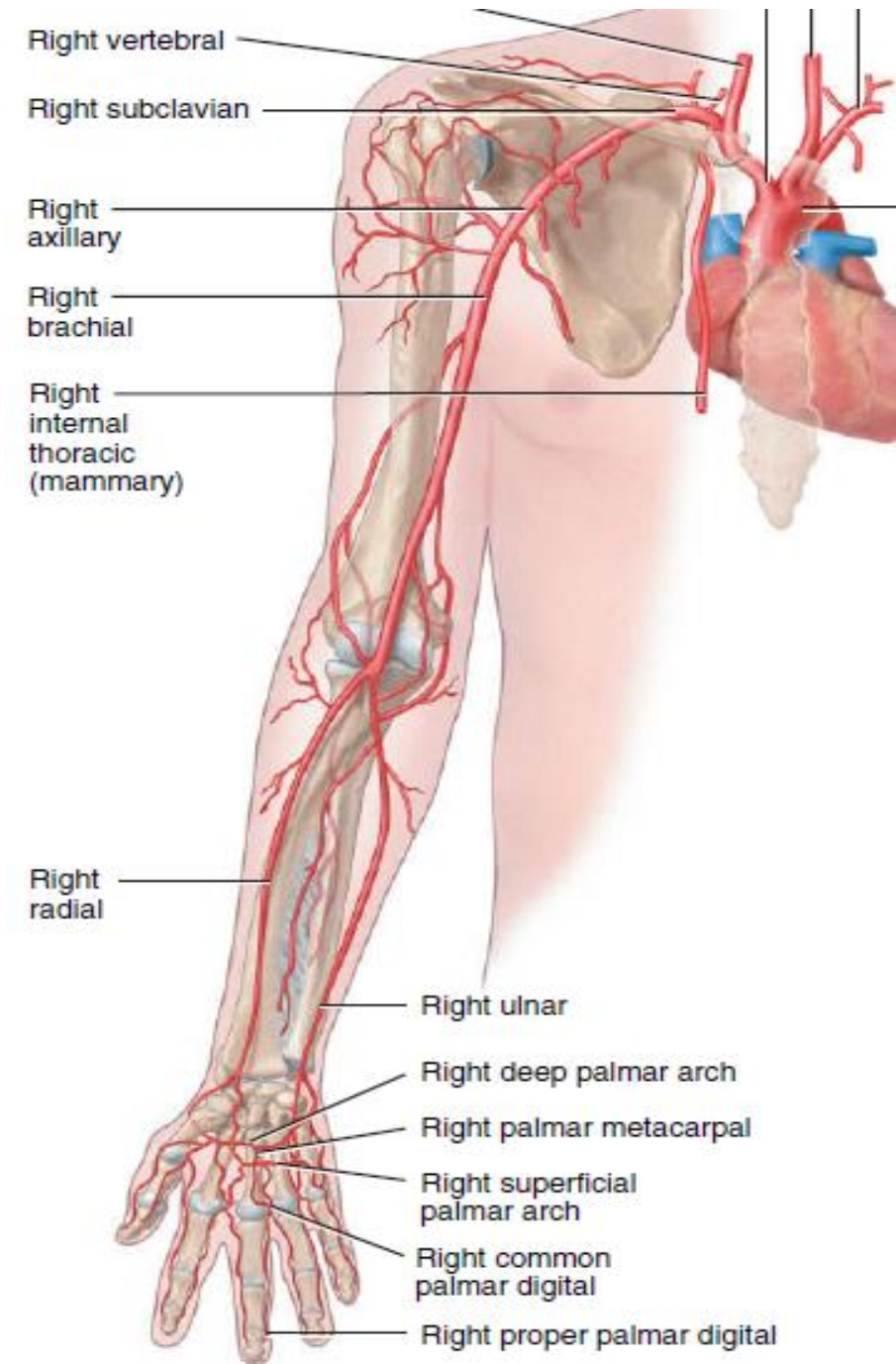
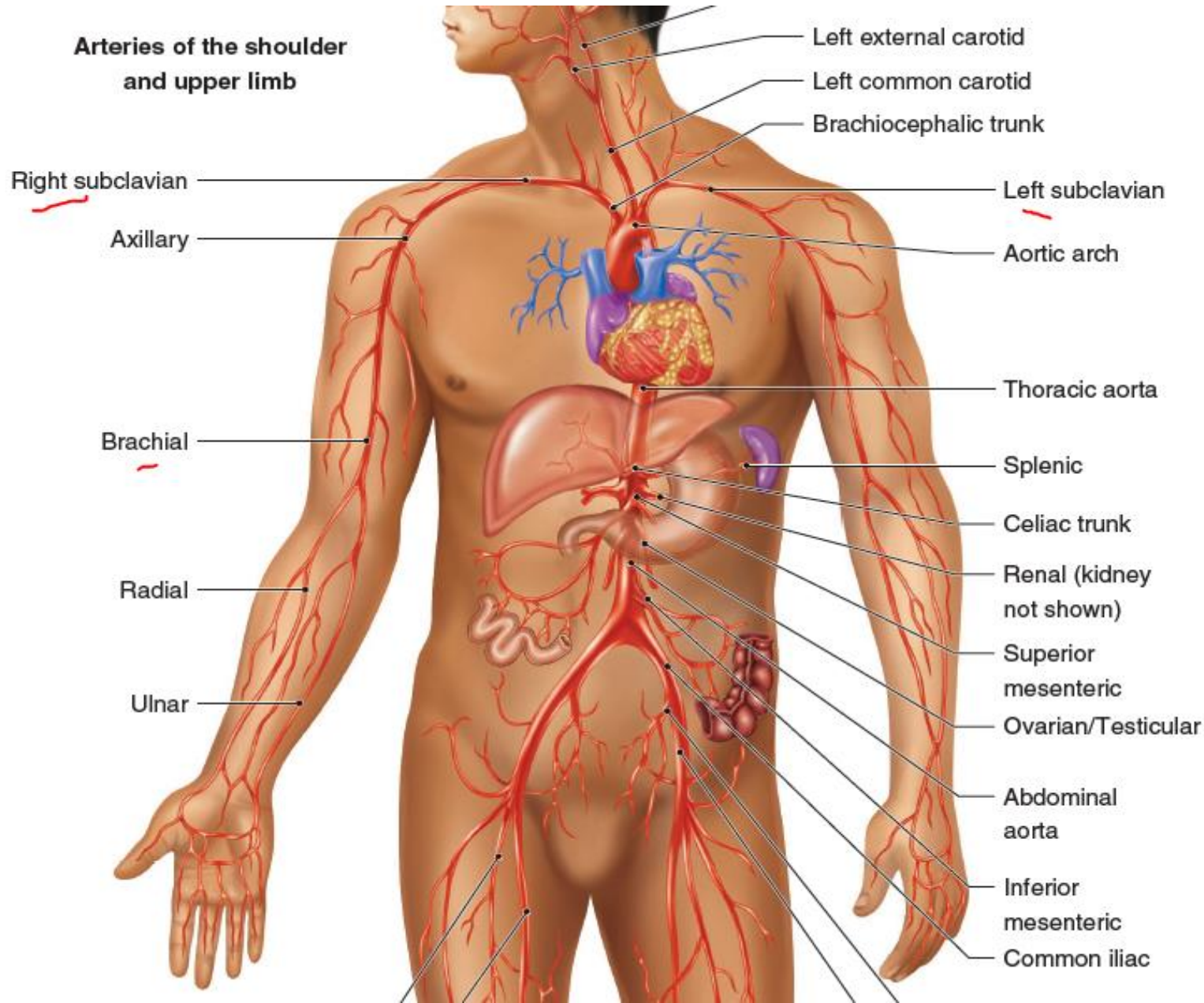
Arteries of the Thorax,,,



Arteries of the Upper Limbs

- The subclavian artery provides branches to the shoulder and passes inferior to the clavicle to become the **axillary artery**, which supplies branches to the thoracic wall and axillary region.
- **The subclavian artery**, located deep to the clavicle, becomes the **axillary artery** in the axilla (armpit), continues into the arm to become the **brachial artery**, which provides branches to serve the arm.
- At the elbow, the brachial artery divides to form a **radial artery** and an **ulnar artery**, which supply the forearm and wrist and merge to form a network of arteries supplying the hand.

Arteries of the Upper Limbs,,

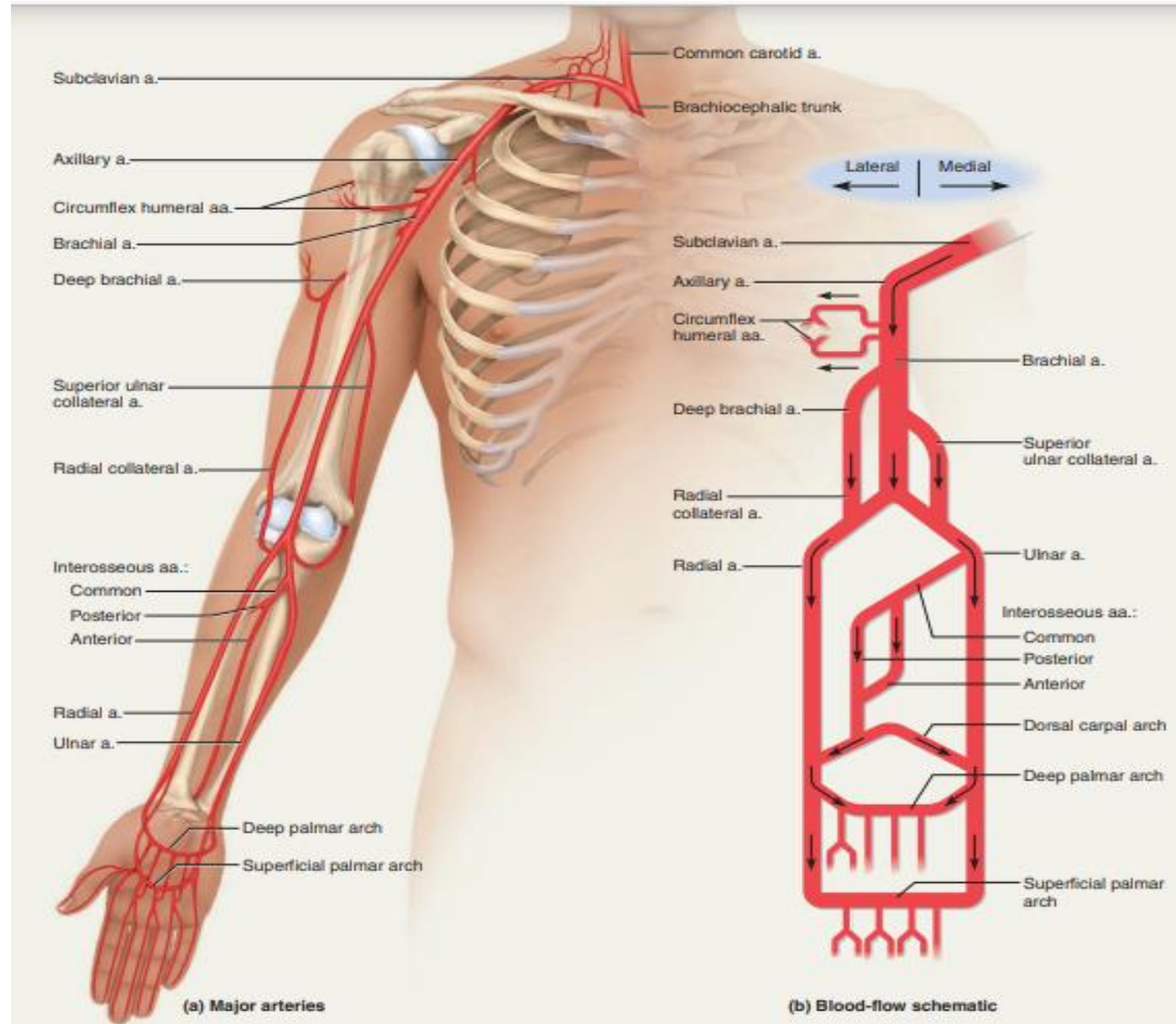


Arteries of the Upper Limbs,,

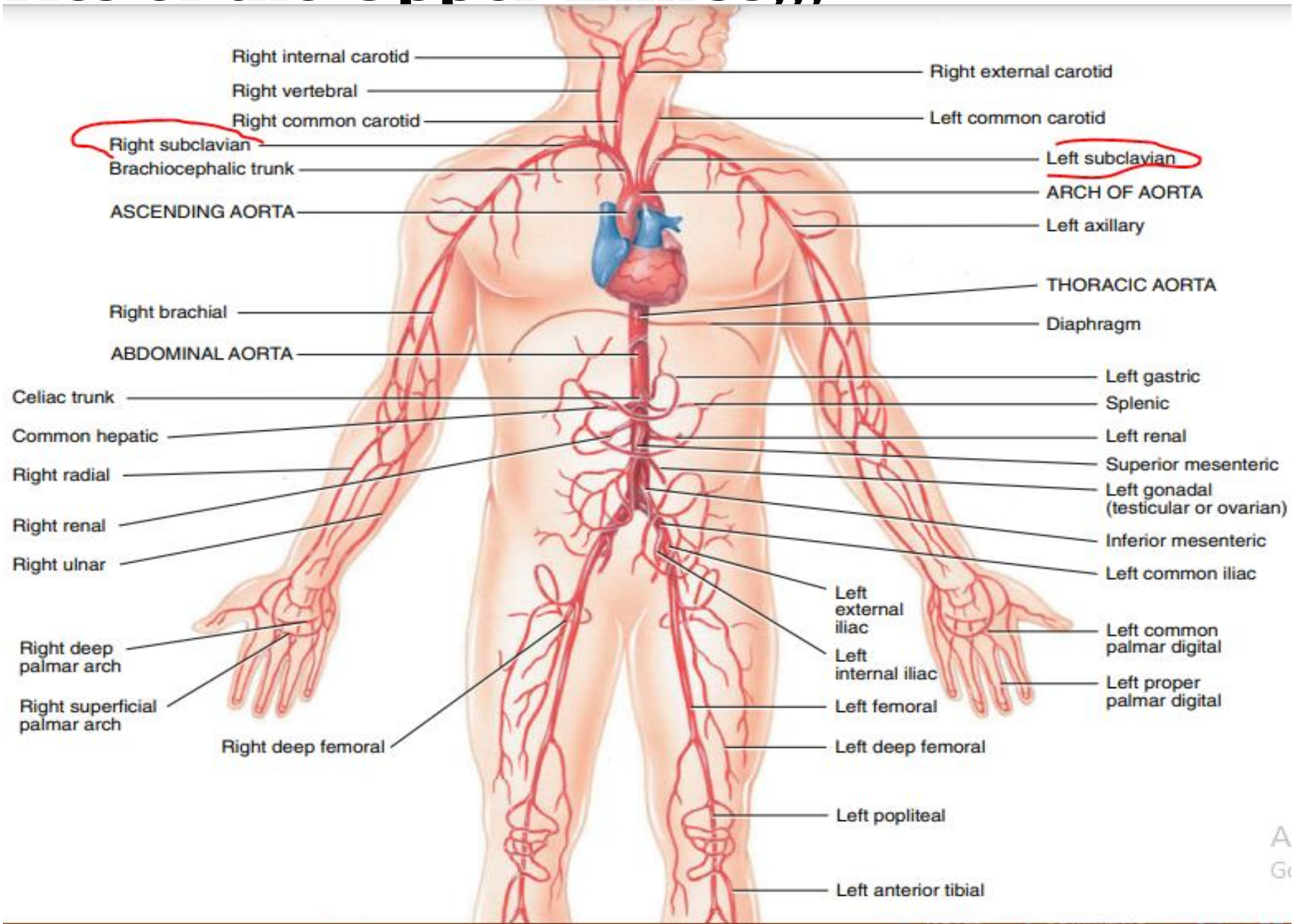
❖ **Right subclavian**- artery from brachiocephalic trunk and

❖ **Left subclavian** artery - from aortic arch.

- It has the several branches in the thorax but becomes the **axillary artery** as it passes in the axilla.
- It becomes the **brachial artery** in the arm.



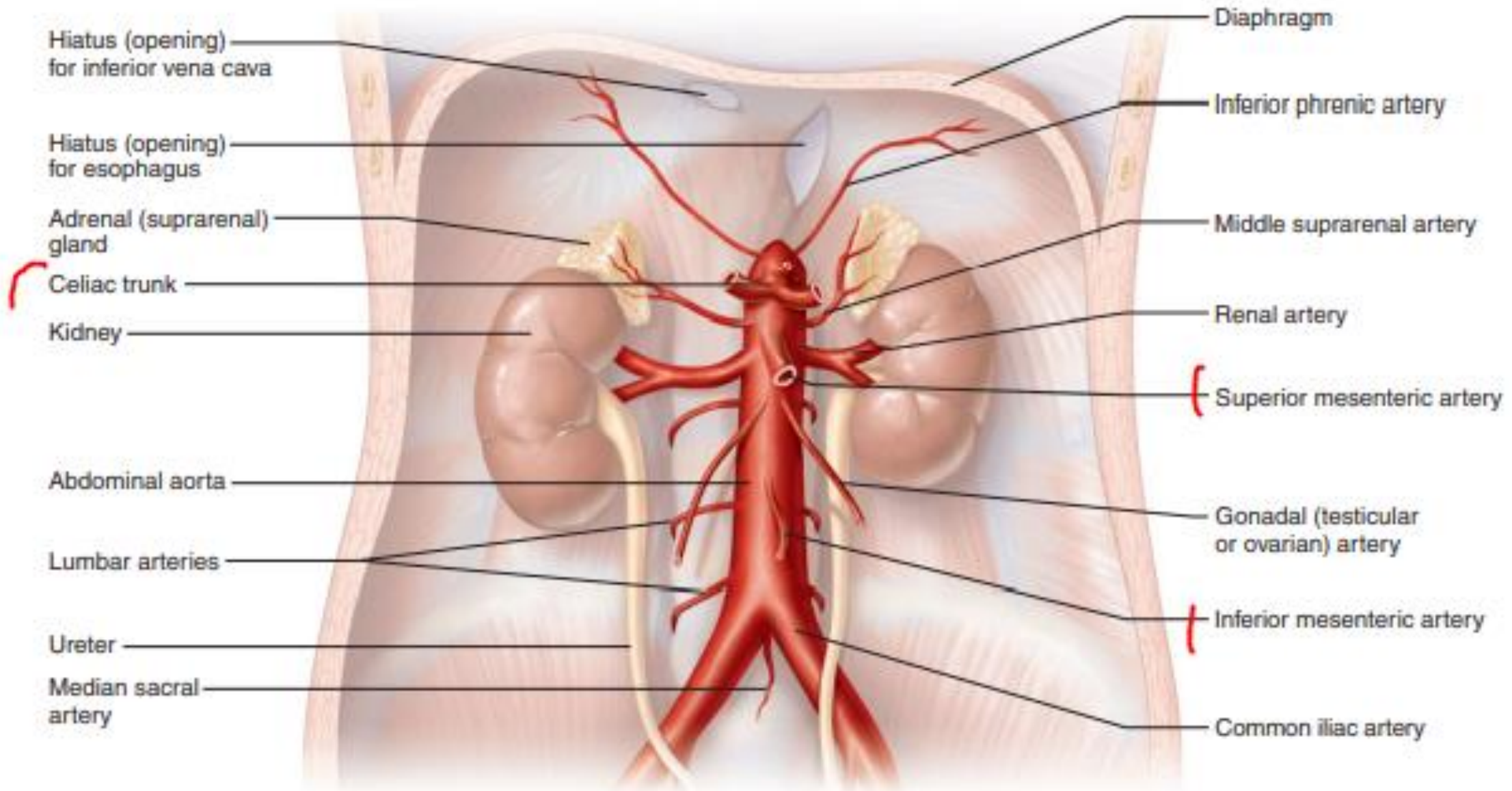
Arteries of the Upper Limbs,,



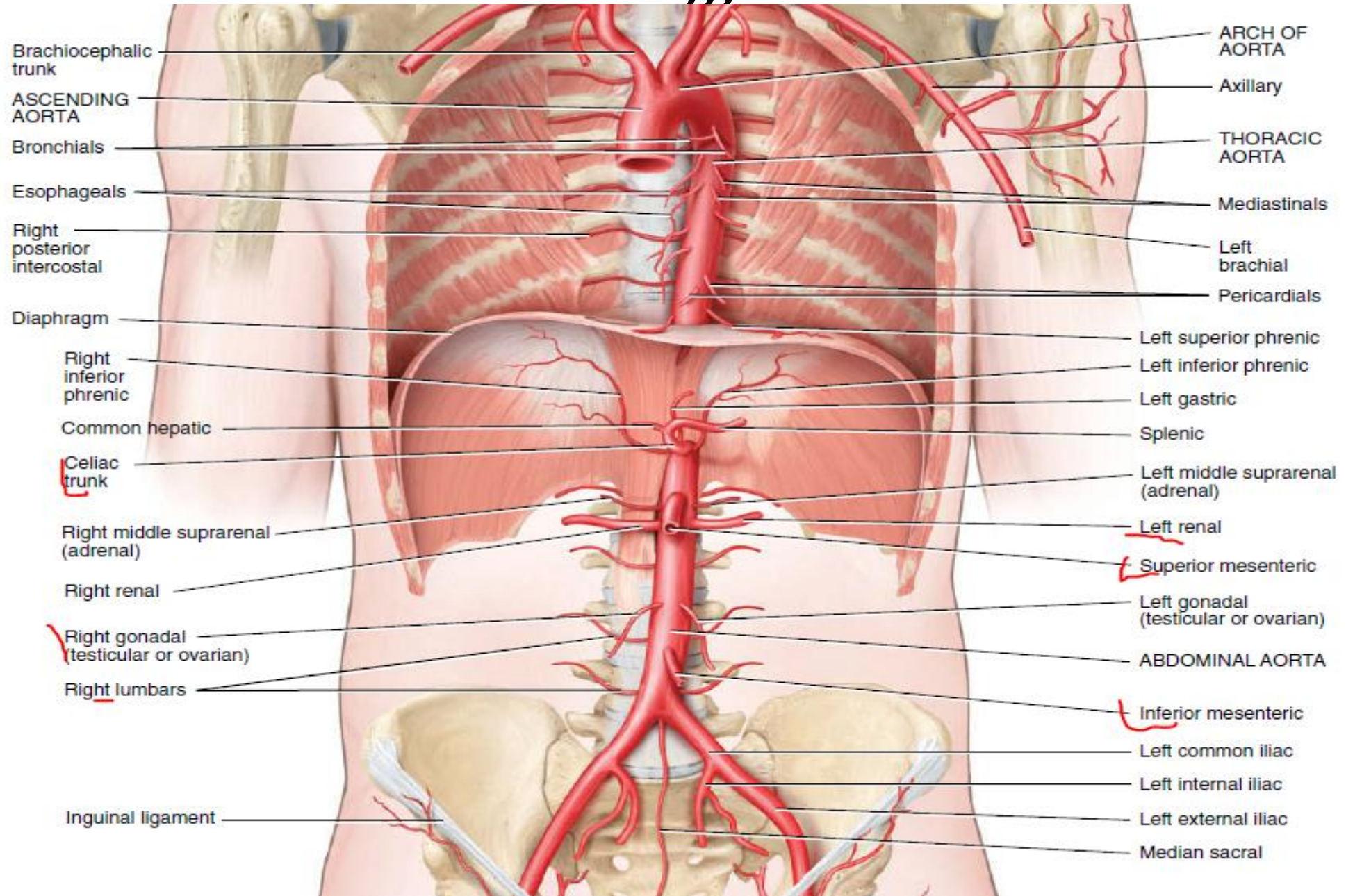
Arteries of the Abdomen

- The arteries to the abdominal organs arise from the abdominal aorta
- ❖ Abdominal aorta is the **segment between diaphragm and L4**.
- In a person at rest, about half of the entire arterial flow is present in these vessels.
- ❖ **Three midline(unpaired)** branches of abdominal aorta **(the celiac trunk, superior mesenteric artery, and inferior mesenteric artery)** bring blood to the digestive tube, the inner tube, abdominal viscera.
- **Paired branches** supply structures of the outer tube (adrenal glands, kidneys, gonads, and abdominal body wall).

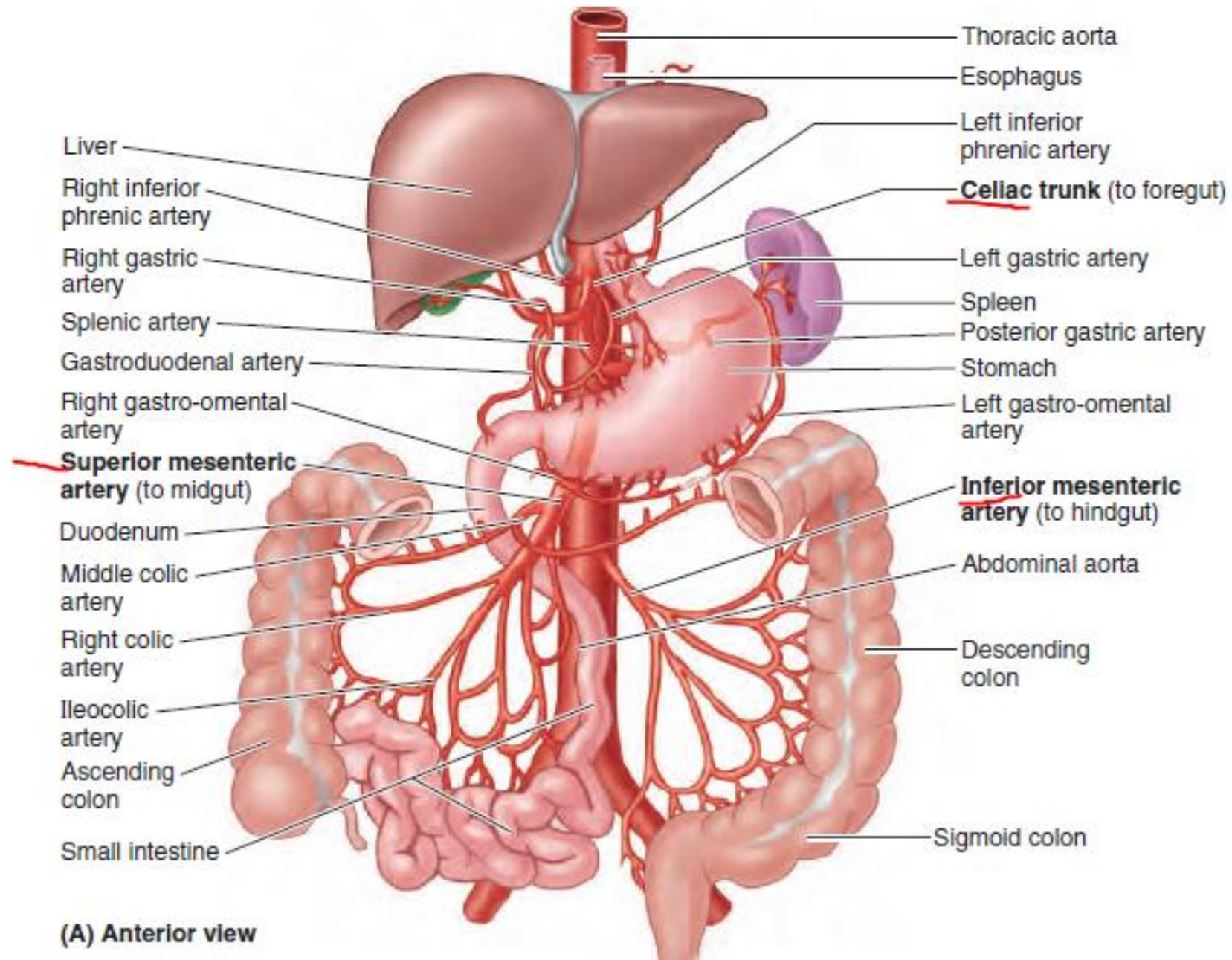
Arteries of the Abdomen,,



Arteries of the Abdomen,,,



Arteries of the Abdomen...



Celiac trunk

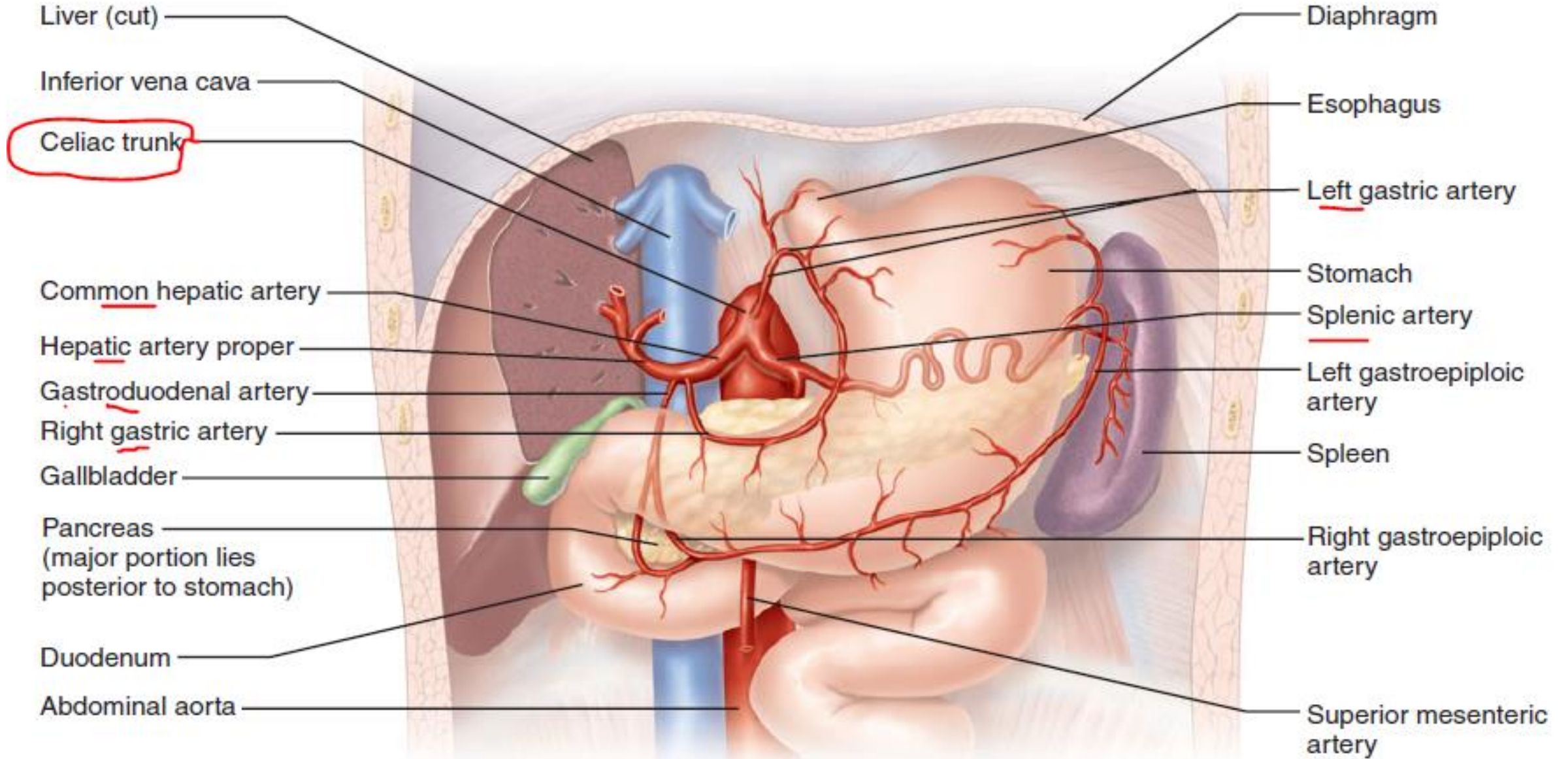
❑ **Celiac trunk** the short, wide, unpaired *celiac trunk* supplies the viscera in the superior part of the abdominal cavity.

- Specifically, it sends branches to the stomach, liver, gallbladder, pancreas, spleen, and a part of the small intestine (duodenum).

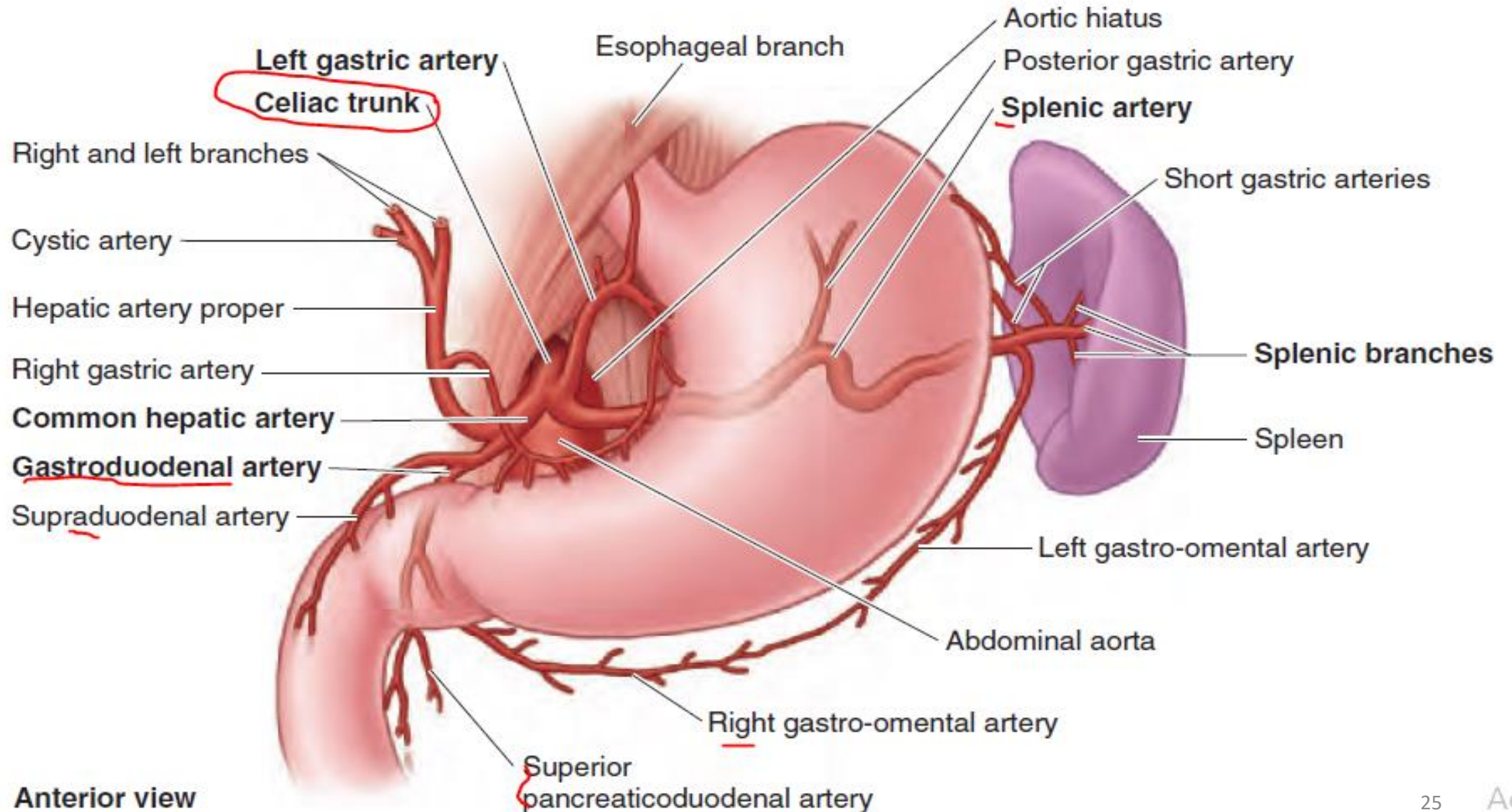
❑ **It emerges midventrally from the aorta at the level of T12 and divides into three branches:**

- ✓ **Splenic artery** (to spleen) supplies the spleen, stomach, and pancreas,
- ✓ **Left gastric artery** (to stomach) supplies the stomach and esophagus
- ✓ **Common hepatic artery** supplies to liver, gallbladder, stomach, duodenum & pancreas..

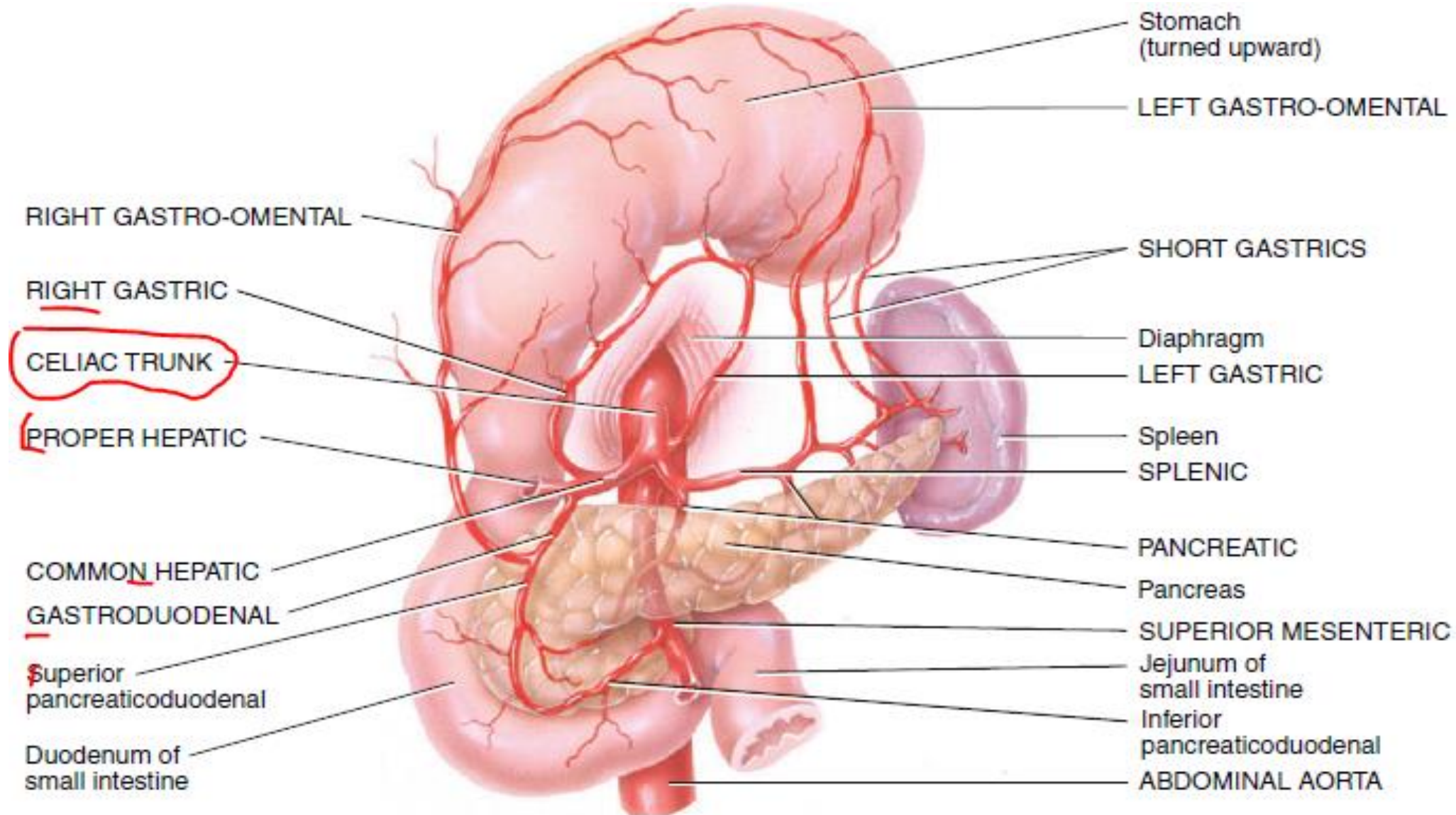
Celiac trunk,,,



Celiac trunk,,,



Celiac trunk,,,

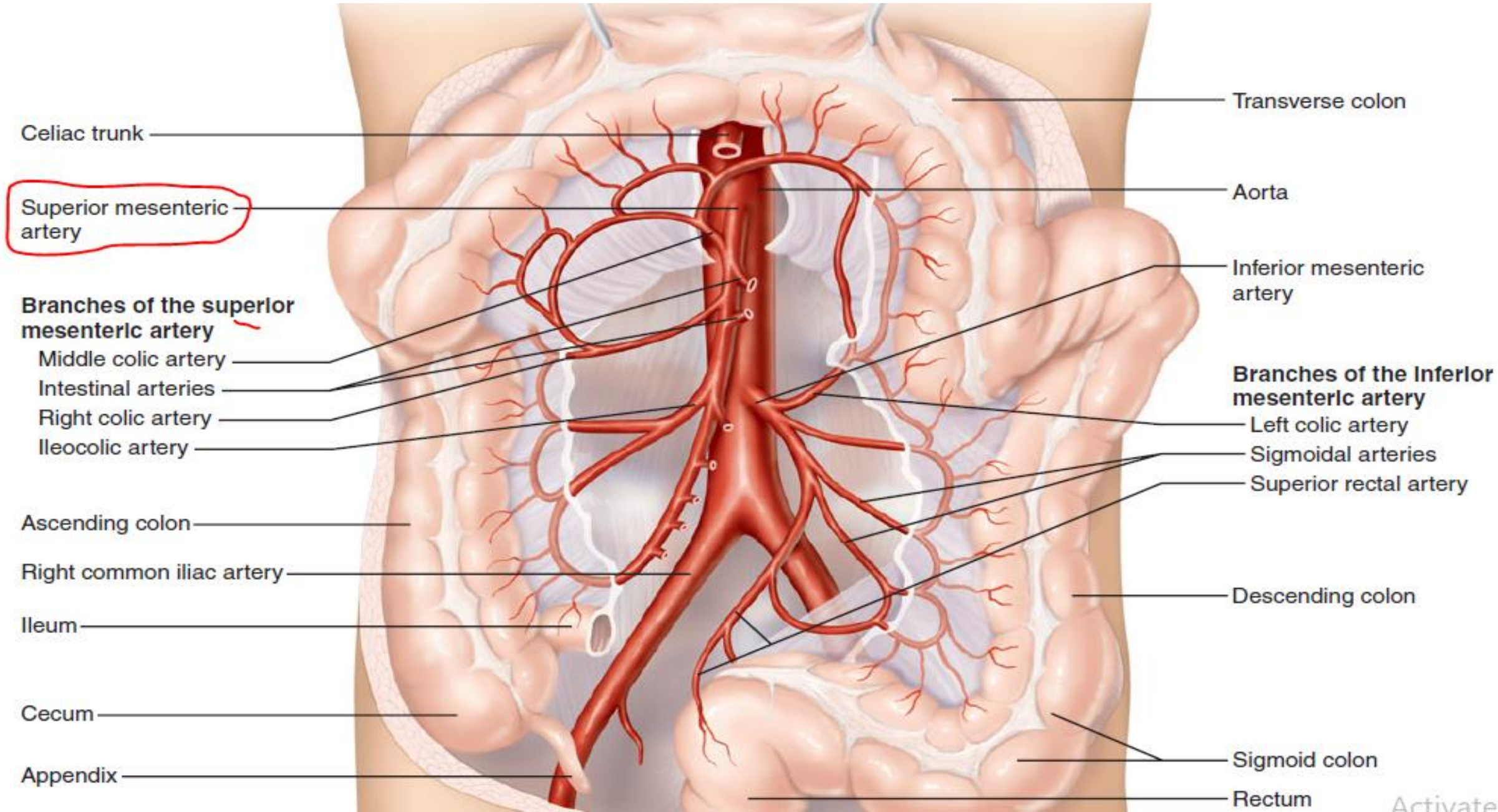


(a) Anterior view of celiac trunk and its branches

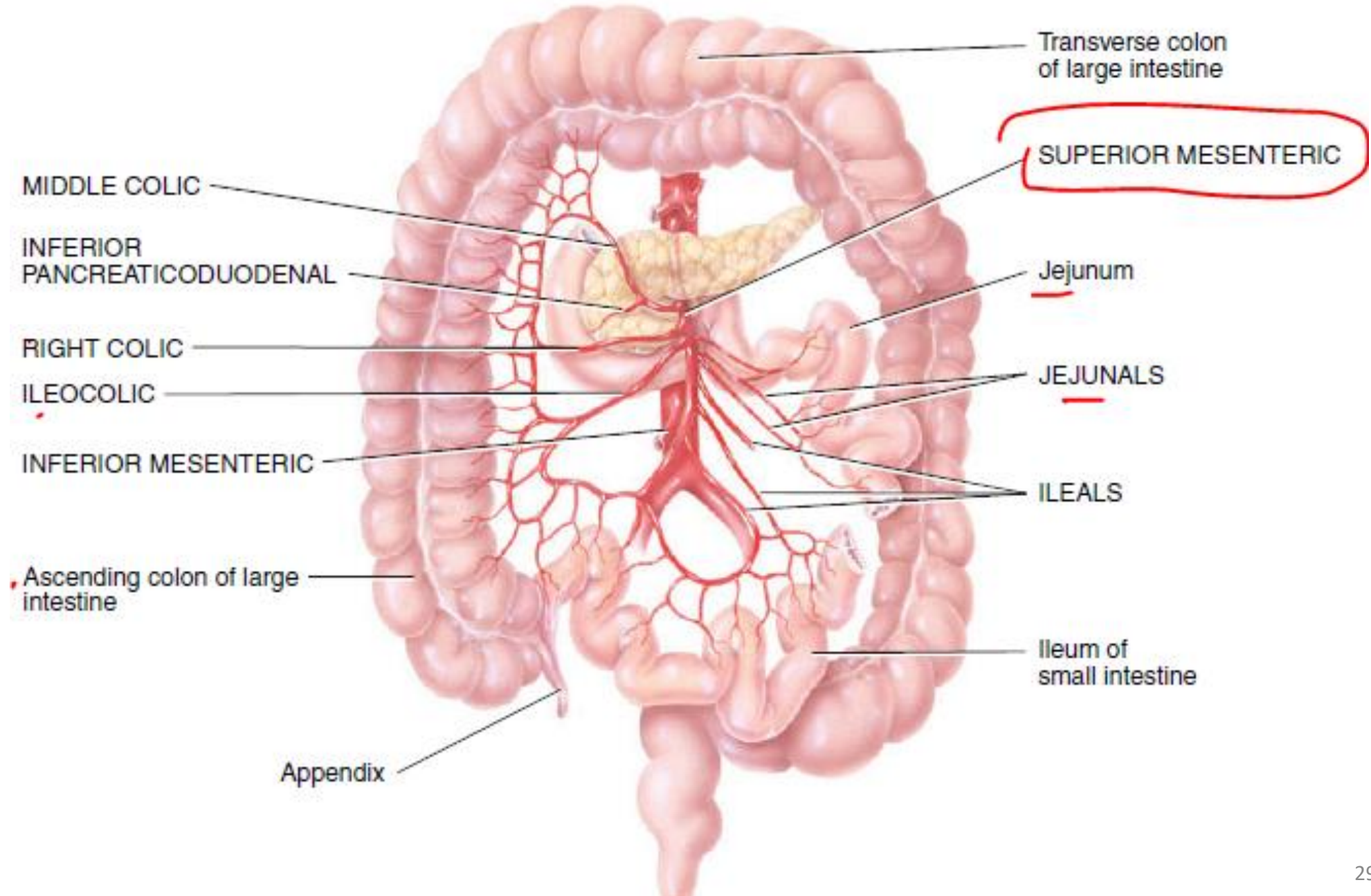
Superior mesenteric artery

- The large, unpaired **superior mesenteric** is the most significant intestinal blood supply serving nearly all of the small intestine and the proximal half of the large intestine.
- It arises from the aorta, posterior to the pancreas at 1st lumbar vertebra(L1).
- The superior mesenteric artery angles gradually to the right as it descends t.
- **From its left** side arise many **intestinal arteries** to the jejunum and ileum.
- **From its right** side emerge branches that supply the proximal half of the large intestine: the ascending colon, cecum, and appendix via the **ileocolic artery**; part of the ascending colon via the **right colic artery**; and part of the transverse colon via the **middle colic artery**.

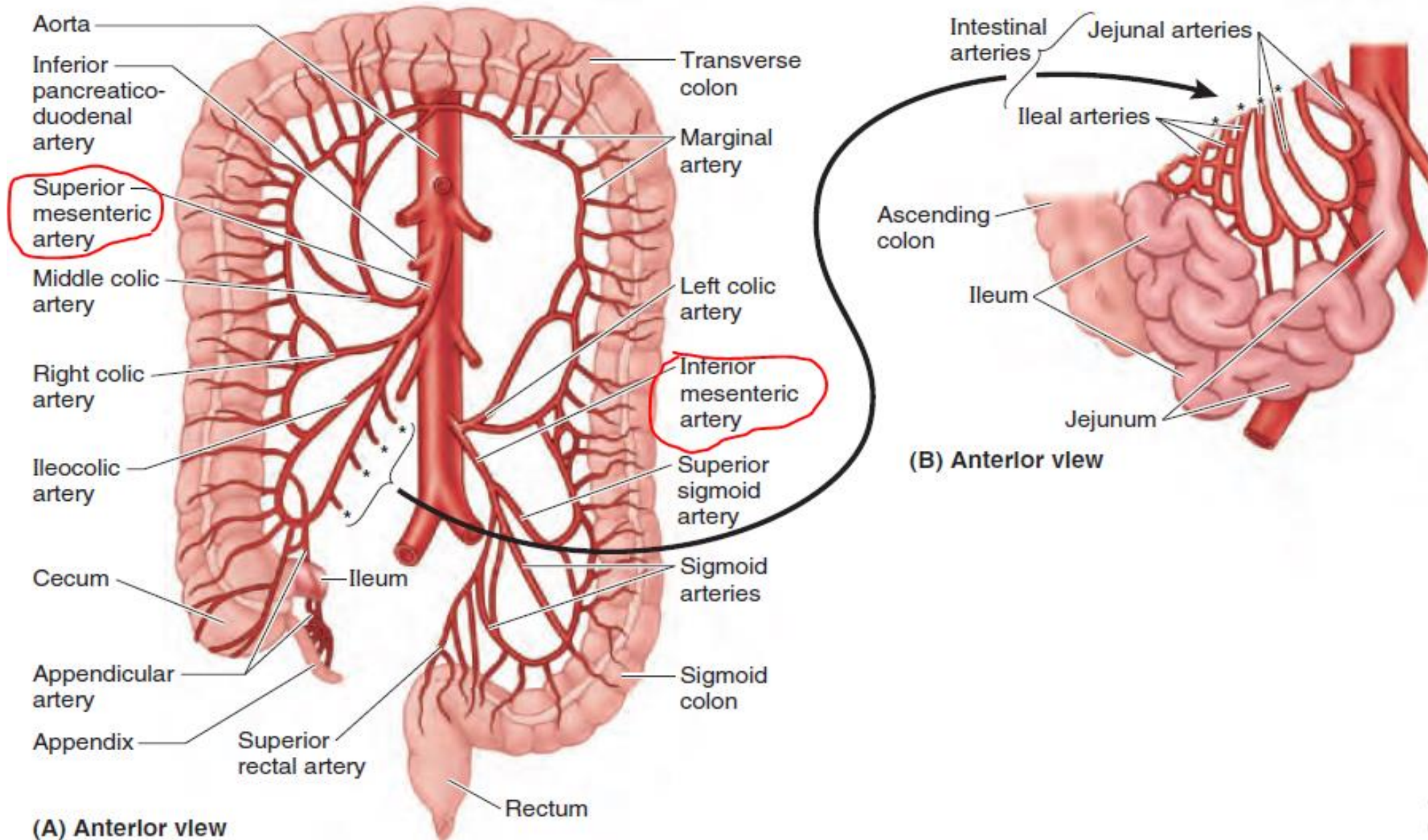
Superior mesenteric artery,,



Superior mesenteric artery,,



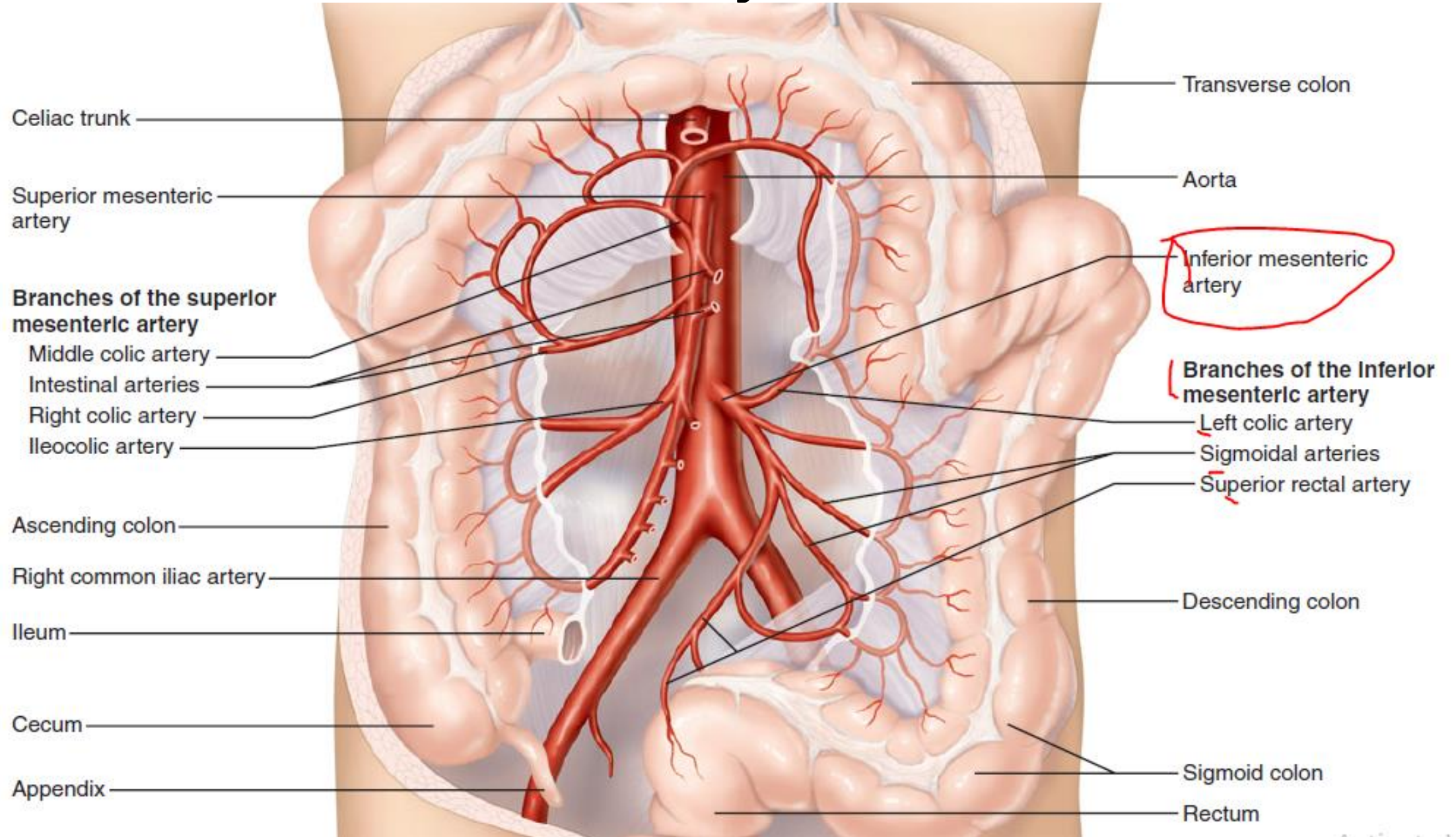
Superior mesenteric artery,,



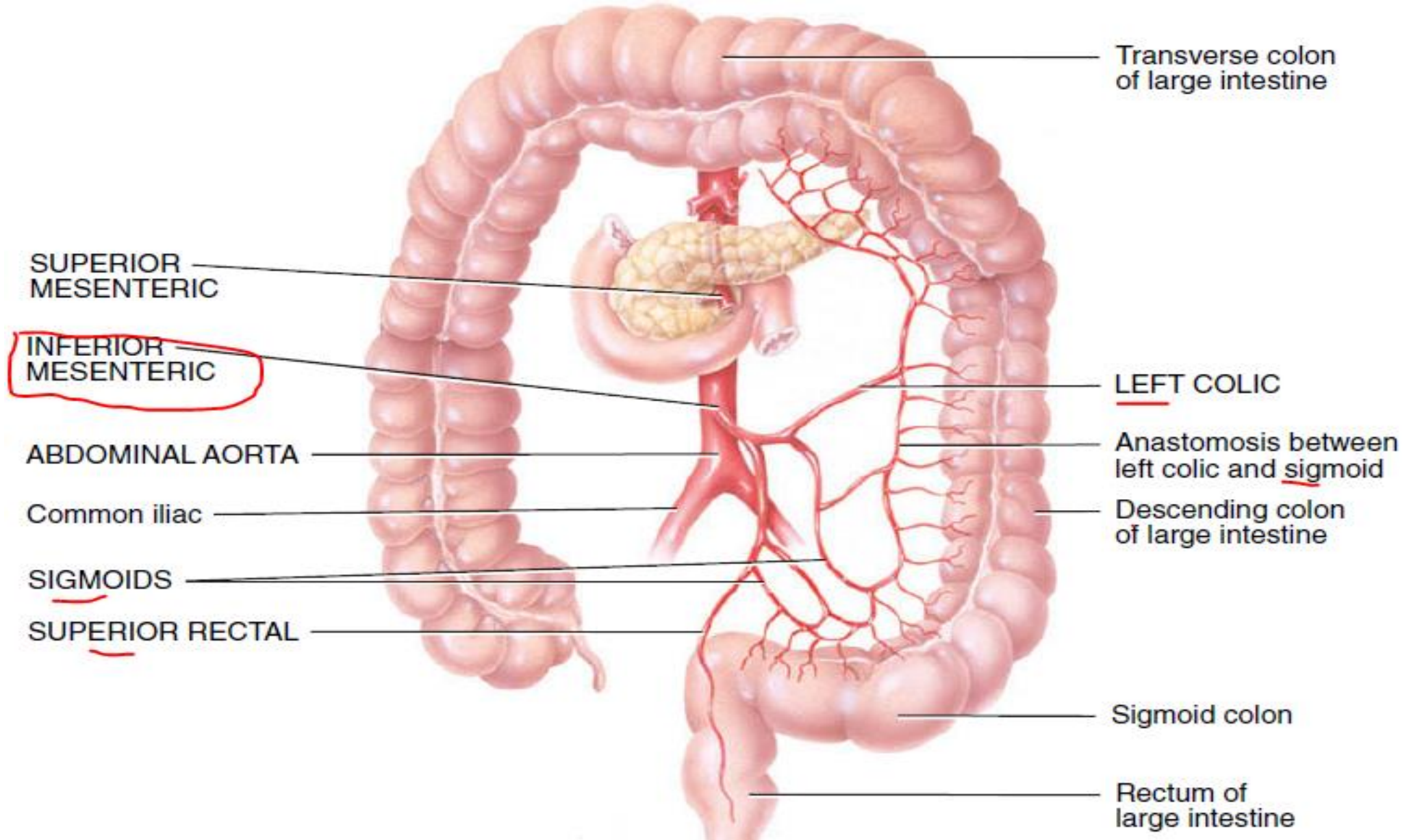
Inferior mesenteric artery

- Inferior Mesenteric Artery The unpaired **inferior mesenteric artery** is the final major branch of the abdominal aorta, arising midventrally at the level of L3.
- It serves the distal half of the large intestine, from the last part of the transverse colon to the middle part of the rectum.
- Its branches are the **left colic** (which joins with the middle colic artery on the transverse colon), **sigmoidal**, and **superior rectal arteries**.

Inferior mesenteric artery,,,



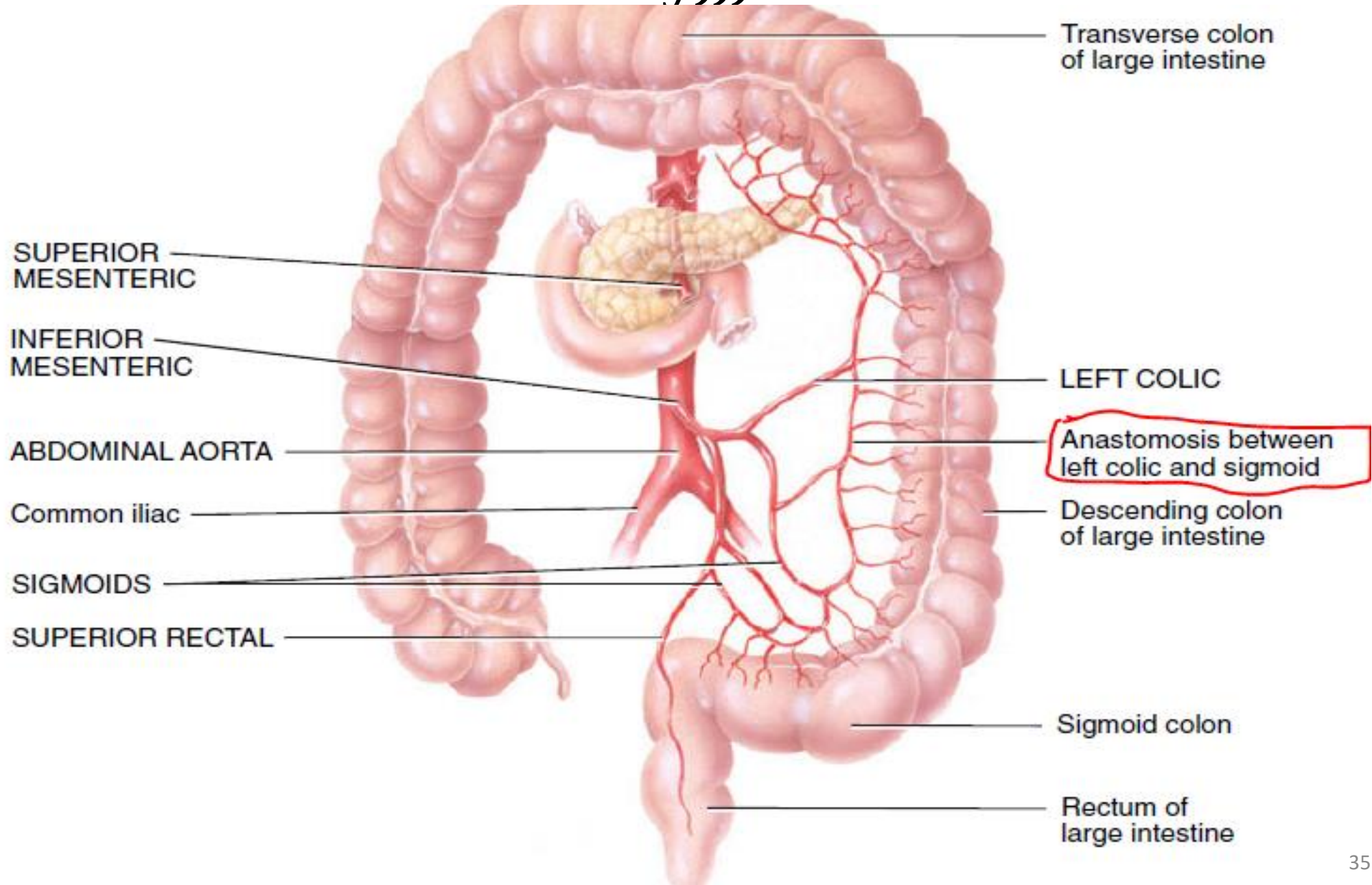
Inferior mesenteric artery,,



Inferior mesenteric artery,,

- Here is typical example of **anastomosis between left colic artery and sigmoid artery.**
- The union of the branches of two or more arteries supplying the same body region is called an **anastomosis.**
- Most tissues of the body receive blood from more than one artery.
- Anastomoses between arteries provide alternative routes for blood to reach a tissue or organ.
- The alternative route of blood flow to a body part through an anastomosis is known as **collateral circulation.**
- Anastomoses may also occur between veins and between arterioles and venules.
- Arteries that do not anastomose are known as **end arteries.**

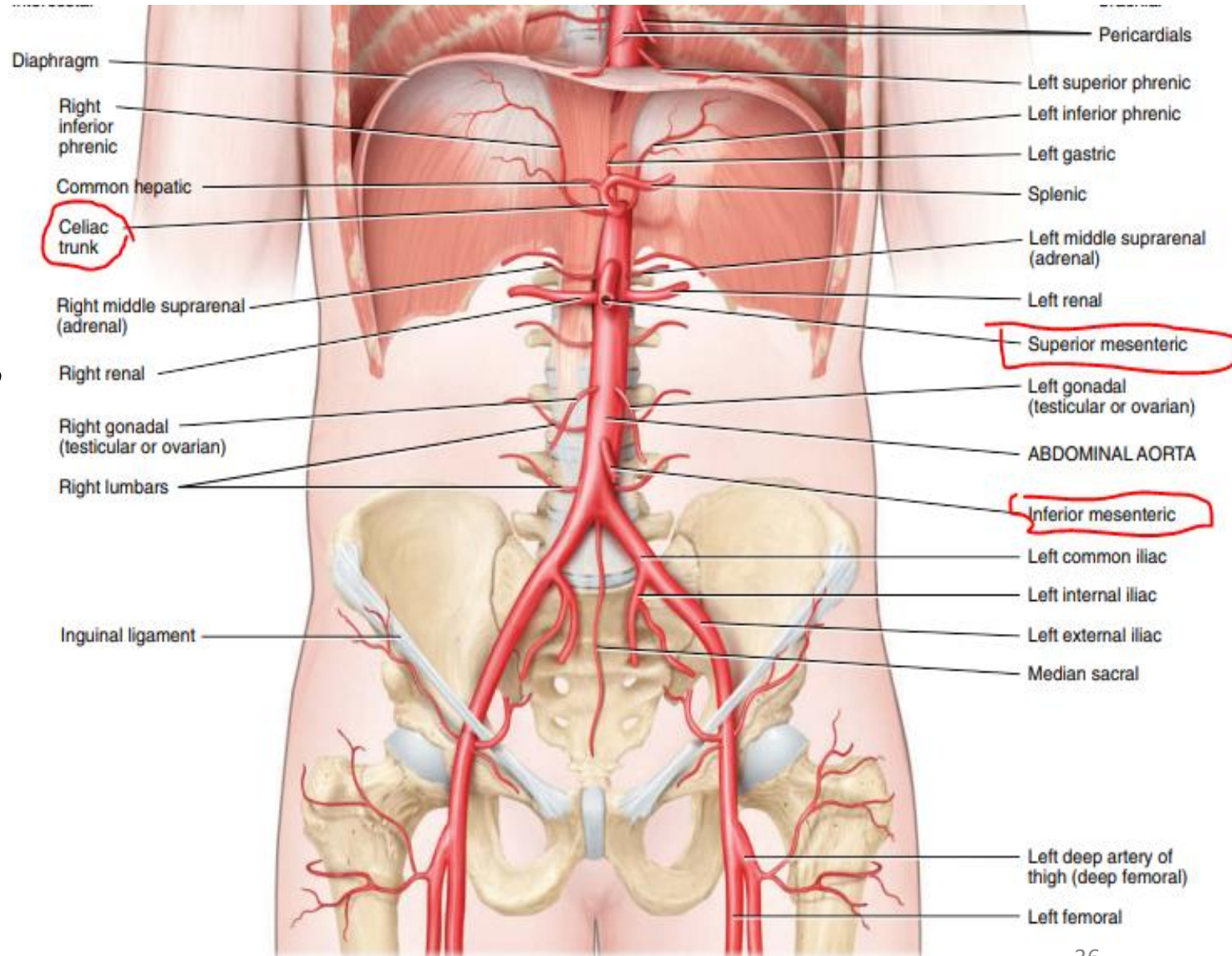
Inferior mesenteric artery,,,



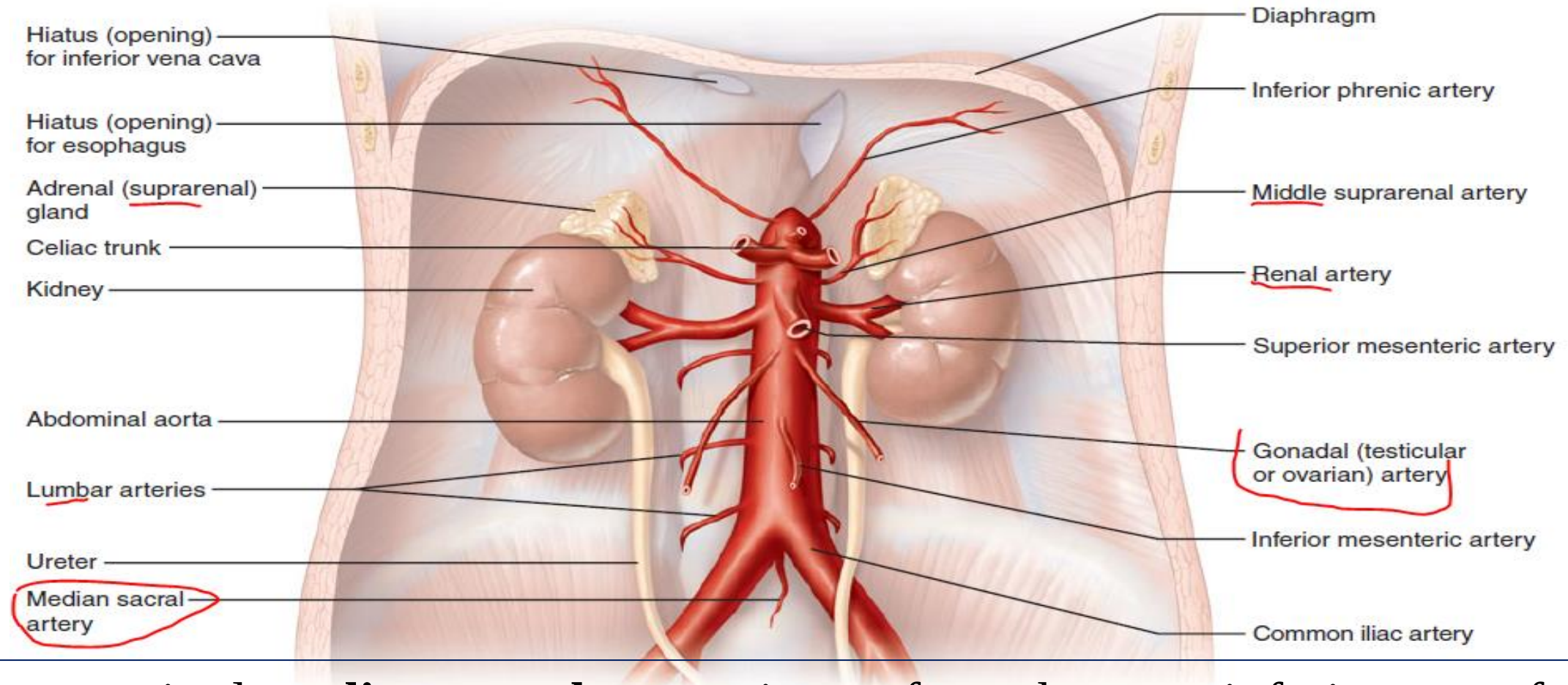
Paired branches of the Abdomen

❖ Paired branches

- ✓ **Renal artery**– to kidney
- ✓ **Suprarenal artery** - to adrenal glands
- ✓ **Gonadal Arteries:** They branch from the aorta at L2, the level where the gonads first develop in the embryo.
 - **Testicular artery** - to testes
 - **ovarian artery** - to ovaries
- ✓ **Four pairs of lumbar arteries** arise from the posterolateral surface of the aorta in the lumbar region.



Paired branches of the Abdomen,,

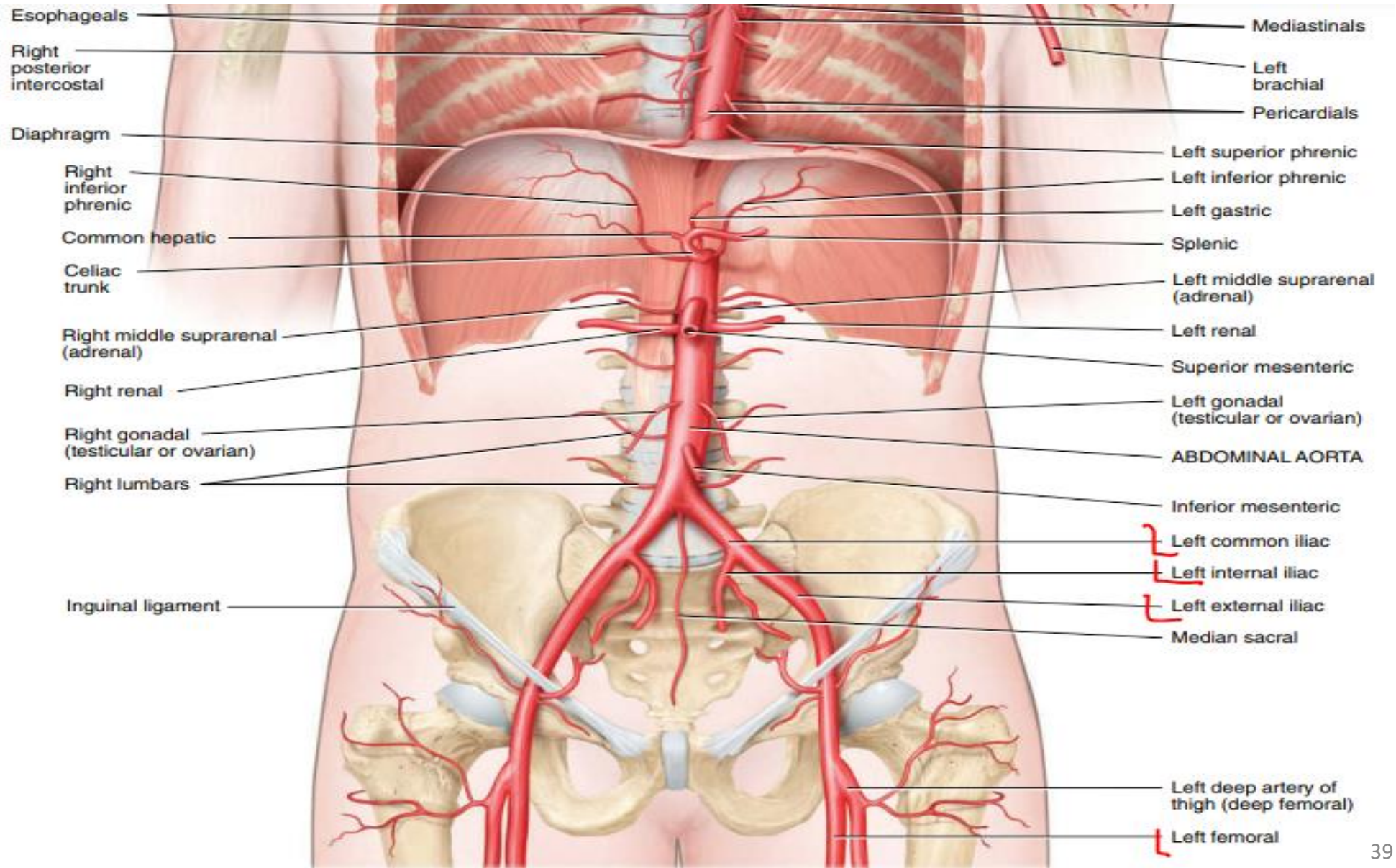


- ❑ The unpaired **median sacral artery** issues from the most inferior part of the aorta.
- ❑ As it descends, this thin artery supplies the sacrum and coccyx along the midline.

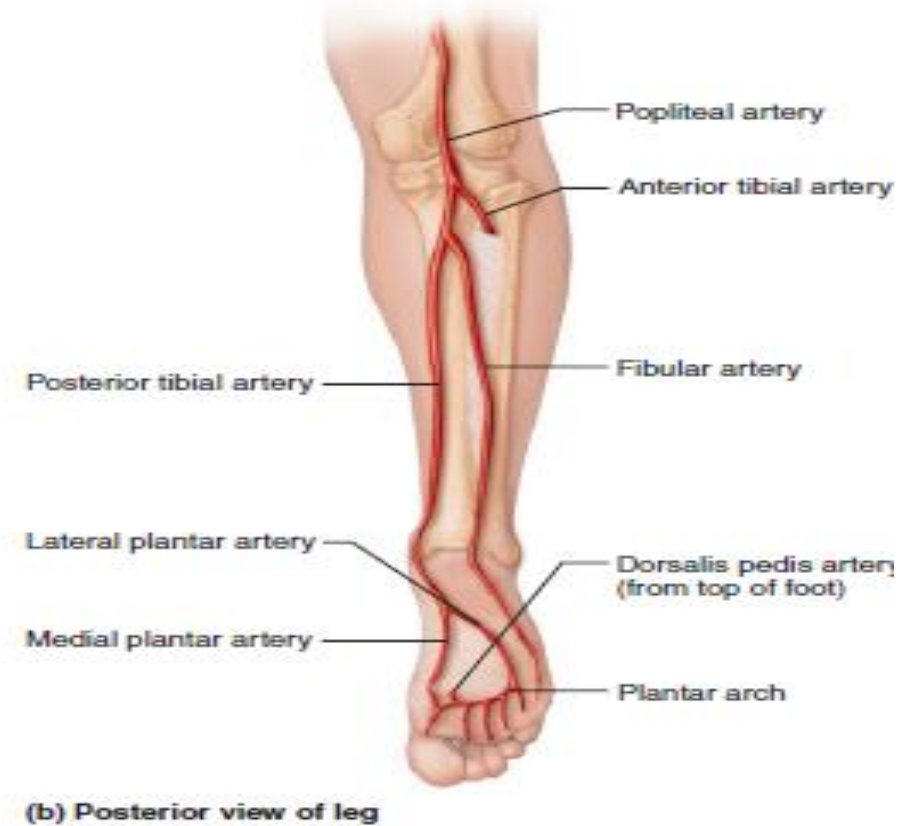
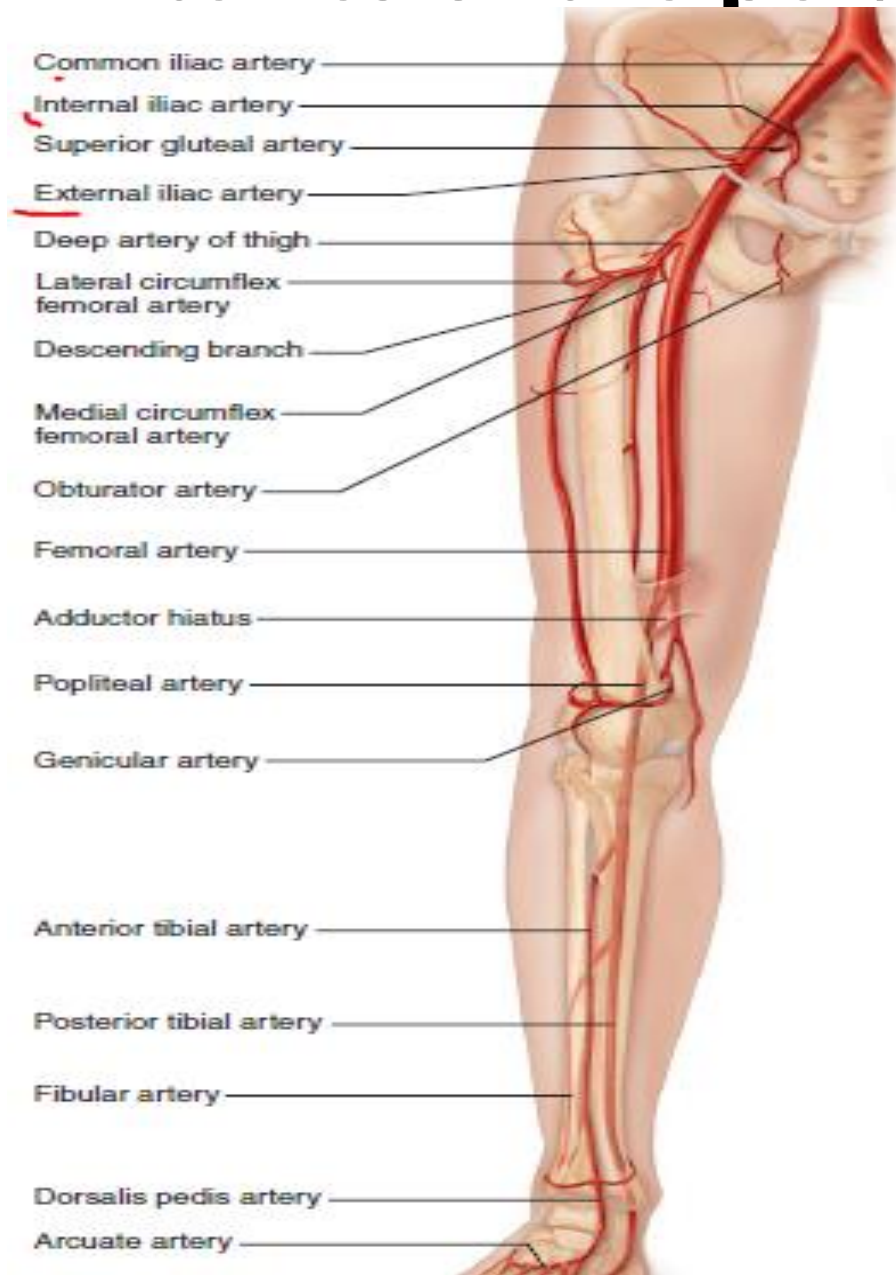
Arteries of the pelvis and lower limbs

- The abdominal aorta divides at the level of the 4th lumbar vertebra into two **common iliac arteries**.
 - ❑ right common iliac arteries
 - ❑ left common iliac arteries
 - ❖ **The common iliac divides into**
 - ✓ the internal iliac and
 - ✓ external iliac.
- The branches of **internal iliac arteries** supply blood to the pelvic walls, pelvic viscera, buttocks, medial thighs, and perineum.
- The right and left **external iliac arteries** carry blood to the lower limbs.

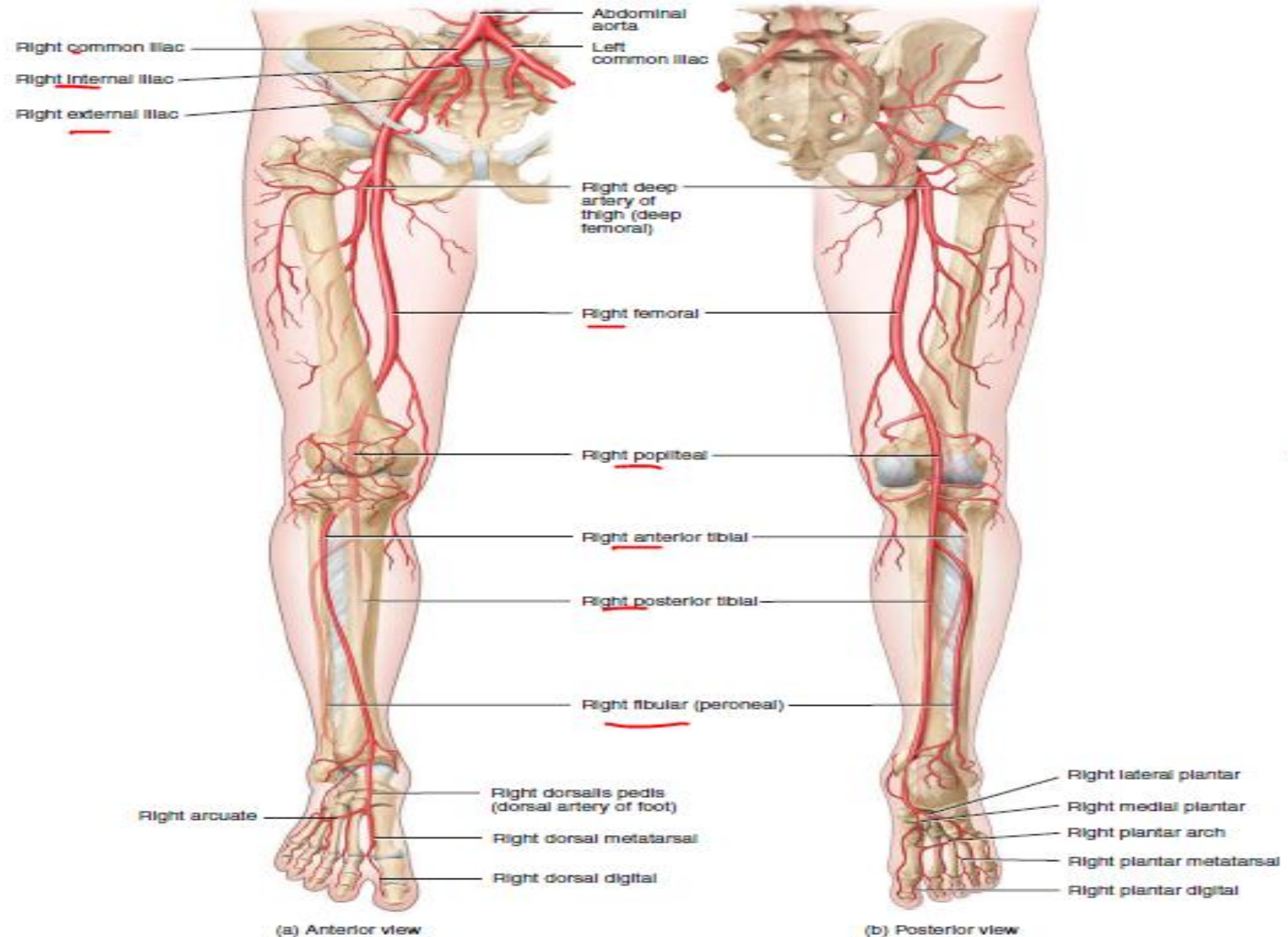
Arteries of the pelvis and lower limbs,,



Arteries of the pelvis and lower limbs,,



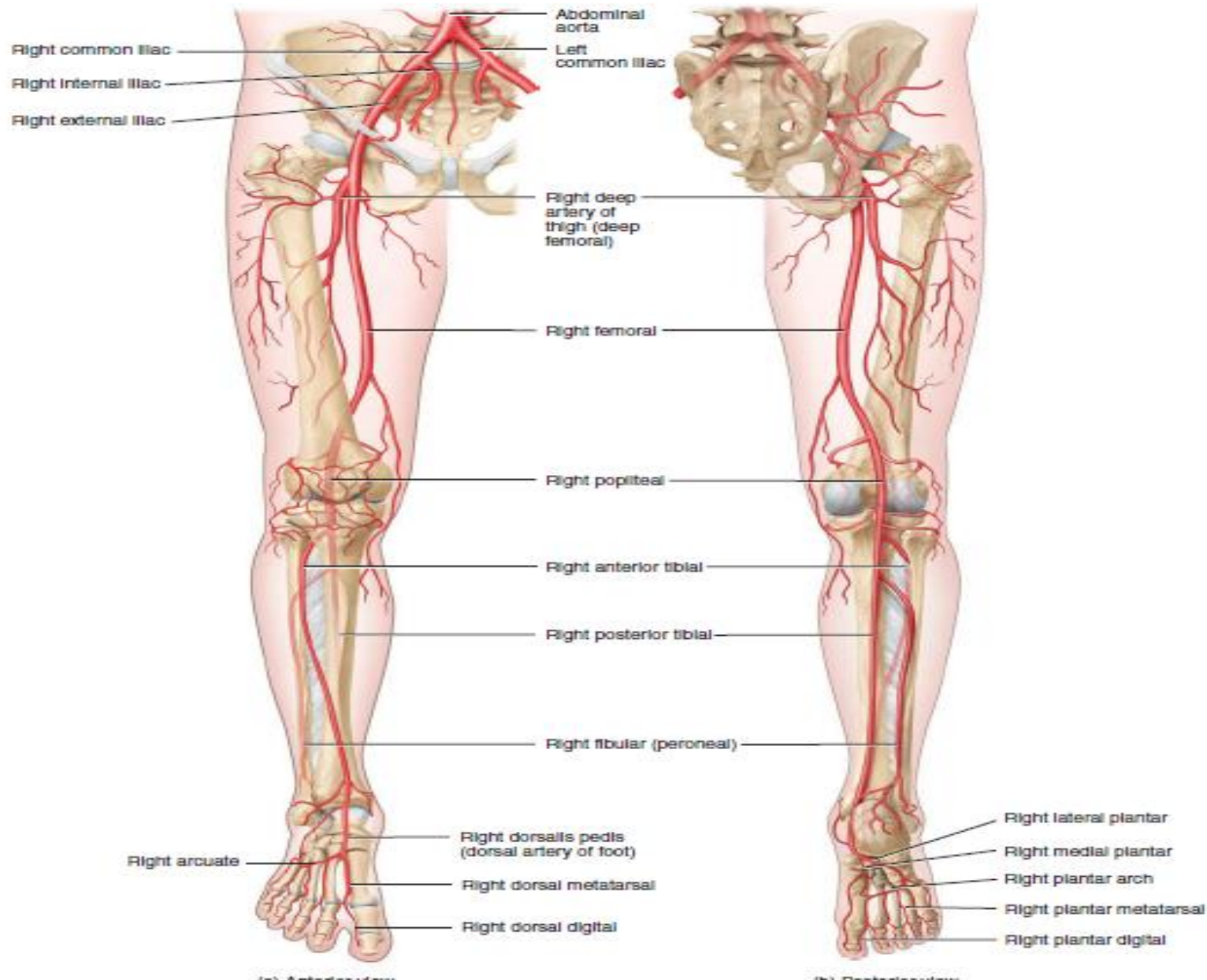
Arteries of the pelvis and lower limbs,,



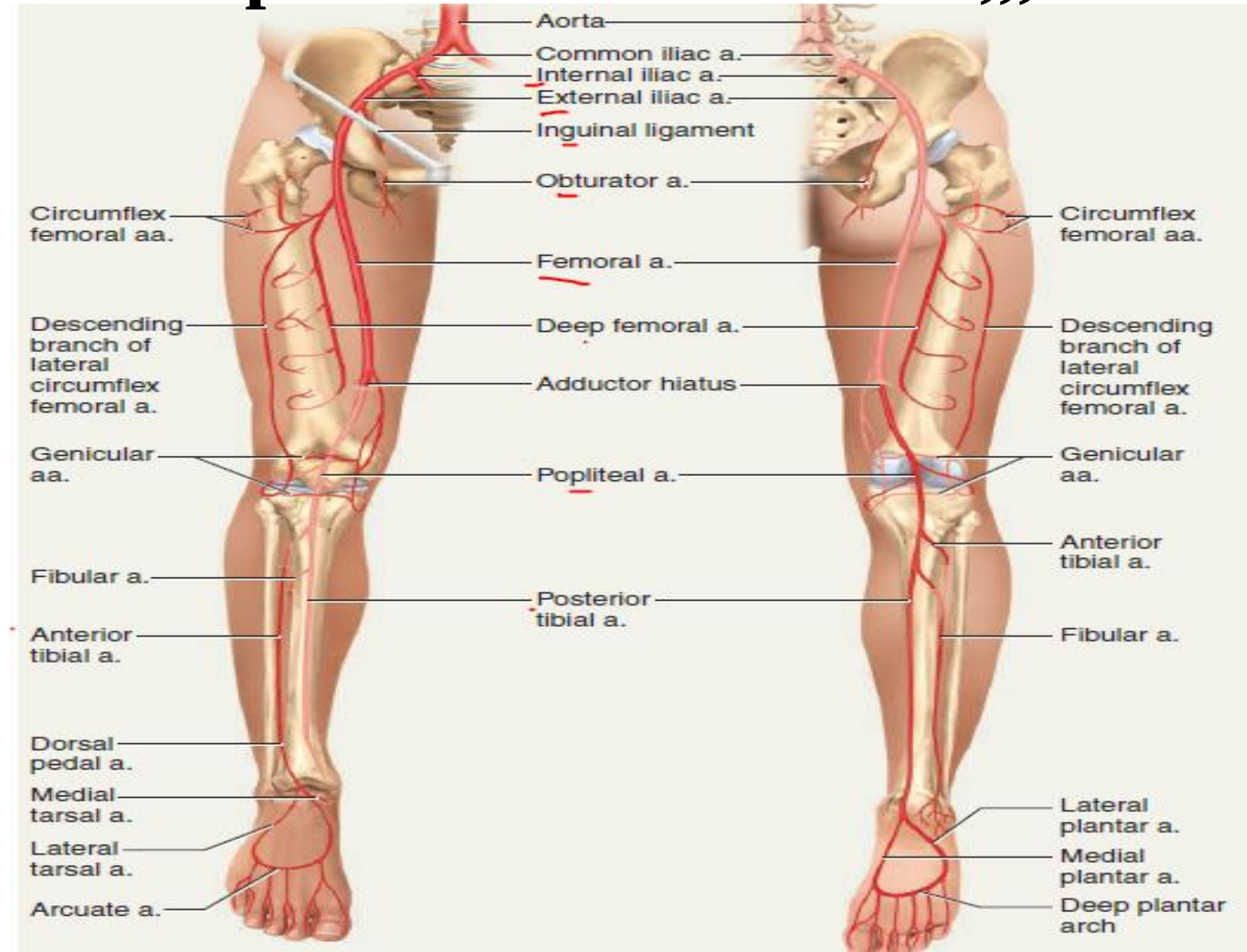
Arteries of the pelvis and lower limbs,,

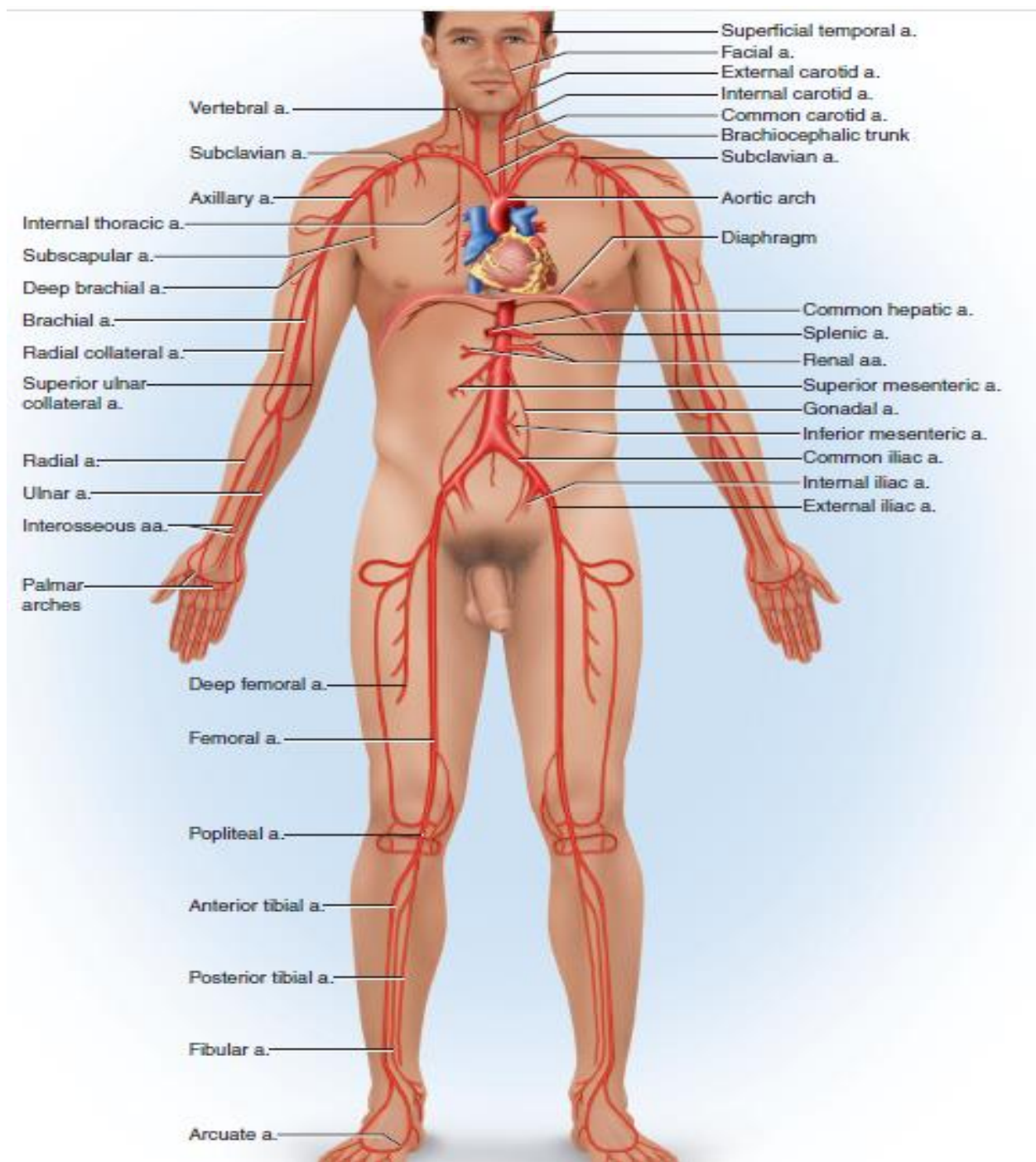
- The external iliac artery in the pelvis becomes the **femoral artery** in the thigh, and it becomes the **popliteal artery** in the popliteal space, which is the posterior region of the knee.
- The popliteal artery branches slightly inferior to the knee to give off the **anterior tibial artery** and the **posterior tibial artery**, both of which give rise to arteries that supply blood to the leg and foot.
- The anterior tibial artery becomes the **dorsalis pedis artery** at the ankle.

Arteries of the pelvis and lower limbs,,



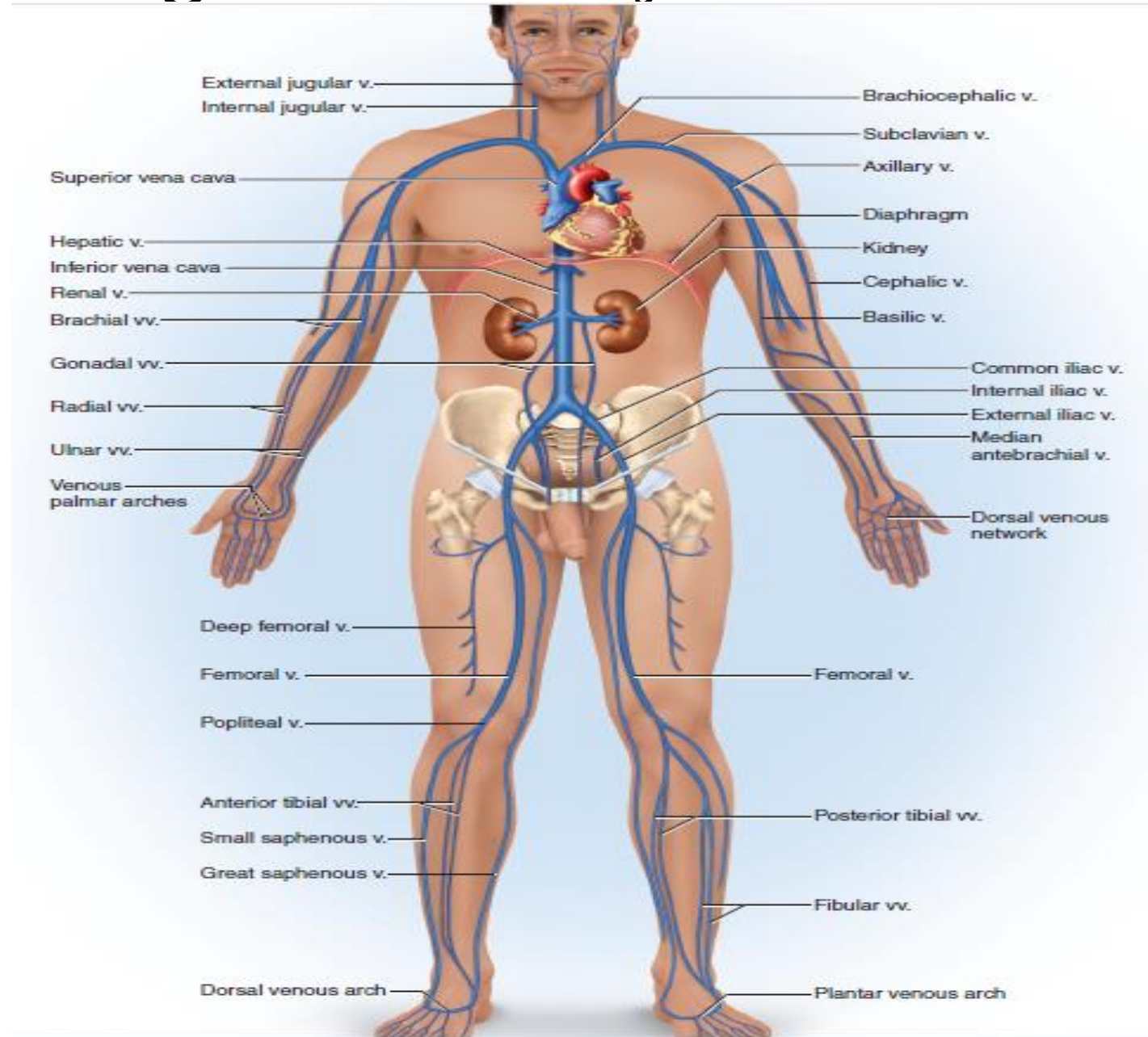
Arteries of the pelvis and lower limbs,,





Venous drainage of the body

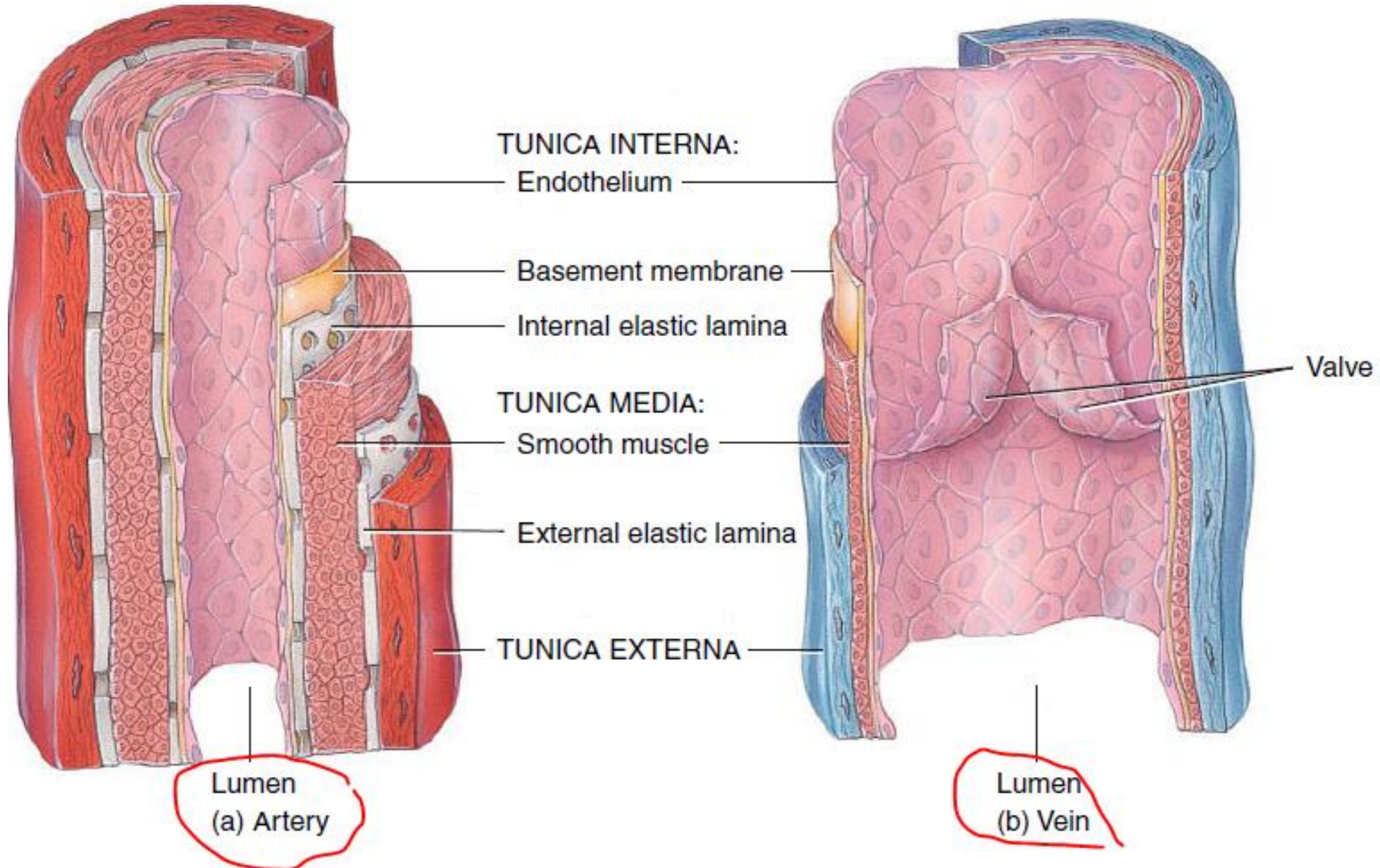
Venous drainage of the body



Venous drainage of the body,,

- **Veins** carry blood from capillaries back to the heart; the smaller veins are called **venules**.
- **The same 3 tissue layers** are present in veins as in the walls of arteries, but there are some differences when compared to the arterial layers.
- **The inner layer of** veins is smooth endothelium, but at intervals this lining is folded to form **valves**.
- **Valves prevent backflow of blood and are most numerous in veins of the legs.**

Venous drainage of the body,,



Venous drainage of the body,,,

- **In general, arteries are deep;** veins may be superficial or deep.
- Superficial veins are located just beneath the skin and can be seen easily.
- Because there are no large superficial arteries, the names of superficial veins do not correspond to those of arteries.
- **Superficial veins are clinically important** as sites for withdrawing blood or giving injections.
- Deep veins generally travel alongside arteries and usually bear the same name.
- **Arteries usually follow definite pathways;** veins are more difficult to follow because they connect in irregular networks in which many tributaries merge to form a large vein.

Venous drainage of the body,,,

- Although most veins run with corresponding arteries, there are some important differences in the distributions of arteries and veins:
 - Whereas just one systemic artery leaves the heart—the aorta exiting the left ventricle, three major veins enter the right atrium of the heart: the superior and inferior venae cavae and the coronary sinus.
 - All large and medium-sized arteries have deep locations for protection. In contrast, many veins lie just beneath the skin, unaccompanied by any arteries; these are called *superficial veins*.

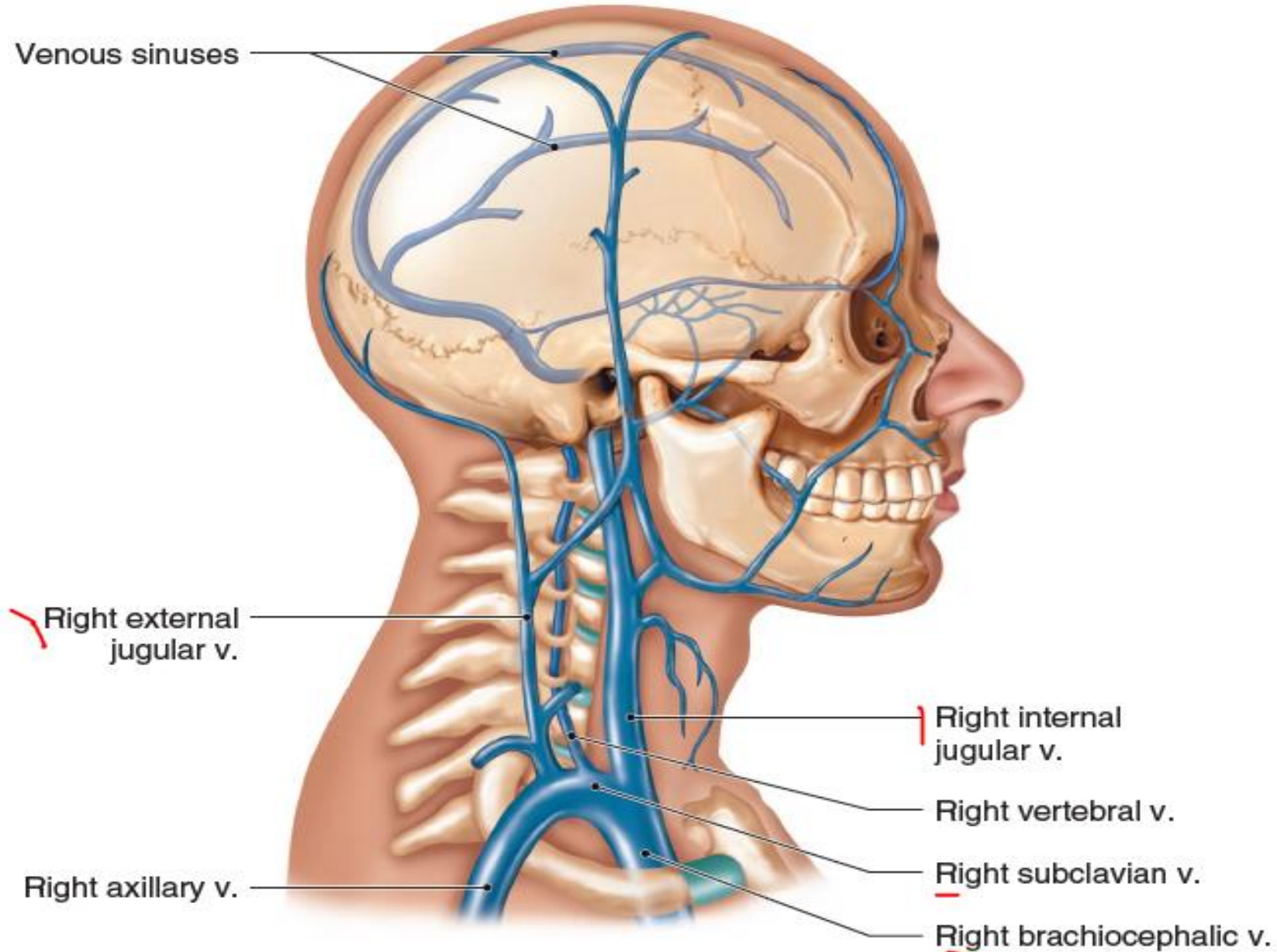
Venous drainage of the body,,

- Several parallel veins often take the place of a single larger vein. Such multivein bundles, including networks of veins forming anastomoses, **are called *venous plexuses*.**
- Two important body areas have unusual patterns of venous drainage.
 - **First, veins from the brain drain into *dural sinuses*,** which are not typical veins but endothelium-lined channels supported by walls of dura mater.
 - **Second, venous blood draining from the digestive organs enters a special subcirculation, the *hepatic portal system*,** and passes through capillaries in the liver before the blood reenters the general systemic circulation.
 - **The dural sinuses and hepatic portal system are considered shortly.**

Veins Draining the neck and head

- ❖ **External jugular vein:** from scalp, portions of face, superficial neck region
 - drain into right and left subclavian vein.
- ❖ **Internal jugular vein:** from brain, meninges, deep regions of face and neck
 - larger and deeper than the external jugular vein.
- ❖ **subclavian vein and internal jugular** unite to form the brachiocephalic vein
 - the two brachiocephalic veins merge to **form the superior vena cava**, which empties into the right atrium.

Veins Draining the neck and head,,



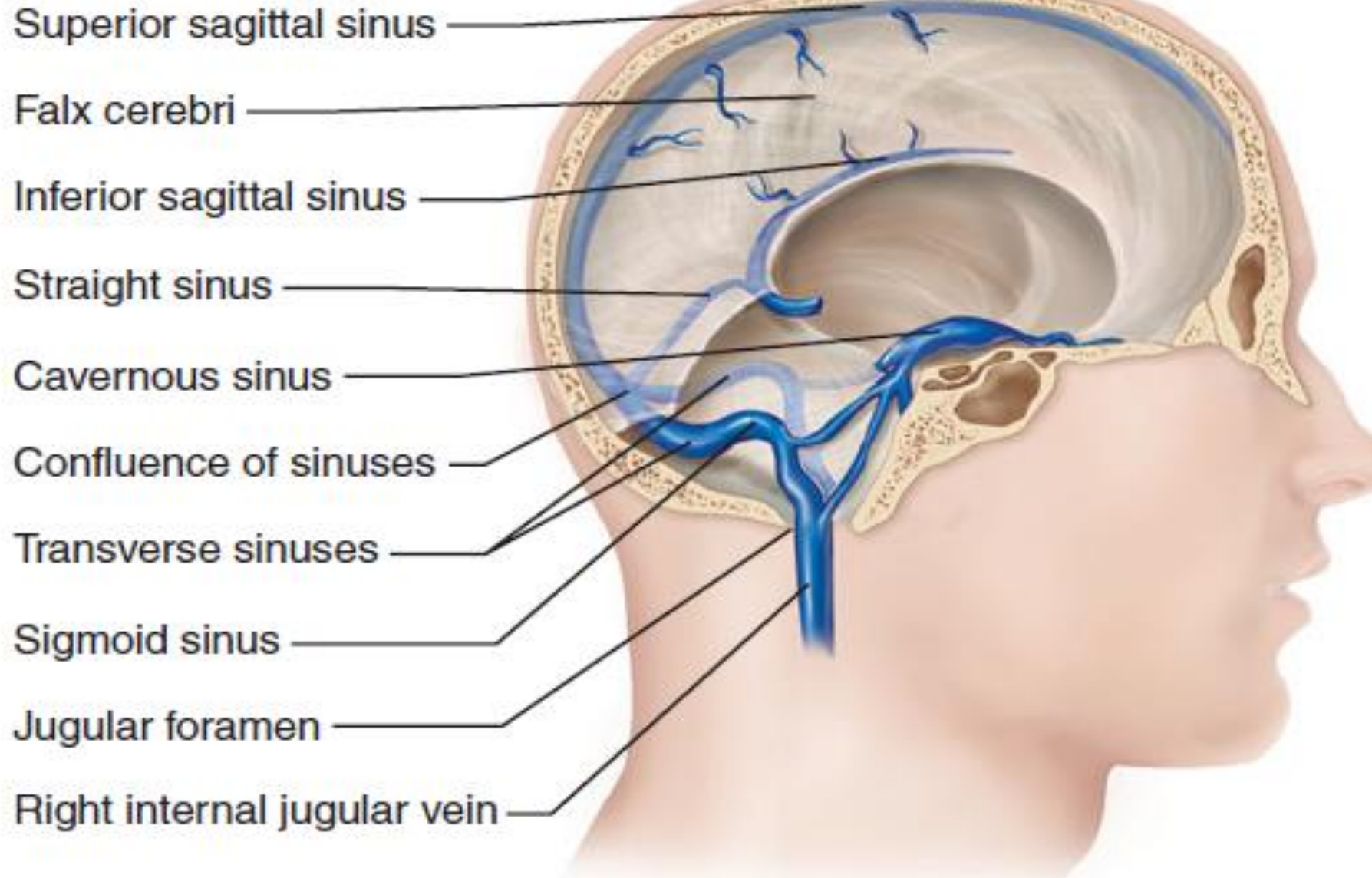
Veins Draining the neck and head,,

- ❖ **Dural Sinuses**; Most veins of the brain drain into the intracranial **dural sinuses**, which form an interconnected series of channels in the skull and lie between the two layers of cranial dura mater.
- The **superior** and **inferior sagittal sinuses** lie in the falx cerebri between the cerebral hemispheres.
- The inferior sagittal sinus drains posteriorly into the **straight sinus**.
- The superior sagittal and straight sinuses then drain posteriorly into the **transverse sinuses**, which run in shallow grooves on the internal surface of the occipital bone.

Veins Draining the neck and head,,

- Each transverse sinus in turn drains into an S-shaped sinus (**sigmoid sinus**), which becomes the **internal jugular vein** as it leaves the skull through the jugular foramen.
- The paired **cavernous sinuses** border the body of the sphenoid bone laterally, and each has an internal carotid artery *running within it*.
- The cavernous sinus drains into the transverse sinus and the internal jugular vein but also communicates with the **ophthalmic vein** of the orbit, which in turn communicates with the facial vein, which drains the nose and upper lip.

Veins Draining the neck and head,,



(b) Dural venous sinuses of the brain

Veins Draining the neck and head,,

Veins of the head and trunk

Dural venous sinuses

External jugular vein

Vertebral vein

Internal jugular vein

Right and left
brachiocephalic veins

Superior vena cava

Great cardiac vein

Hepatic veins

Splenic vein

Hepatic portal vein

Renal vein

Superior mesenteric vein

Inferior mesenteric vein

Inferior vena cava

Common iliac vein

Internal iliac vein

Veins that drain the upper limb

Subclavian vein

Axillary vein

Cephalic vein

Brachial vein

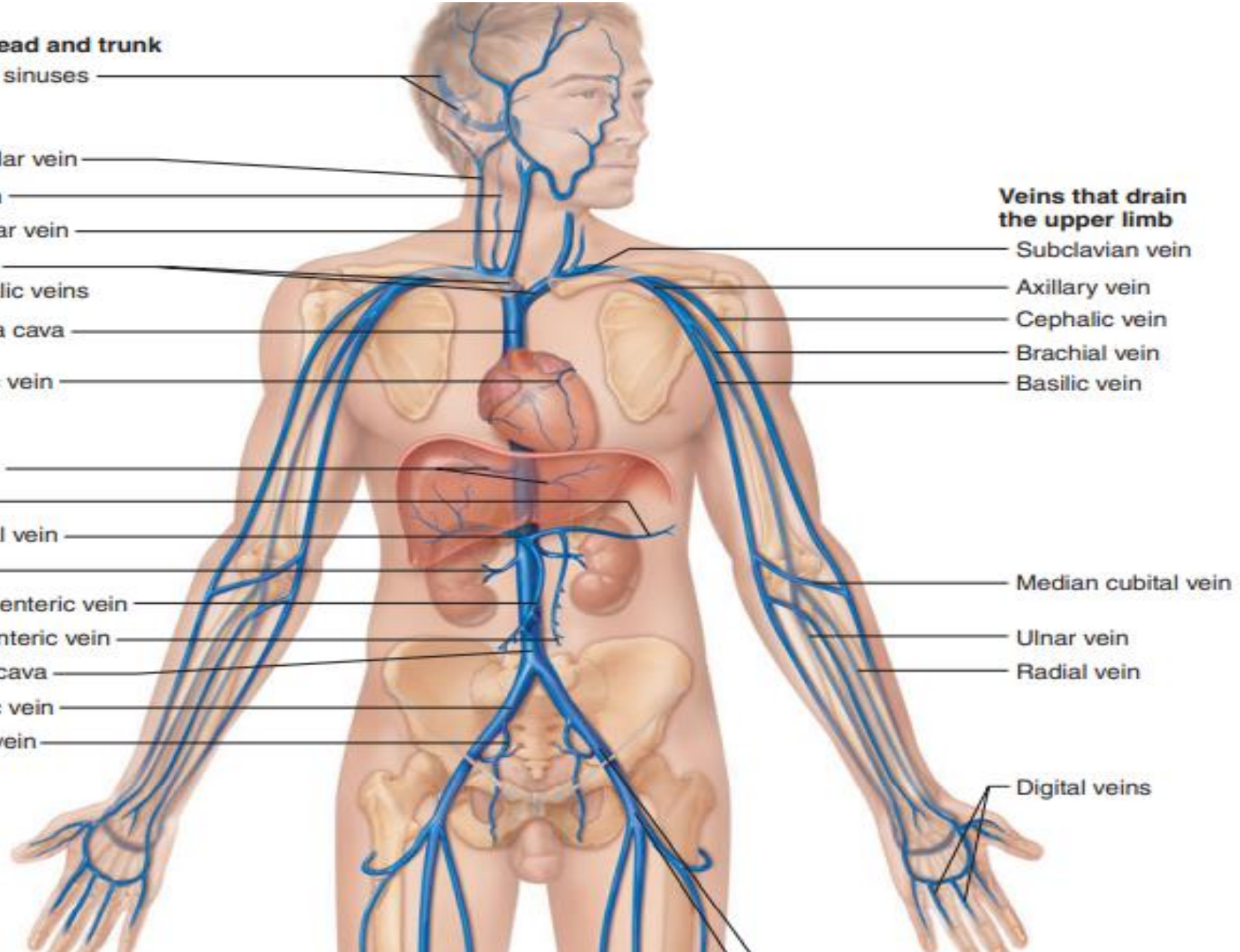
Basilic vein

Median cubital vein

Ulnar vein

Radial vein

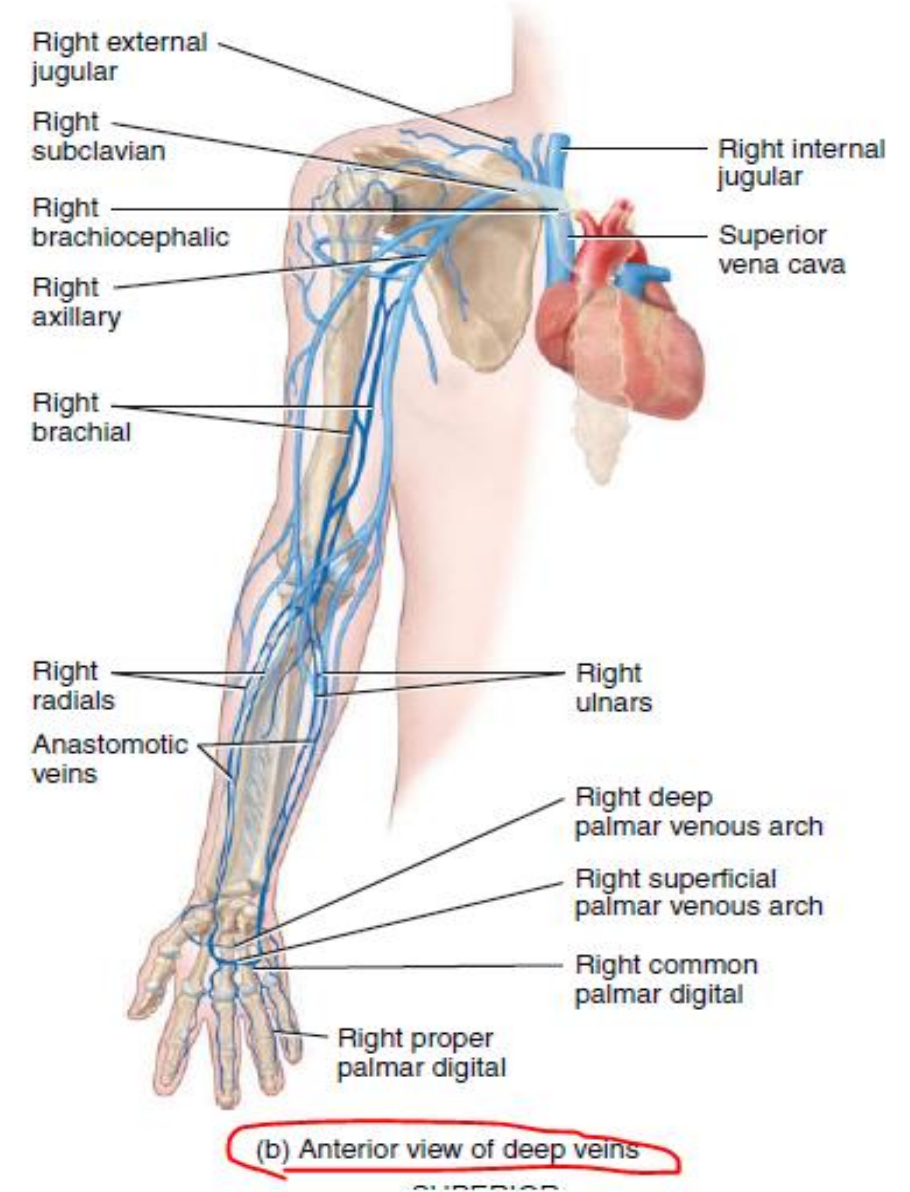
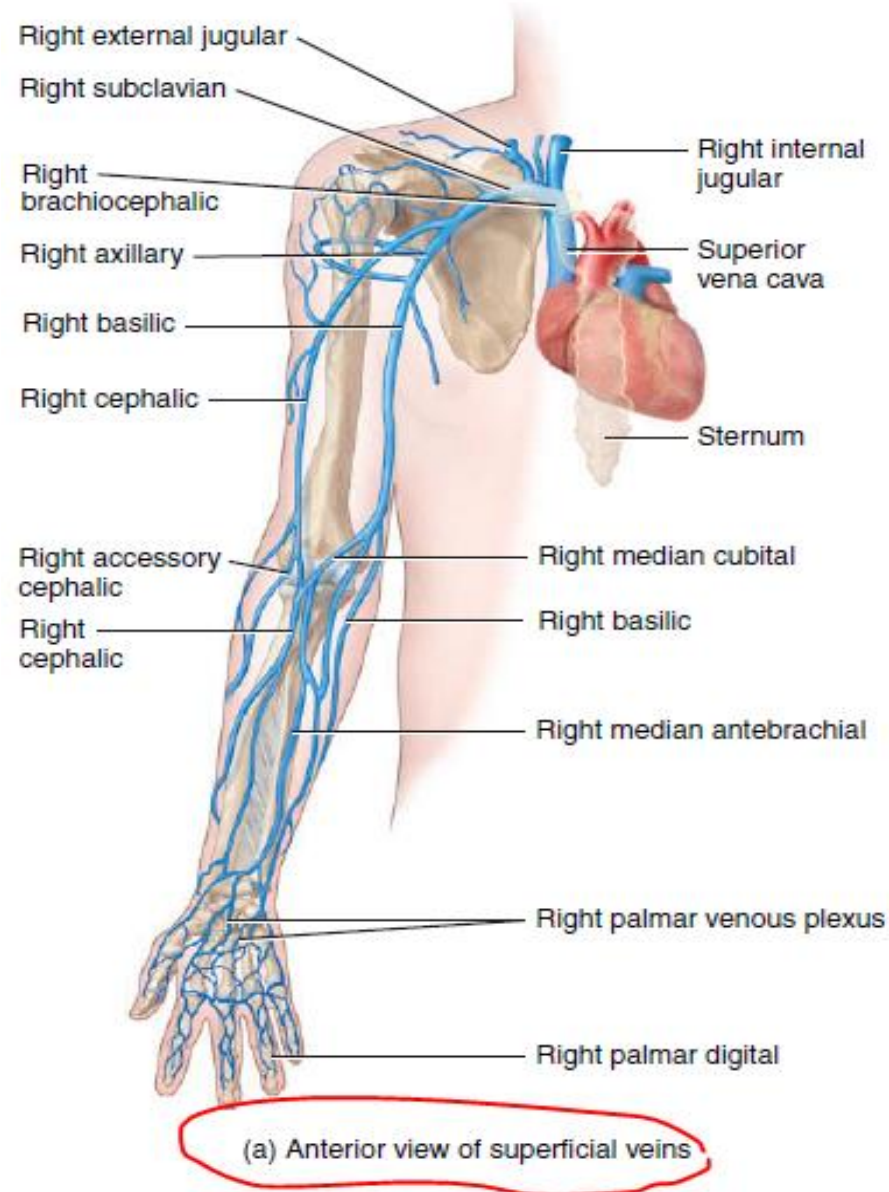
Digital veins



Veins of the upper extremity

- Consists of superficial and deep venous drainage.
- **Deep veins** accompany the arteries and bear their names / region
 - radial vein & ulnar vein
 - both drain from deep and superficial palmar arches
 - radial and ulnar veins join in the cubital fossa to form the brachial vein, which continues up on the medial side of the arm.
 - Brachial vein -axillary → subclavian → internal jugular → brachiocephalic.

Veins of the upper extremity ,,



Veins of the upper extremity ,,

➤ **superficial veins** **Veins of the upper extremity:**

☐ are the **basilic and cephalic veins**

✓ **basilic vein** drains blood **from ulnar side** of forearm, medial side of arm

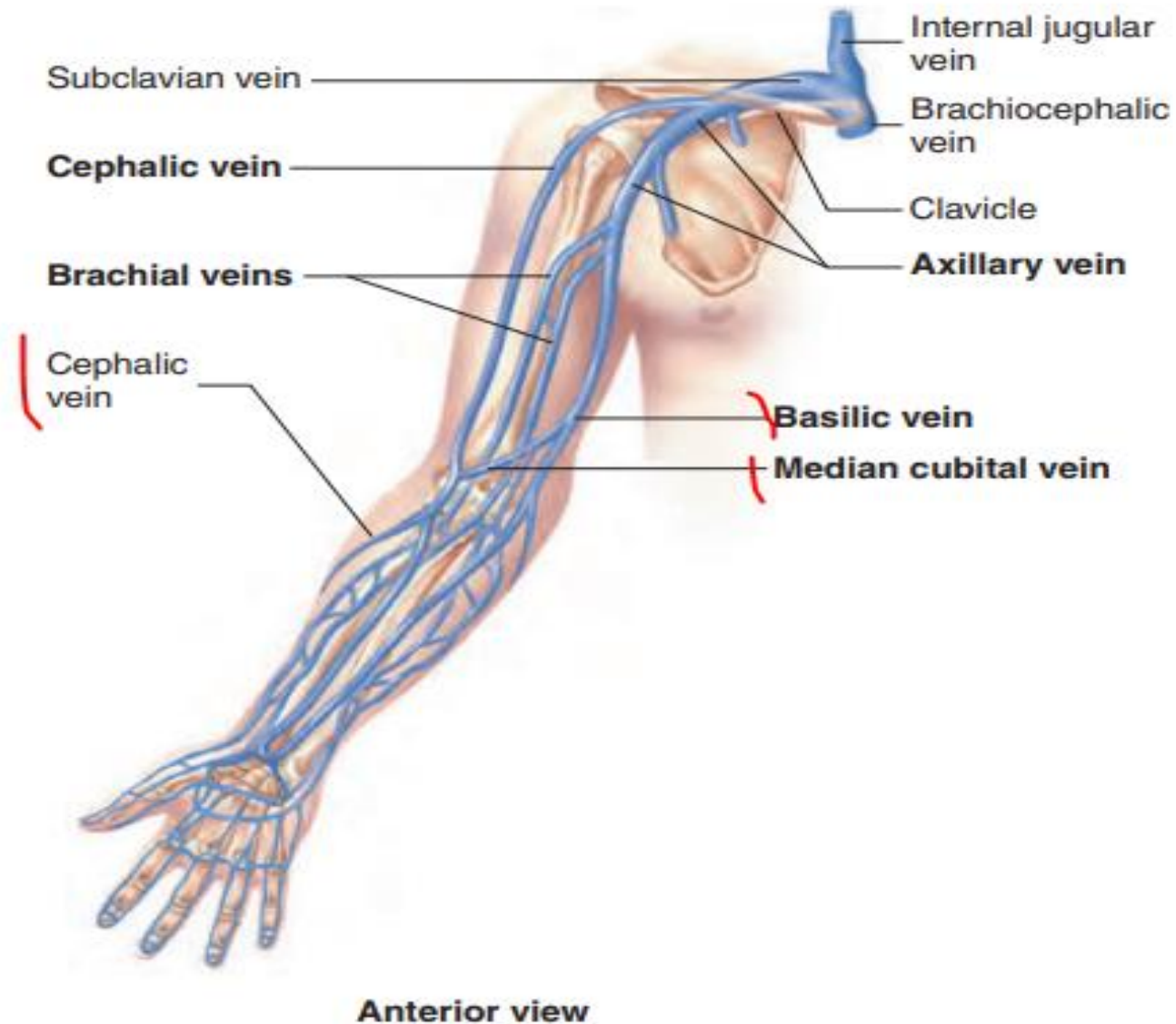
- merges **with brachial vein near the head of the humerus** to form the axillary vein

✓ **cephalic vein** drains superficial region of hand and forearm on radial side

- joins axillary vein in the shoulder region.
- **median** cubital vein ascends from the cephalic vein to join basilic vein on radial side.
- It is a site of **venipuncture**

Veins of the upper extremity ,,

superficial veins Veins of the upper extremity



Veins of the head and trunk

Dural venous sinuses

External jugular vein

Vertebral vein

Internal jugular vein

Right and left
brachiocephalic veins

Superior vena cava

Great cardiac vein

Hepatic veins

Splenic vein

Hepatic portal vein

Renal vein

Superior mesenteric vein

Inferior mesenteric vein

Inferior vena cava

Common iliac vein

Internal iliac vein

Veins that drain the upper limb

Subclavian vein

Axillary vein

Cephalic vein

Brachial vein

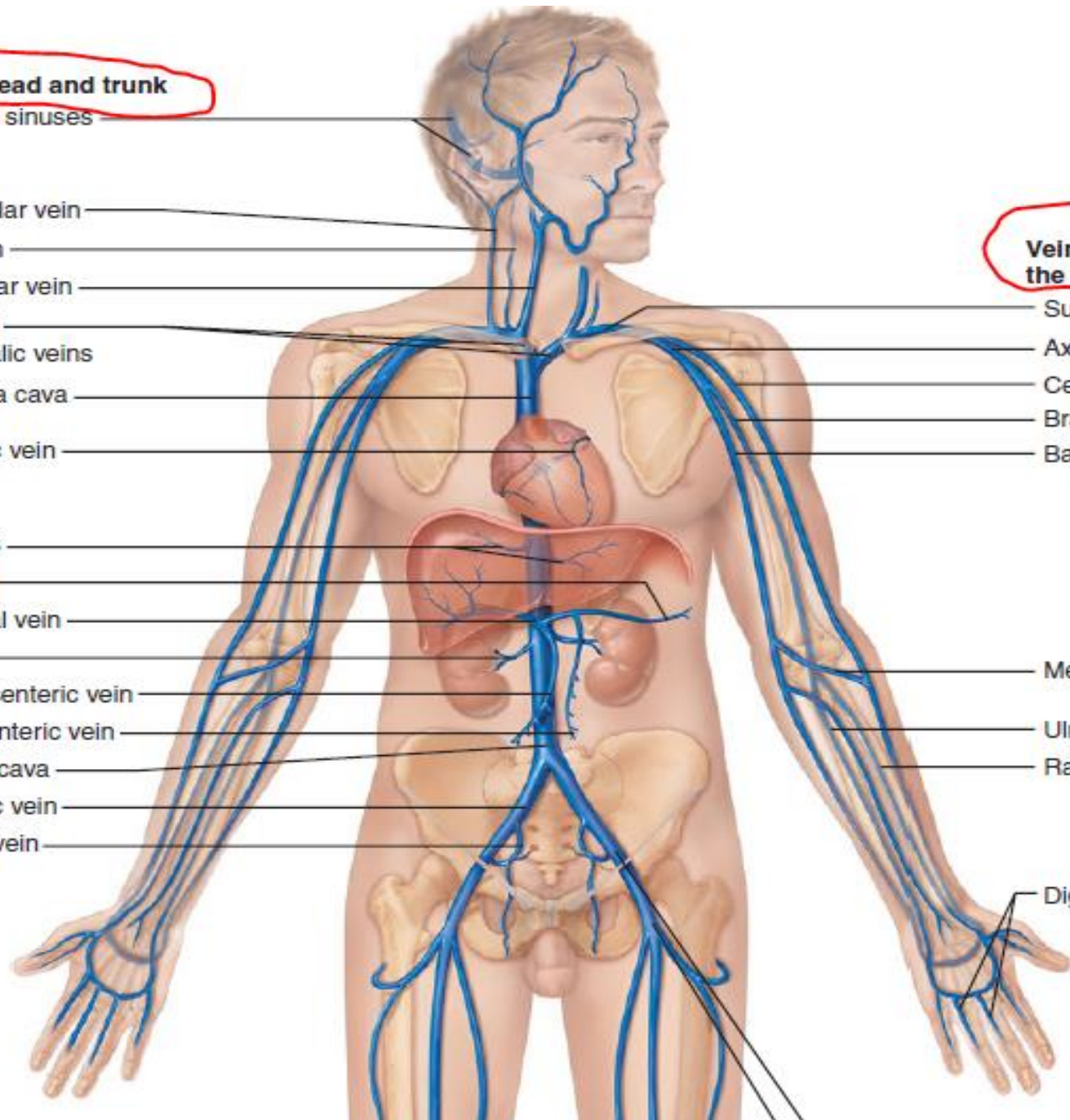
Basilic vein

Median cubital vein

Ulnar vein

Radial vein

Digital veins



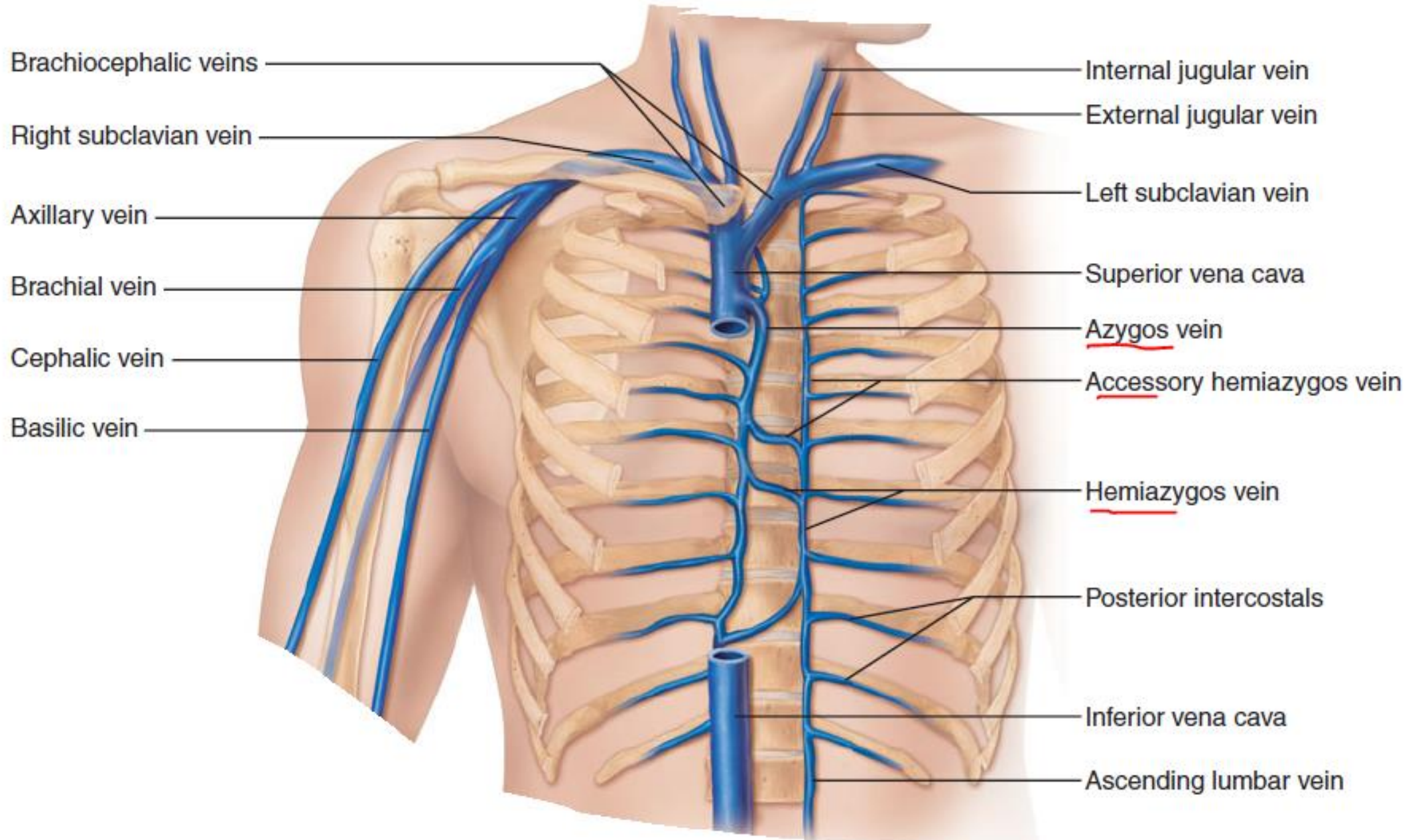
Veins of the thorax

- Blood draining from upper some intercostal spaces enters the brachiocephalic veins.
- Blood from the other intercostal spaces, as well as from some of the thoracic viscera, drains into a group of veins called the *azygos system*.
- This group of veins, ultimately empty into the superior vena cava, consists of the *azygos vein*, the *hemiazygos vein*, and the *accessory hemiazygos vein*.

❑ **The azygos vein**, ascends along the right or the center of the thoracic vertebral bodies.

- It receives all of the right **posterior intercostal veins** (except the first).

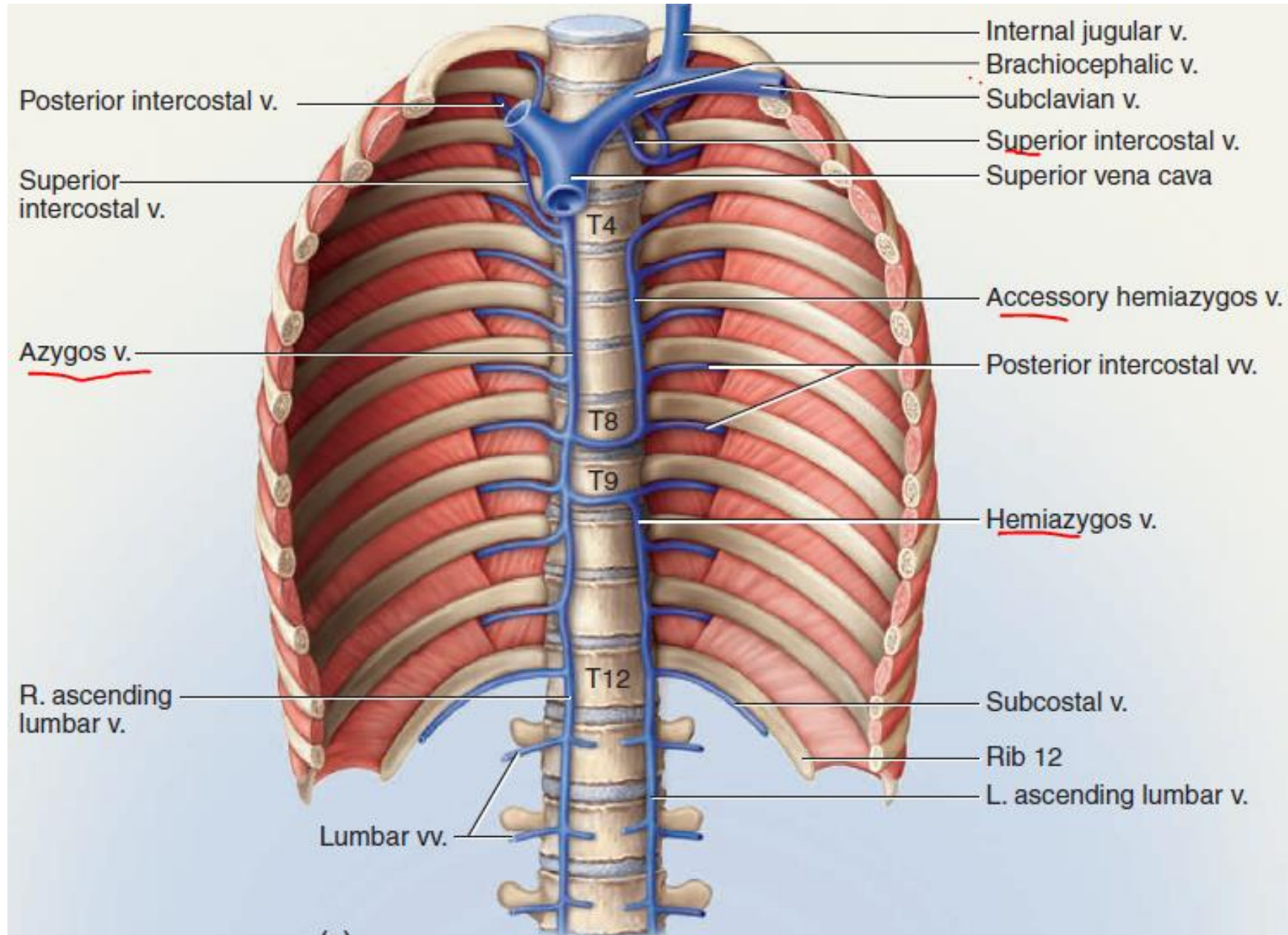
Veins of the thorax,,,



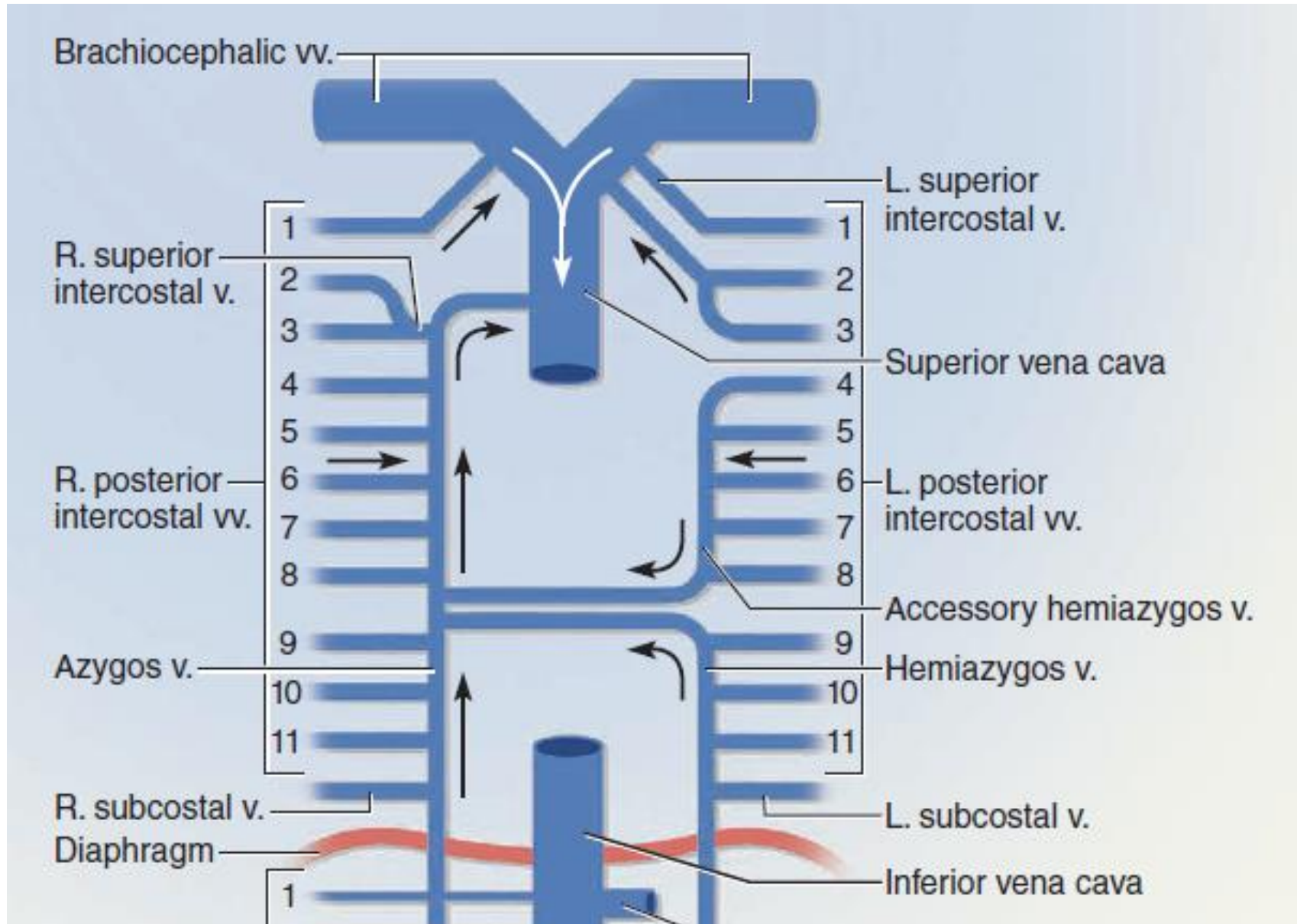
Veins of the thorax,,,

- ❑ The **hemiazygos vein**, which ascends on the left side of the vertebral column, corresponds to the inferior half of the azygos on the right (*hemiazygos* half the azygos).
- At about midthorax, the hemiazygos runs roughly horizontally across the vertebrae and joins the azygos vein.
- ❑ The **accessory hemiazygos vein** can be thought of as a superior continuation of the hemiazygos, receiving the fourth (or fifth) through the eighth left posterior intercostal veins; it also courses to the right to join the azygos.
- ❖ The azygos system, can serve as a bypass for the inferior vena cava, which drains blood from the lower body.

Veins of the thorax,,



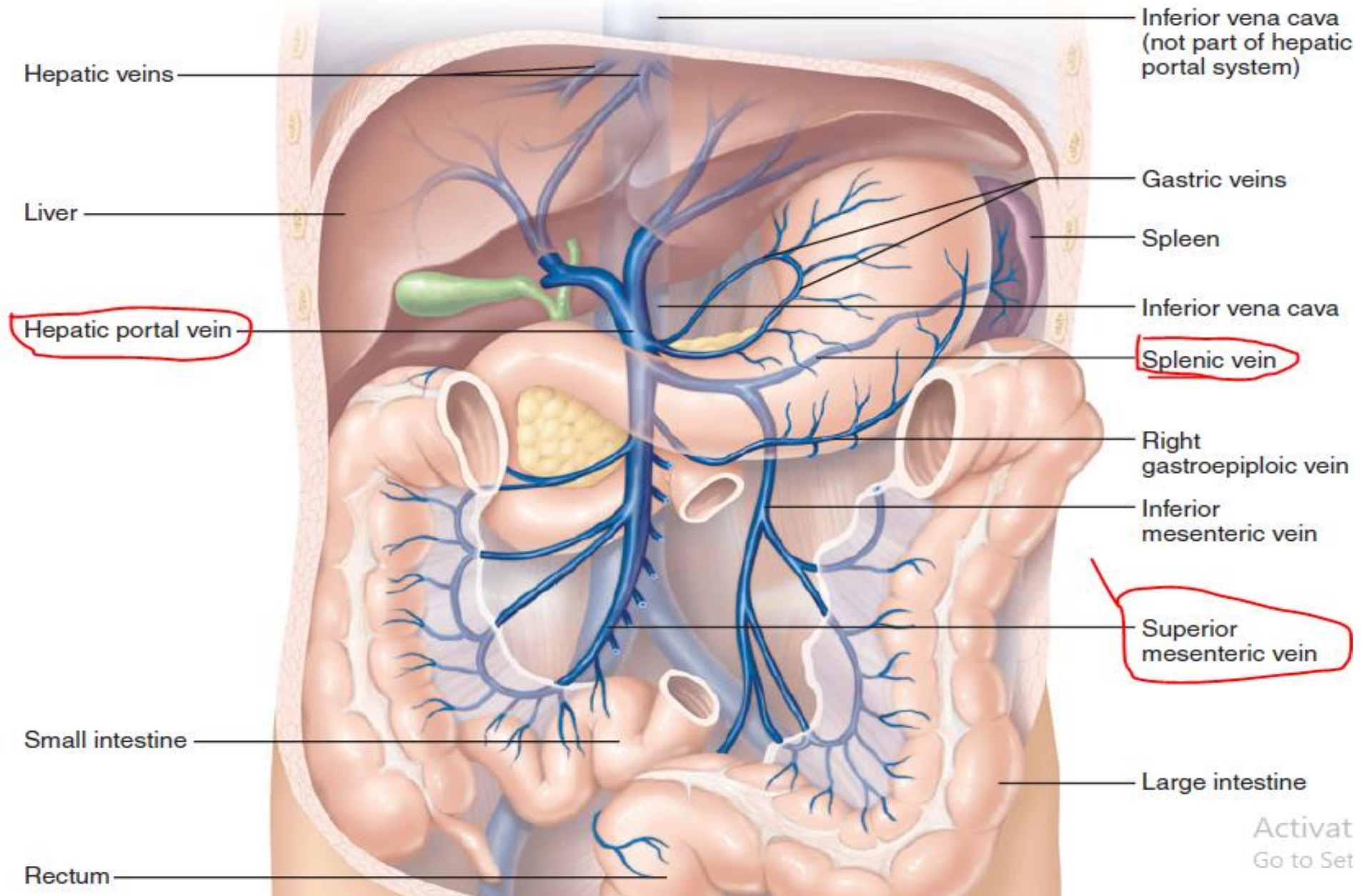
Veins of the thorax,,,



Veins of the Abdomen

- Blood returning from the abdominopelvic viscera and the abdominal wall reaches the heart via the inferior vena cava.
- The inferior vena cava does not receive veins directly from the gastrointestinal tract, spleen, pancreas, and gallbladder.
- Blood from the digestive organs returns via the hepatic portal system.
- These organs pass their blood into a common vein, the **hepatic portal vein**, which delivers the blood to the liver.
- The superior mesenteric and splenic veins unite to form the hepatic portal vein
- *After passing through the liver for processing, blood drains into the hepatic veins, which empty into the inferior vena cava.*

Veins of the Abdomen,,



Veins of the lower extremity

- have a deep and a superficial group

❖ The deep veins accompany the corresponding arteries

□ These include:

- ✓ posterior and anterior tibial veins originate in the foot and ascend upwards in front of the tibia to the back of the knee where they merge to form the popliteal vein.
- ✓ Popliteal vein: it becomes the femoral vein above the knee
- ✓ the femoral vein :receives blood from the deep femoral vein and above this, receives from the great saphenous vein, then becomes the external iliac vein (as it passes under the inguinal ligament).
- ✓ the external iliac vein merges with the internal iliac vein to form the **common iliac vein**
- ✓ **The left and right common iliac** veins merge to form the **inferior vena cava**, which returns blood to the right atrium of the heart.

Veins of the lower extremity,,

Veins of the head and trunk

Dural venous sinuses

External jugular vein

Vertebral vein

Internal jugular vein

Right and left
brachiocephalic veins

Superior vena cava

Great cardiac vein

Hepatic veins

Splenic vein

Hepatic portal vein

Renal vein

Superior mesenteric vein

Inferior mesenteric vein

Inferior vena cava

Common iliac vein

Internal iliac vein

Veins that drain the upper limb

Subclavian vein

Axillary vein

Cephalic vein

Brachial vein

Basilic vein

Median cubital vein

Ulnar vein

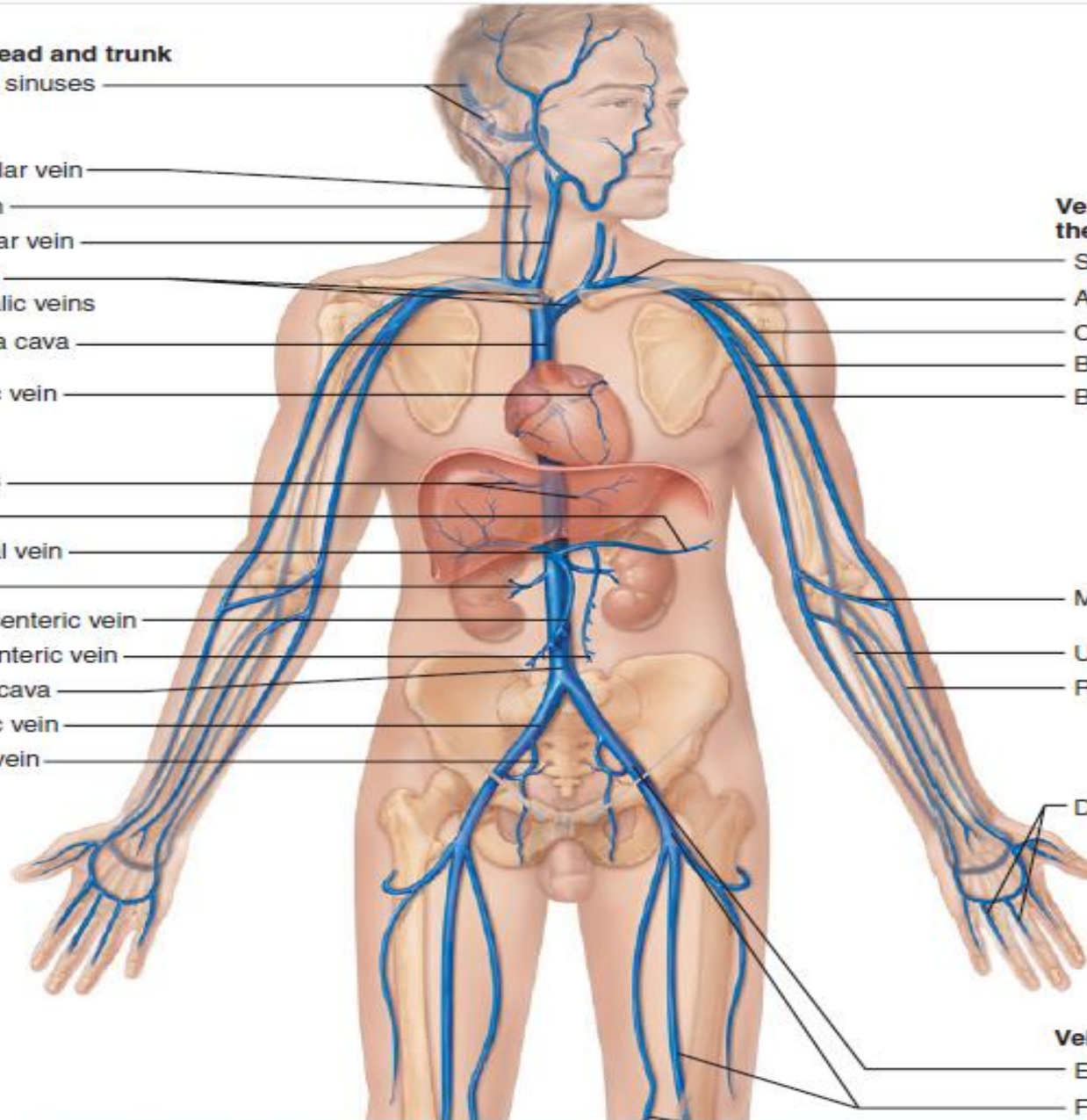
Radial vein

Digital veins

Veins that drain the lower limb

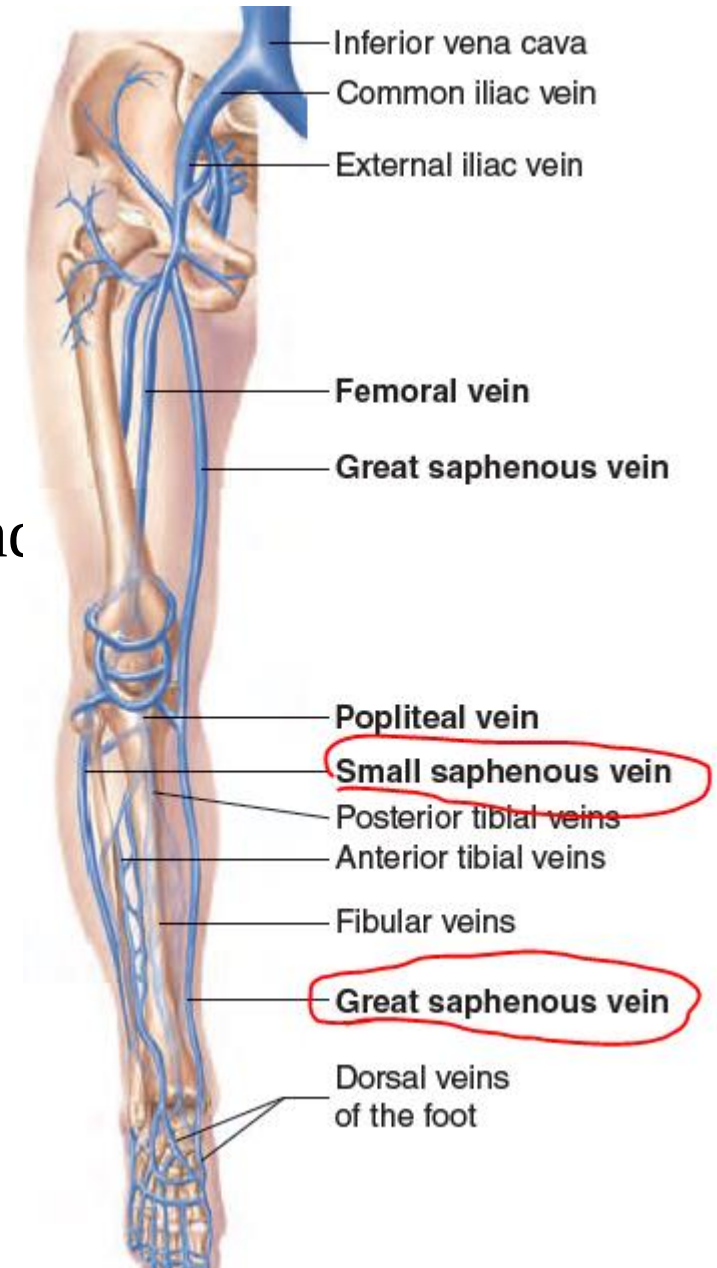
External iliac vein

Femoral vein

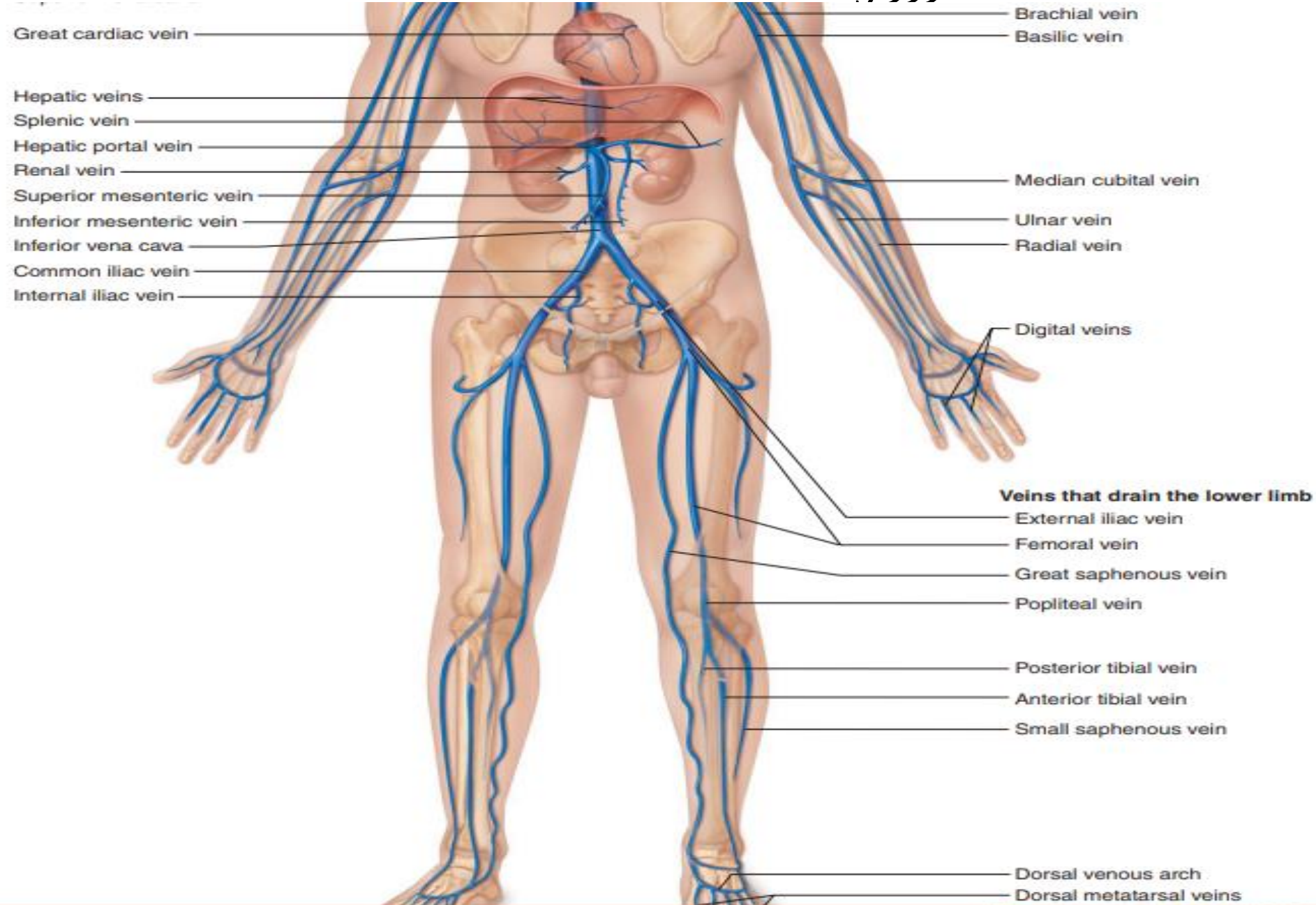


Veins of the lower extremity,,

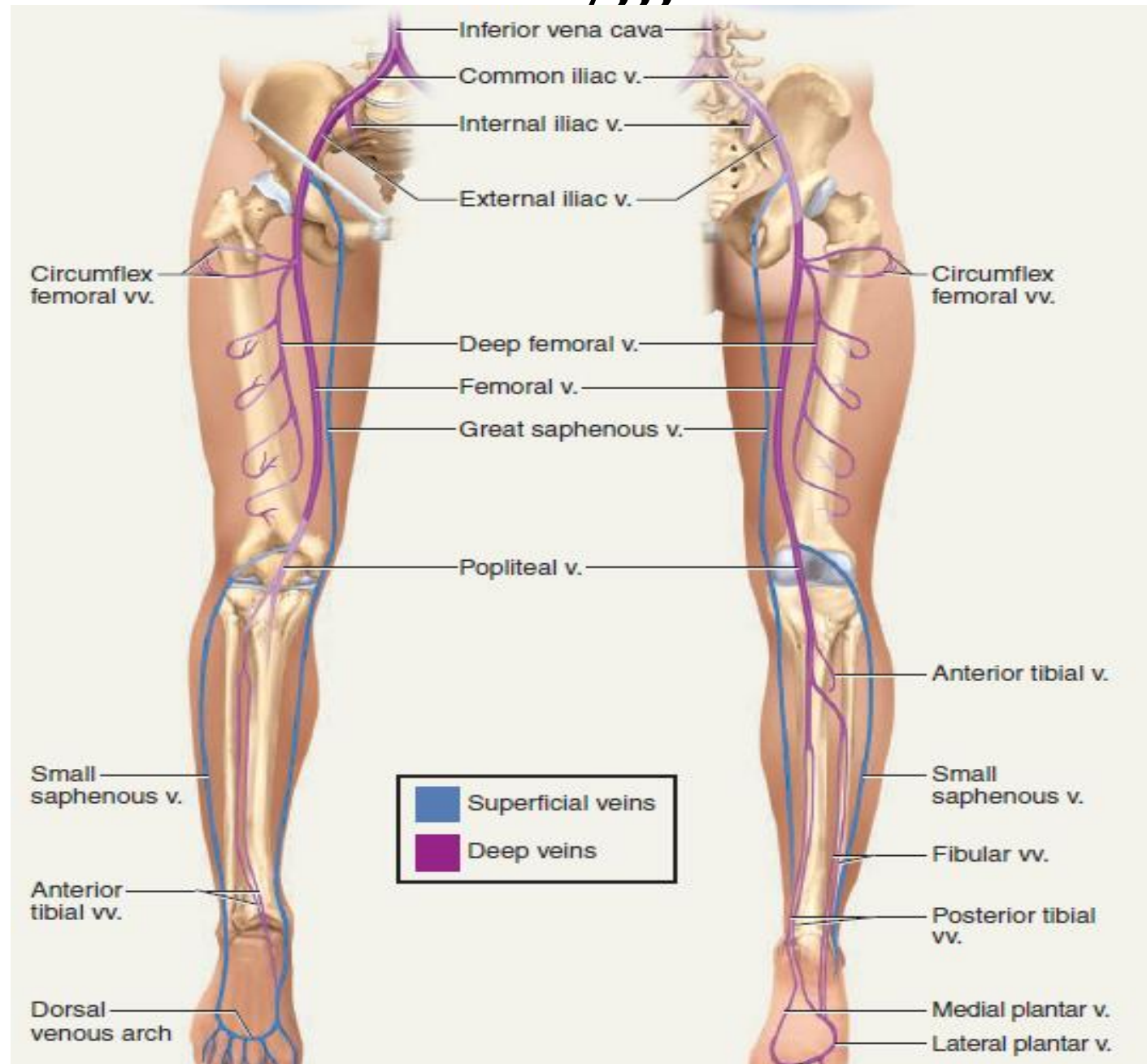
- The superficial vein lower extremity include
 - small saphenous vein
 - arises from the lateral side of the foot, courses posteriorly along the surface of the calf of the leg and enters deep into the popliteal vein behind the knee.
 - Great saphenous vein
 - longest vessel in the body, originates at the arch of the foot and ascends superiorly along the medial aspect of the leg and thigh before draining into the femoral vein.



Veins of the lower extremity,,

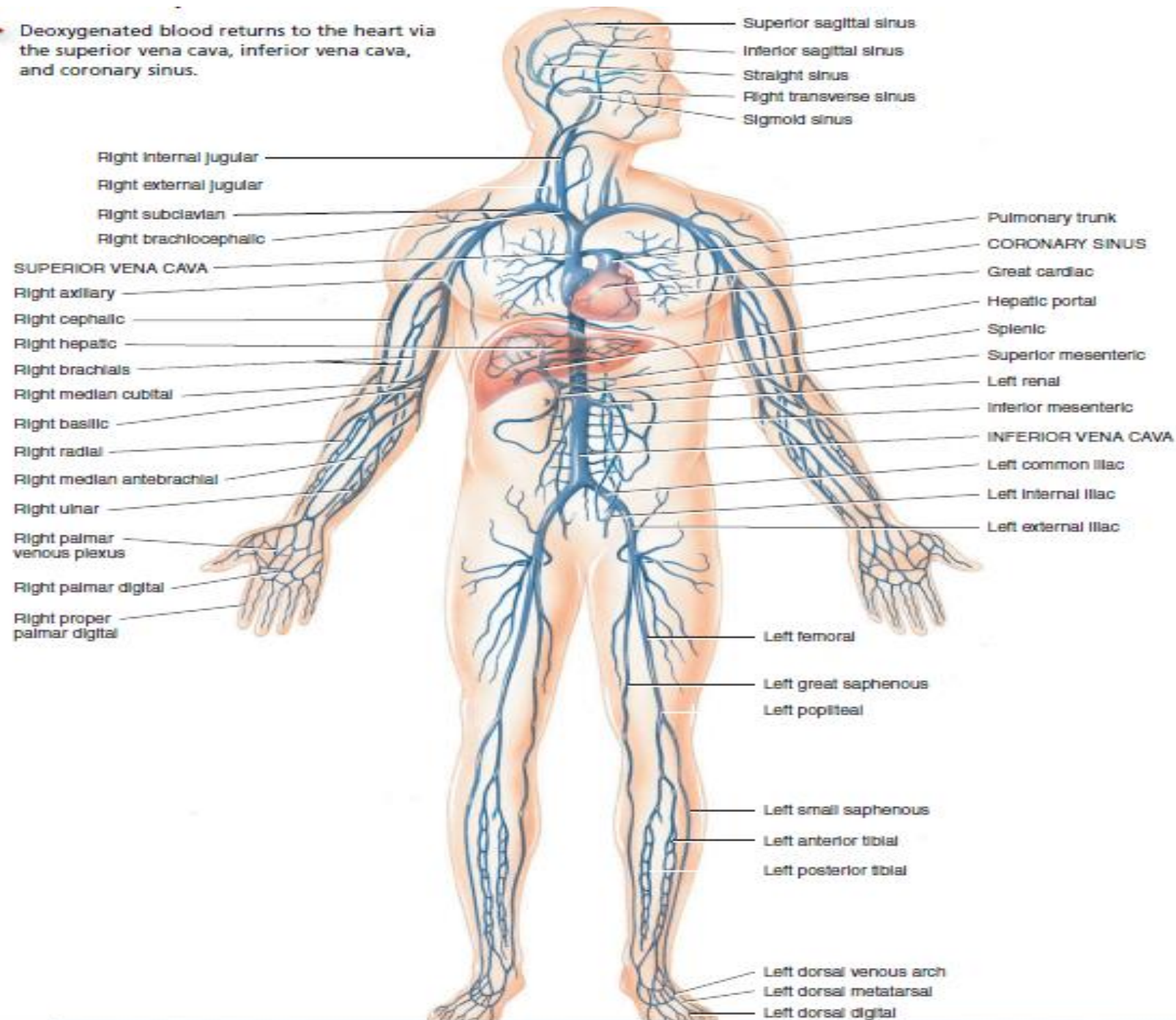


Veins of the lower extremity,,,





Deoxygenated blood returns to the heart via the superior vena cava, inferior vena cava, and coronary sinus.



Circulatory Routes!

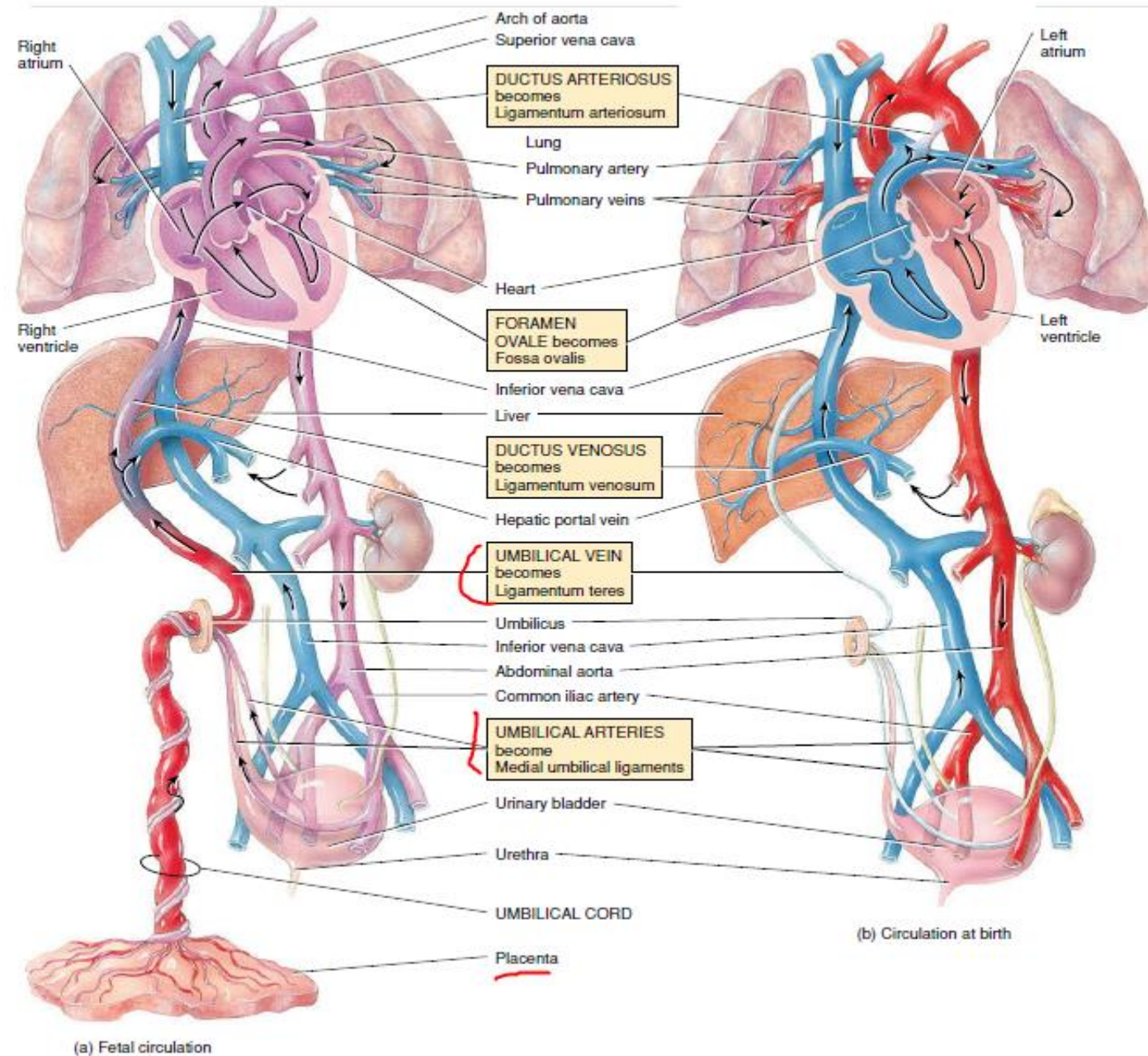
The Fetal Circulation

- The circulatory system of a fetus, called the **fetal circulation**, exists only in the fetus and contains special structures that allow the developing fetus to exchange materials with its mother.
- **Circulation** patterns are clearly established by the fourth week of embryonic life.
- **It differs from the postnatal (after birth) circulation because the lungs, kidneys, and gastrointestinal organs do not begin to function until birth.**
- The fetus obtains O₂ and nutrients **from maternal blood** and eliminates CO₂ and other wastes into the maternal blood.
- The exchange of materials between **fetal and maternal circulations occurs** through the **placenta**, which forms inside the mother's uterus and attaches to the umbilicus (navel) of the fetus by the **umbilical cord**.

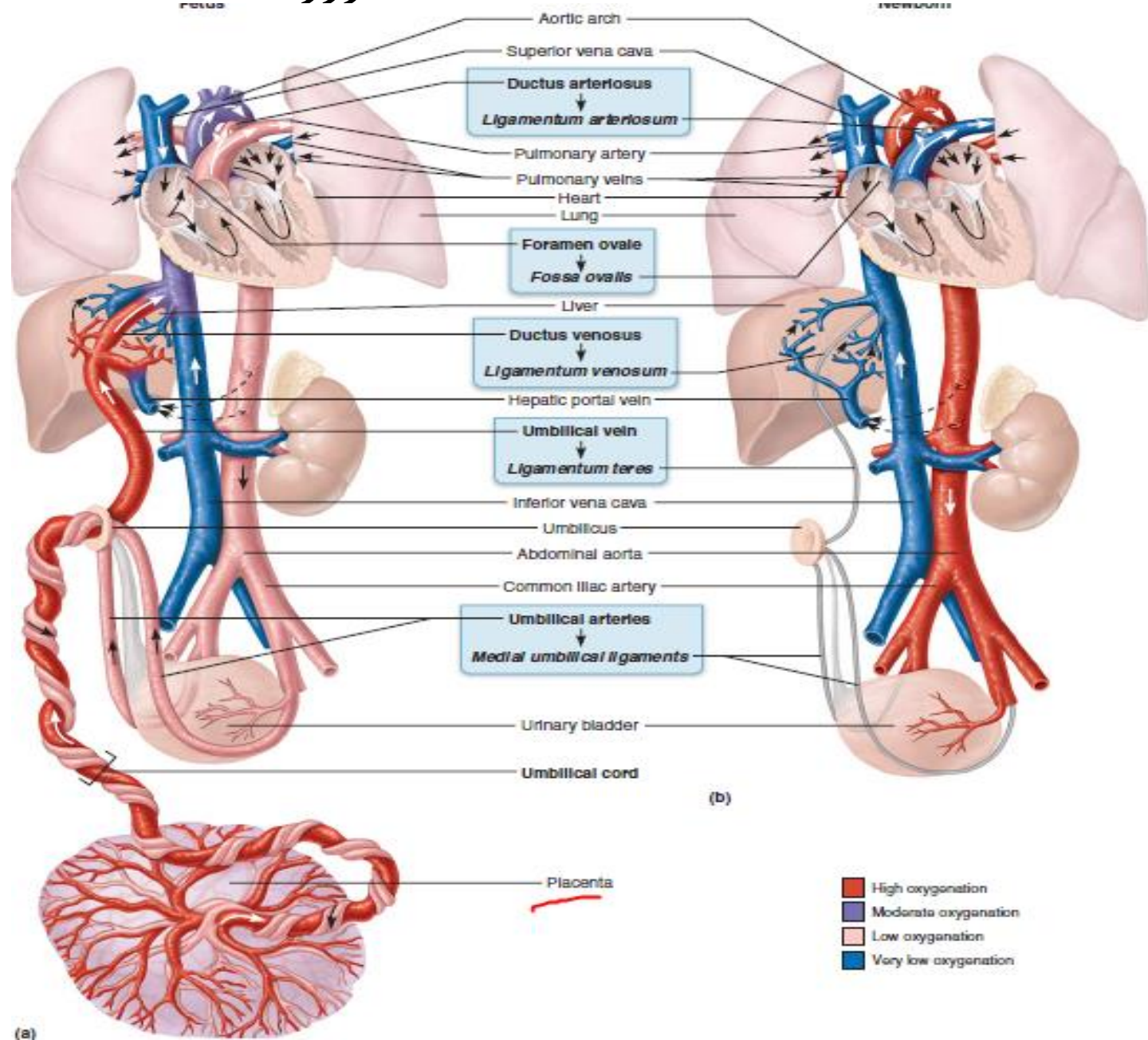
The Fetal Circulation,,,

- Normally, there is no direct mixing of maternal and fetal blood because all exchanges occur by diffusion through capillary walls.
- **Blood passes from** the fetus to the placenta via two **umbilical arteries**.
- At the placenta, fetal blood picks up O₂ & nutrients and eliminates CO₂ & wastes.
- ❖ **The oxygenated blood returns** from the placenta via a single **umbilical vein**.
- Most of the fetal blood does not pass from the right ventricle to the lungs, as it does in postnatal circulation, because an opening called the **foramen ovale** exists in the septum between the right and left atria.

The Fetal Circulation,,,



The Fetal Circulation,,



The Fetal Circulation,,

❖ **There are two major differences between fetal and postnatal circulation:**

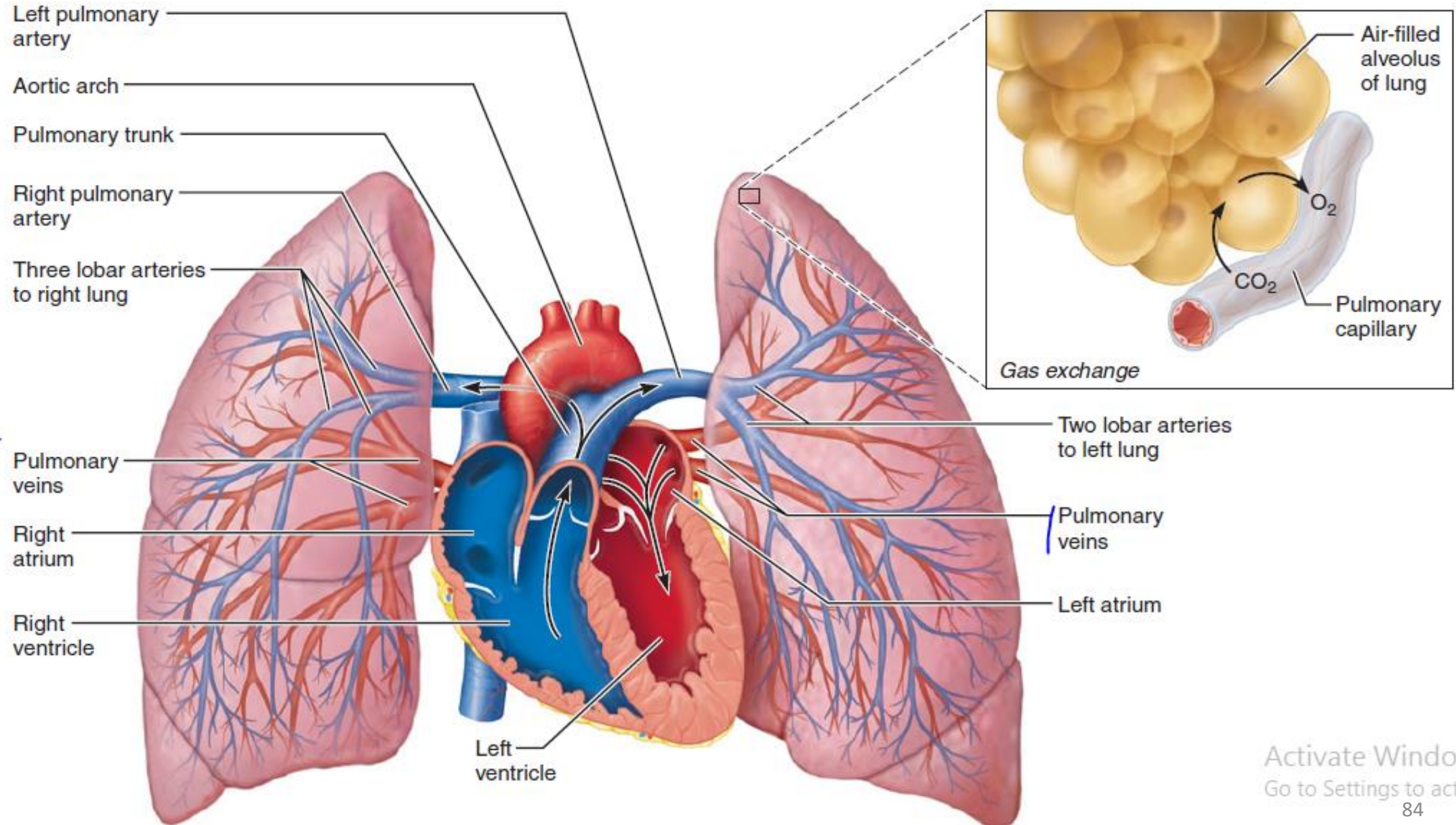
- I. **First**, the fetus must supply blood to the *placenta*, a pancakeshaped organ at the end of the umbilical cord through which oxygen and nutrients are obtained from the mother's uterus.
- II. **Second**, because the fetal respiratory organ is the placenta and the fetus does not breathe, its lungs do not need much blood. Therefore, the fetus sends very little blood around the pulmonary circuit.

❖ **Anatomical structures in the fetal circulation includes** paired **umbilical arteries** branch from the internal iliac arteries and carry blood to **the placenta** to pick up oxygen and nutrients and single **umbilical vein** returns oxygen rich blood to the fetus.

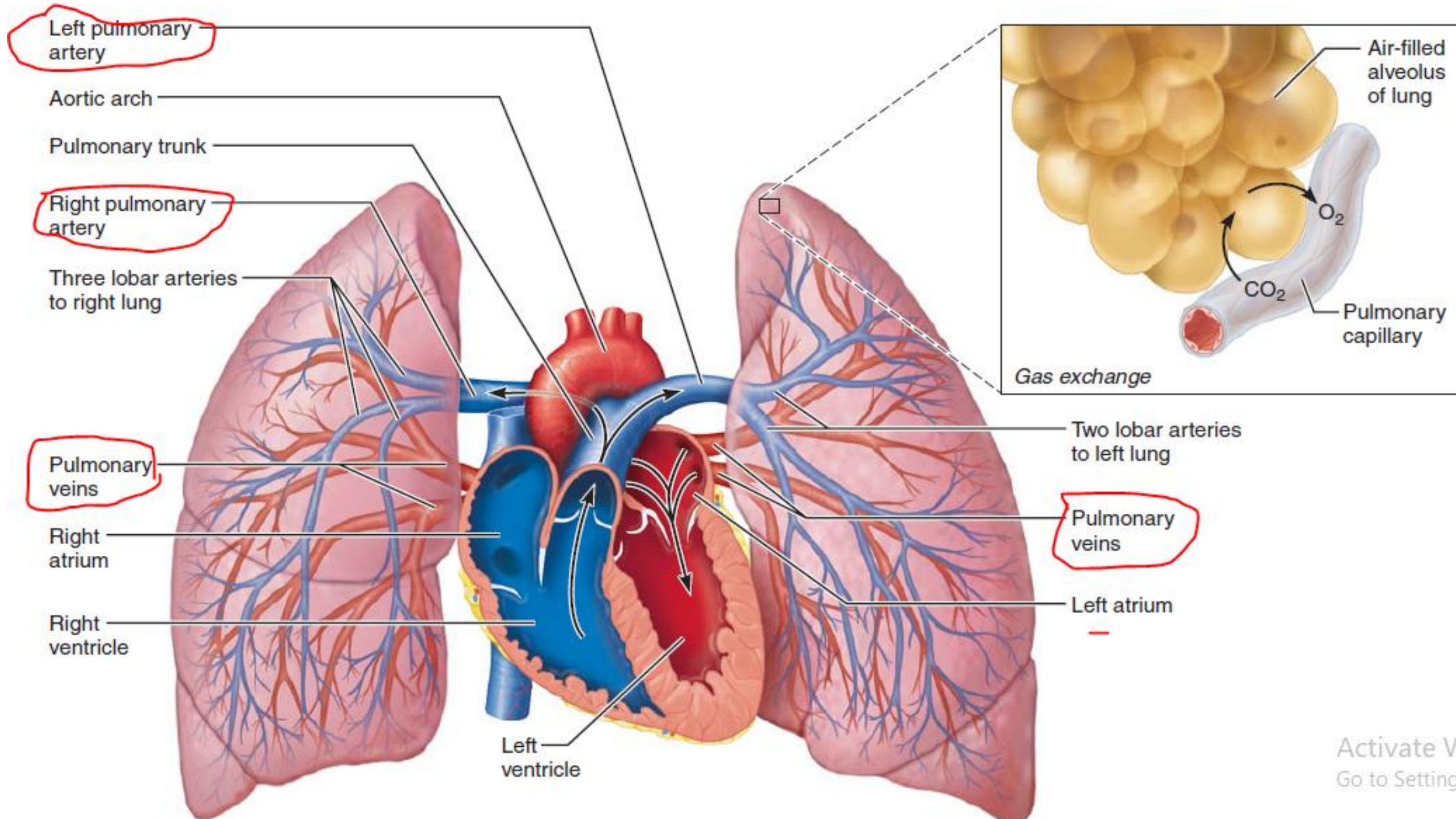
Anatomy of the Pulmonary

- The heart is a double pump that serves two distinct circulation pathways: **the pulmonary and systemic circuits.**
- The **pulmonary circulation** is the system of blood vessels that carries blood from the right ventricle of the heart to the lungs and back to the left atrium of the heart
- Blood from the right ventricle is pumped into **pulmonary trunk.**
- The pulmonary trunk then branches into the **right and left pulmonary arteries.**
- These arteries carry oxygen-poor blood to the pulmonary capillaries in the lungs, where the blood takes up O₂ and releases CO₂.
- Blood rich in O₂ flows from the lungs to the left atrium.
- Four **pulmonary veins** carry the oxygen-rich blood to the left atrium.

Anatomy of the Pulmonary,,



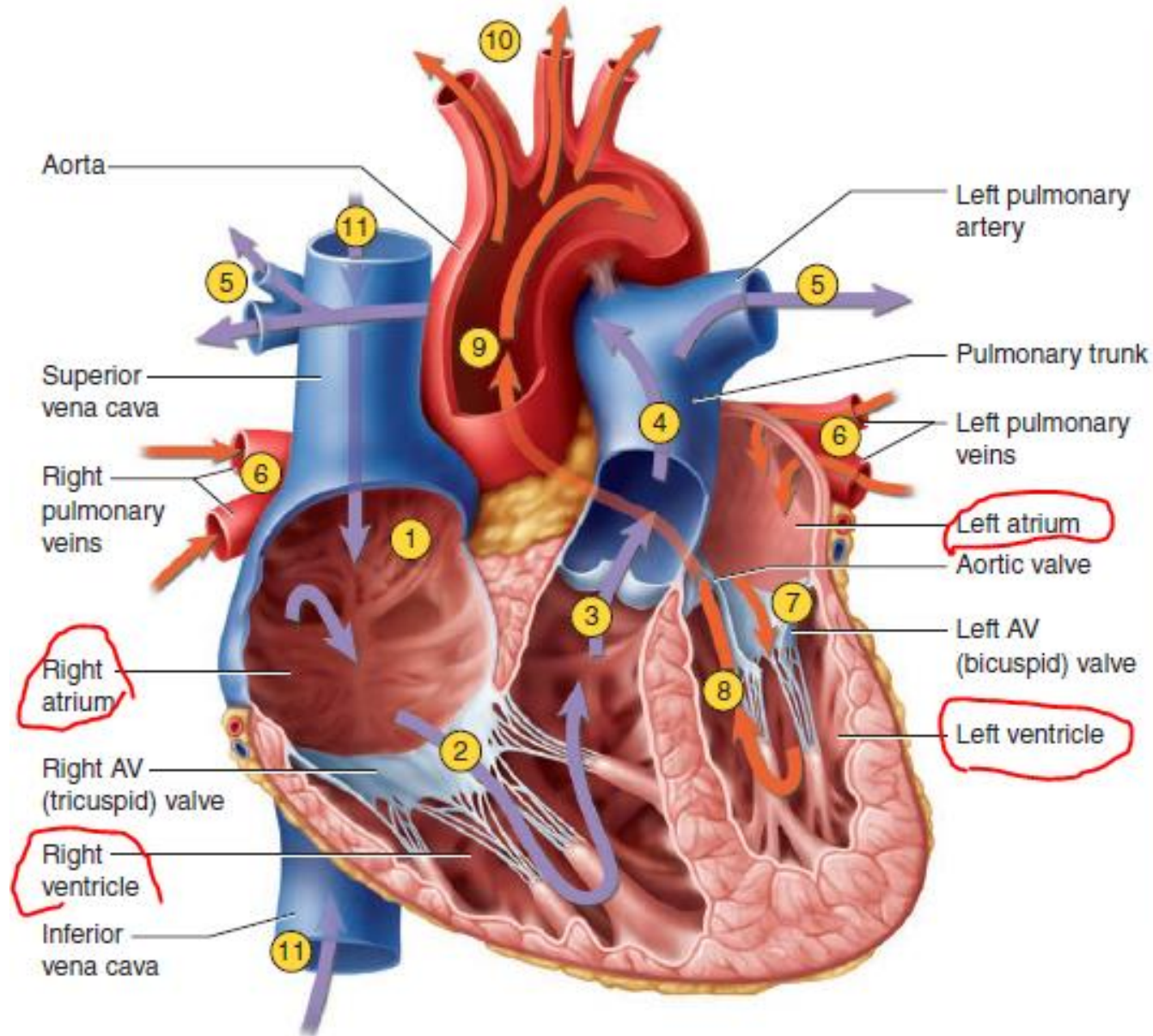
Anatomy of the Pulmonary,,



The systemic circulation

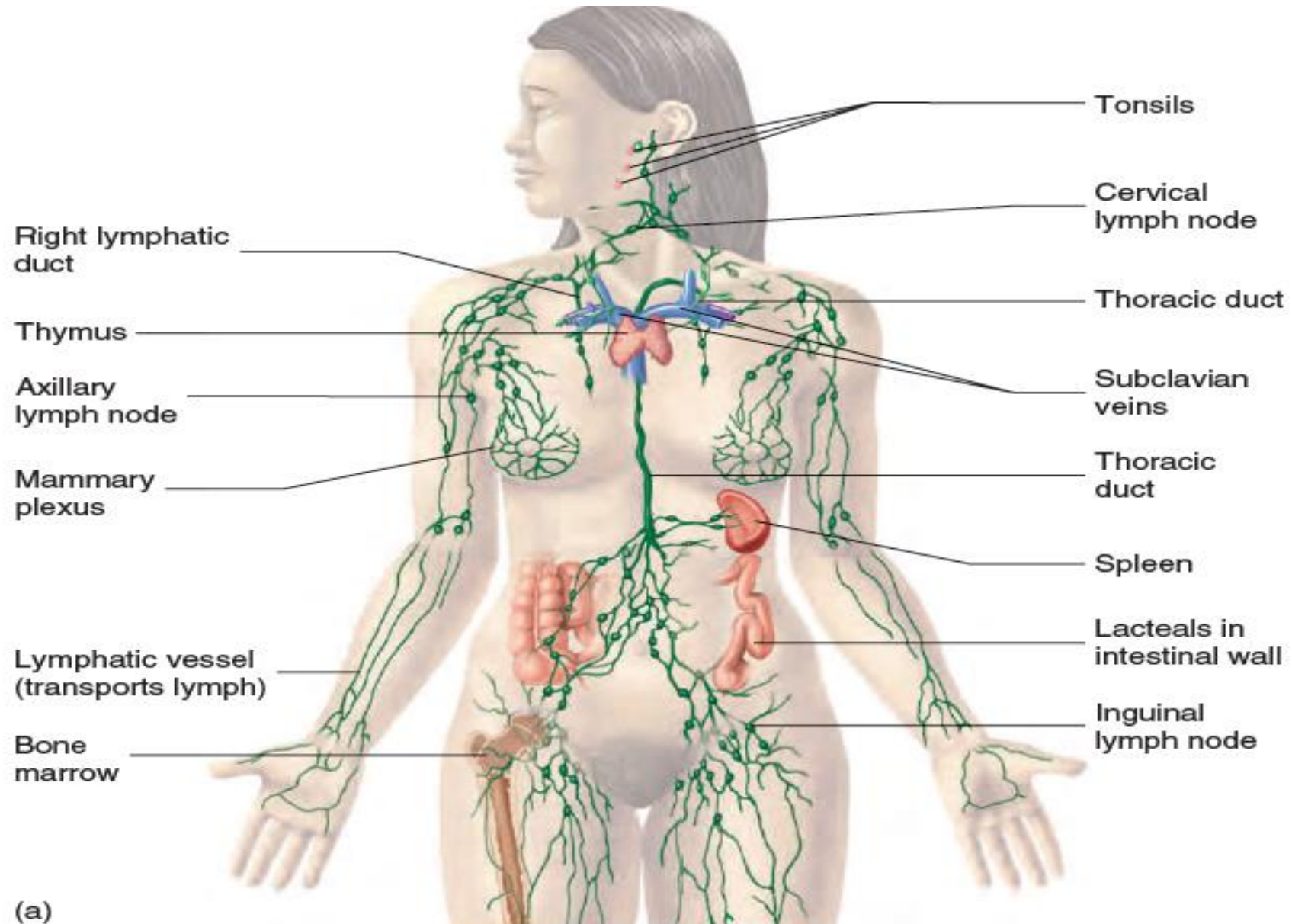
- The **systemic circulation** is the system of blood vessels that carries blood from the left ventricle of the heart to the tissues of the body and back to the right atrium.
- Oxygen-rich blood from the pulmonary veins passes from the left atrium into the left ventricle and from the left ventricle into the aorta.
- Arteries distribute blood from the aorta to all portions of the body

The systemic circulation,,



- 1 Blood enters right atrium from superior and inferior venae cavae.
- 2 Blood in right atrium flows through right AV valve into right ventricle.
- 3 Contraction of right ventricle forces pulmonary valve open.
- 4 Blood flows through pulmonary valve into pulmonary trunk.
- 5 Blood is distributed by right and left pulmonary arteries to the lungs, where it unloads CO_2 and loads O_2 .
- 6 Blood returns from lungs via pulmonary veins to left atrium.
- 7 Blood in left atrium flows through left AV valve into left ventricle.
- 8 Contraction of left ventricle (simultaneous with step 3) forces aortic valve open.
- 9 Blood flows through aortic valve into ascending aorta.
- 10 Blood in aorta is distributed to every organ in the body, where it unloads O_2 and loads CO_2 .
- 11 Blood returns to heart via venae cavae.

Anatomy of the Lymphatic System



Anatomy of the Lymphatic System

- The main function of the **lymphatic system** is to return excess tissue fluid back to the blood vascular system.
- **The lymphatic system, unlike the circulatory system,** does not circulate fluid to and from tissues.
- *Instead, the lymphatic system carries fluid in one direction, from tissues to the circulatory system.*
- The lymphatic system assists in circulating body fluids and helps defend the body against disease-causing agents.
- ❖ **The lymphatic system** is closely related to the cardiovascular system, **both structurally and functionally.**

The components of the lymphatic system

❖ The components of the lymphatic system are;

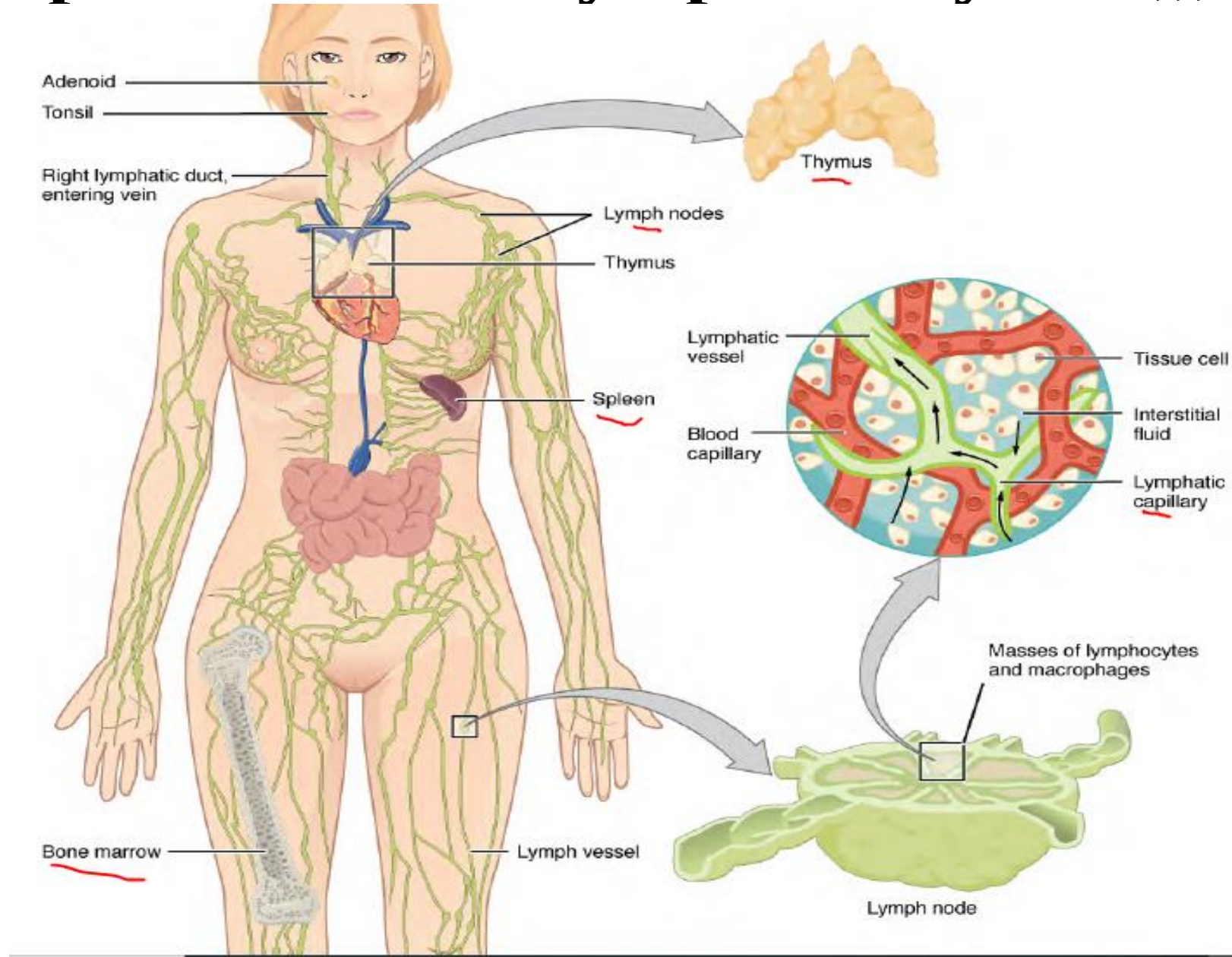
- ✓ **Lymphatic plexuses**, networks of lymphatic capillaries that originate blindly in the extracellular (intercellular) spaces of most tissues.
- ✓ **Lymphatic vessels** (lymphatics), a nearly body wide network of thin-walled vessels that **have abundant lymphatic valves**. Lymphatic capillaries and vessels occur almost everywhere blood **capillaries are found, except for example, teeth, bone, bone marrow, and the entire central nervous system.**

The components of the lymphatic system,,,

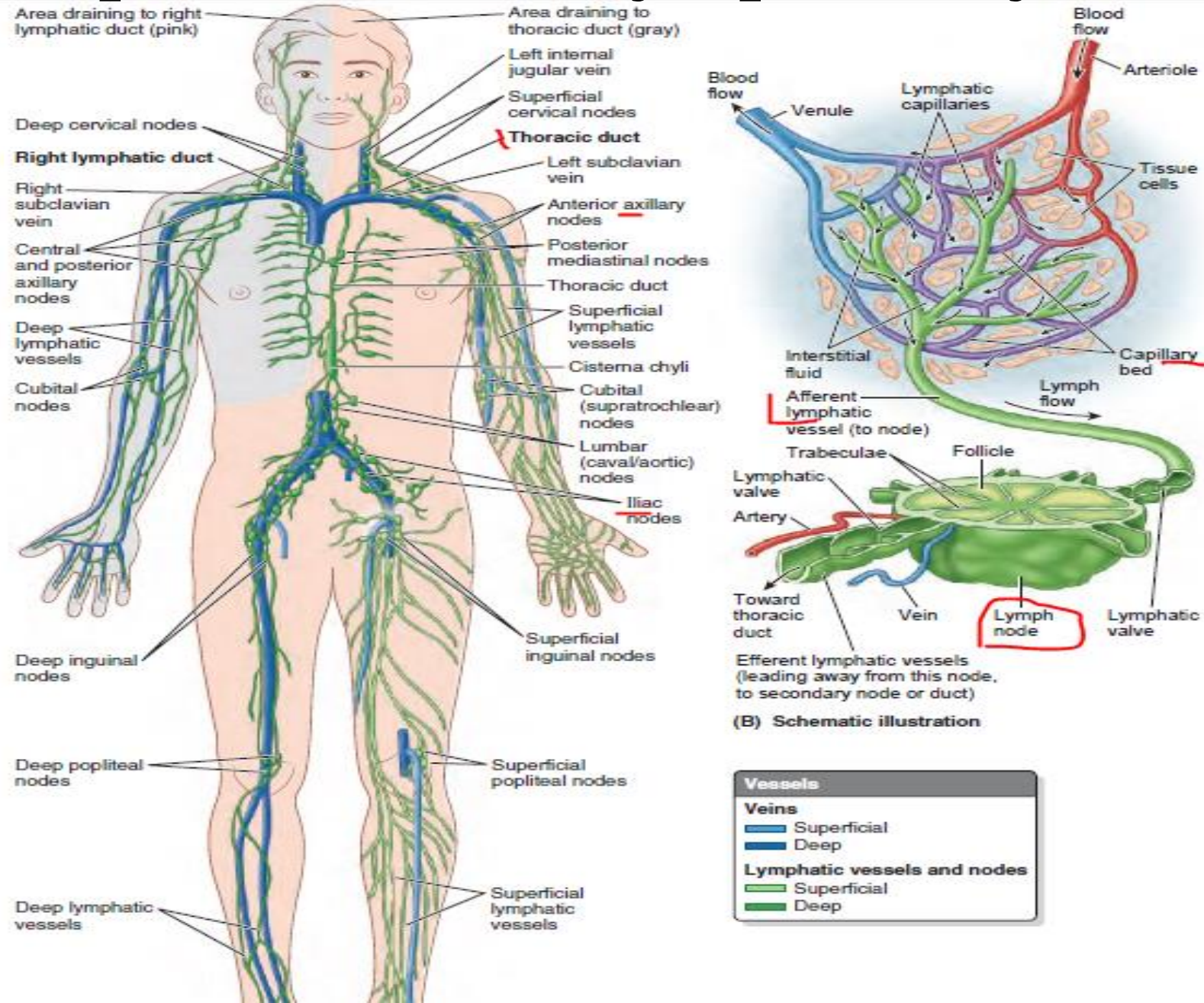
❖ The components of the lymphatic system are;,,,

- ✓ **Lymph nodes**, small masses of lymphatic tissue located along the course of lymphatic vessels through which lymph is filtered on its way to the venous system.
- ✓ **Lymphocytes**, circulating cells of the immune system that react against foreign materials.
- ✓ **Lymphoid organs**, parts of the body that produce lymphocytes, such as the **thymus, red bone marrow, spleen, tonsils, and the solitary and aggregated lymphoid nodules in the walls of the alimentary tract and appendix.**

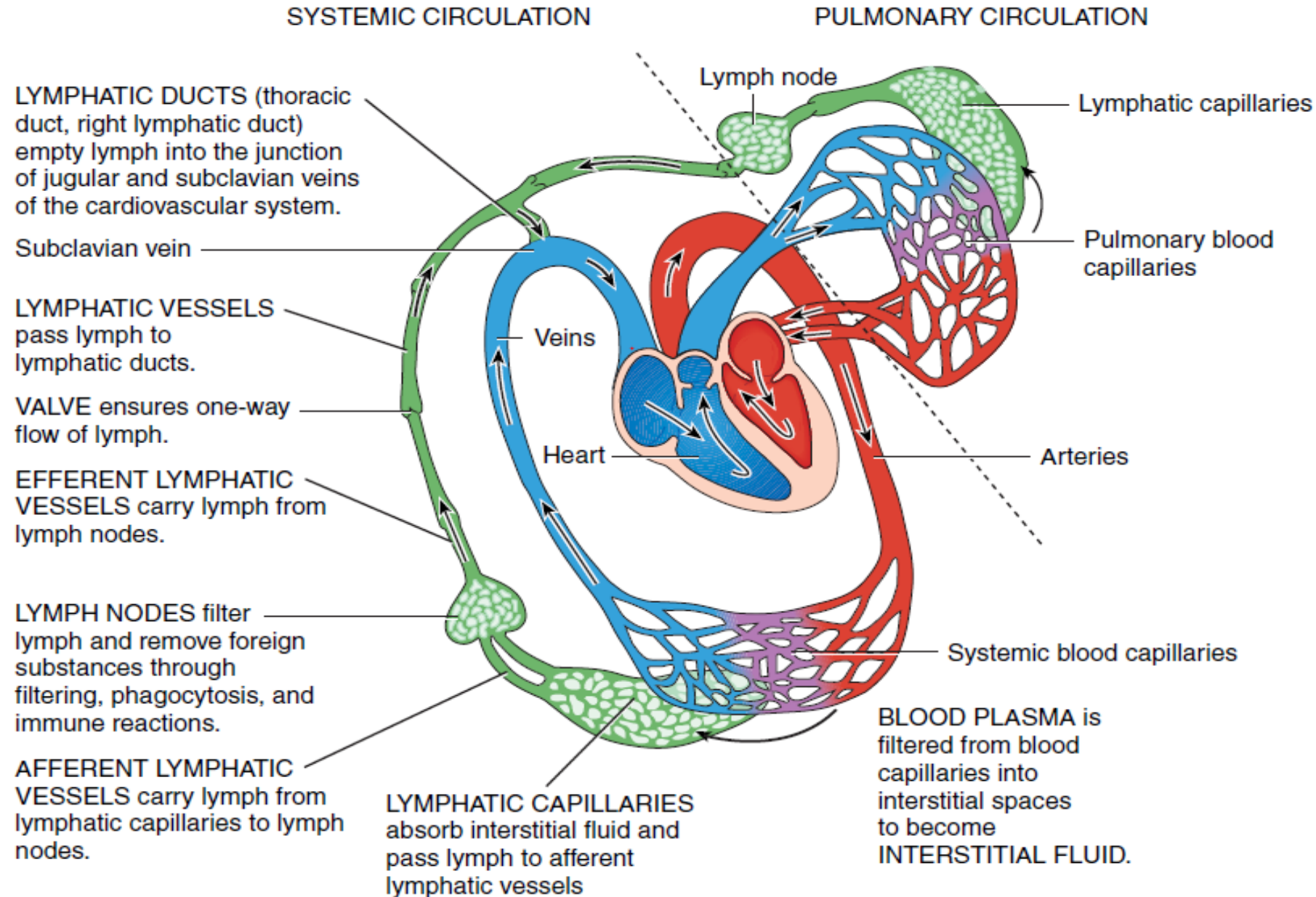
The components of the lymphatic system,,,



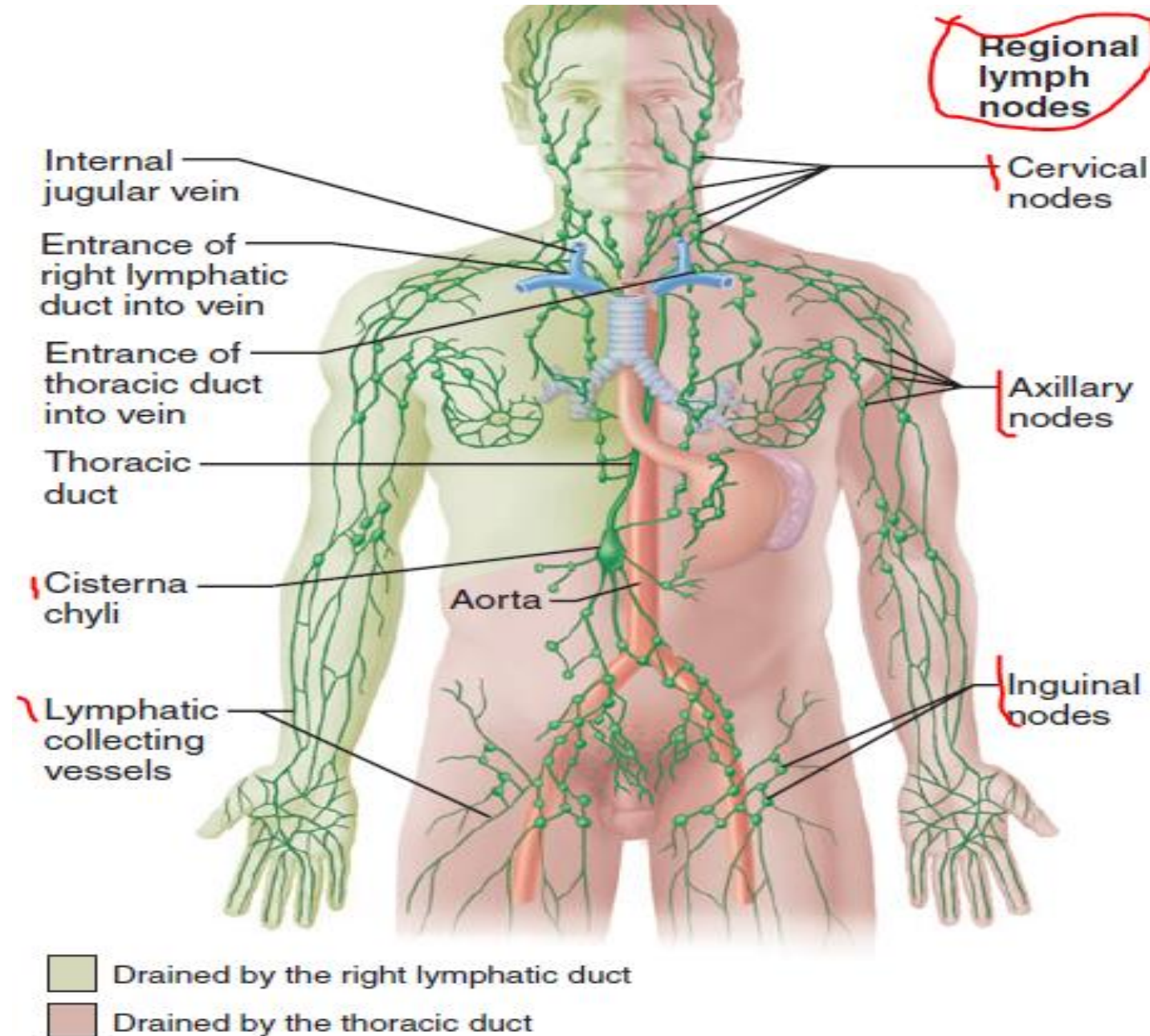
The components of the lymphatic system,,,



The components of the lymphatic system,,,



The components of the lymphatic system,,



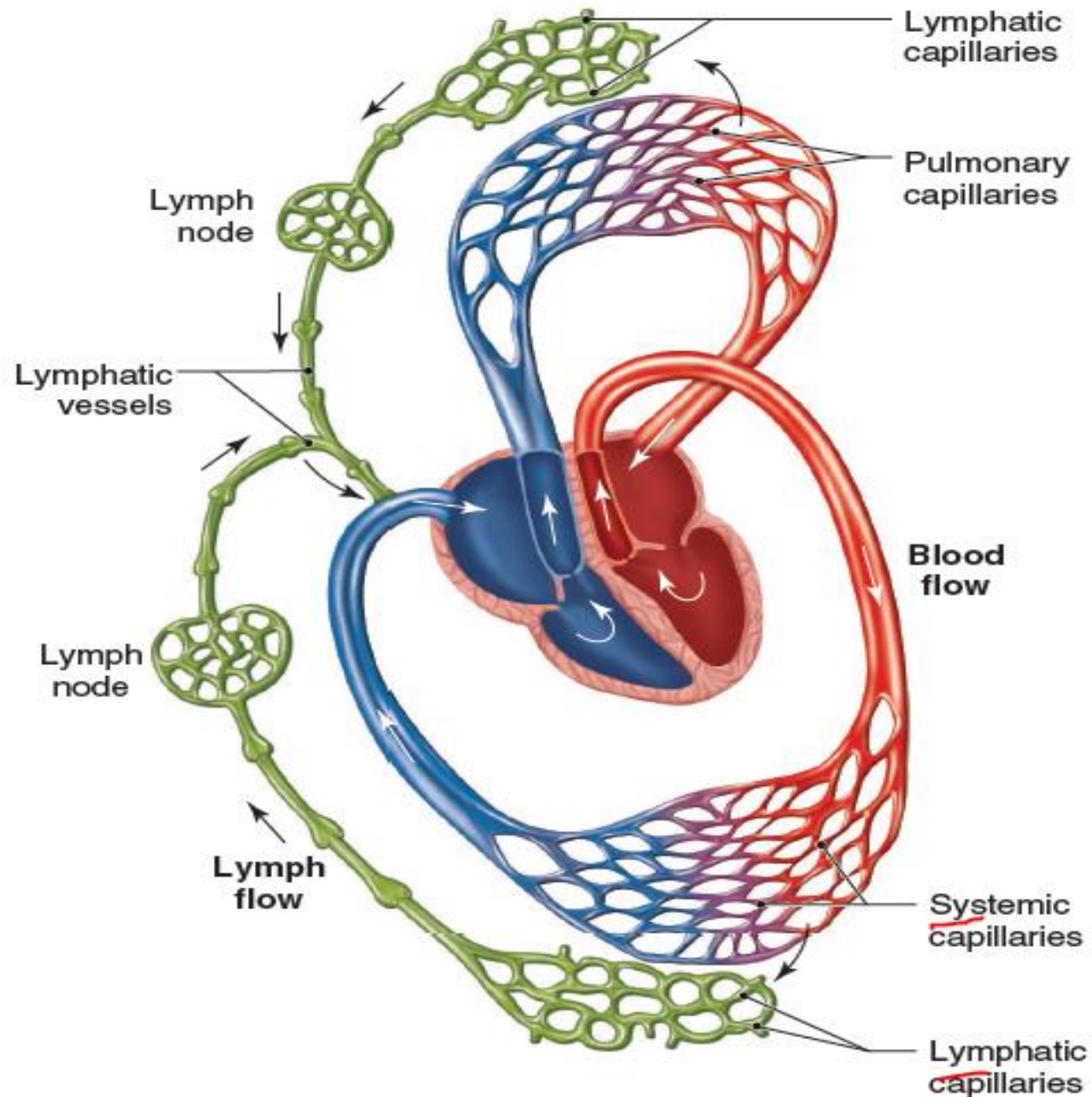
Functions of the Lymphatic System

- The **lymphatic system** is the system of vessels, cells, and organs that carries excess fluids to the bloodstream and filters pathogens from the blood.
- ❑ **The lymphatic system has three main functions:**
 - ❖ ***Drains excess interstitial fluid.*** Lymphatic vessels drain excess interstitial fluid from tissue spaces and return it to the bloodstream.
 - ❖ **Lipid absorption.** In the small intestine, special lymphatic vessels **called *lacteals*** absorb dietary lipids that are not absorbed by the blood capillaries.
 - ❖ ***Carries out immune responses.*** Pathogens, such as microorganisms and other foreign substances, are filtered from lymph by **lymph nodes** and from blood by **the spleen.**

lymphatic capillaries

- **Lymphatic capillaries**, also called the **terminal lymphatics**, are vessels where interstitial fluid enters the lymphatic system to become lymph fluid.
- The smallest vessels, that first receive lymph, are **the *lymphatic capillaries***.
- ✓ Lymphatic capillaries are present in most tissues of the body
- ✓ The lymphatic network of vessels begins with the microscopic **lymphatic capillaries**.
- ✓ Lymphatic capillaries have greater permeability than blood capillaries.
- ✓ Lymphatic capillaries are also slightly larger in diameter than blood capillaries.
- ✓ One set of lymphatic capillaries, **called *lacteals*, has a unique function**.

lymphatic capillaries,,,



- **Blood capillaries lose fluid to the tissue spaces.**
- **The lymphatic system picks up excess tissue fluid and returns it to the bloodstream.**

Larger Lymphatic Vessels Trunks, and Ducts

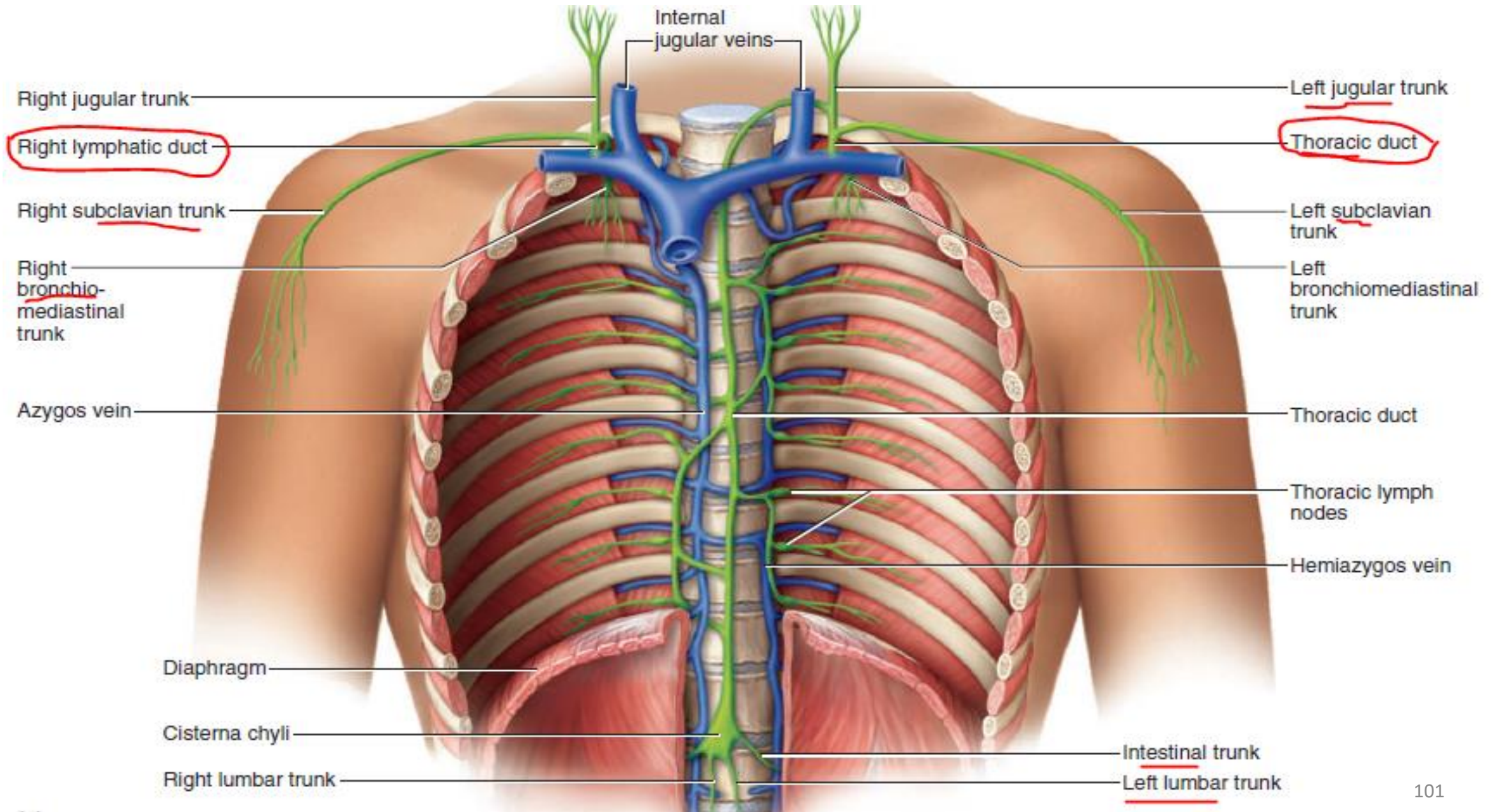
❑ **Lymphatic vessels** (lymphatics), network of thin-walled vessels that have abundant lymphatic valves.

- Lymphatic capillaries unite to form larger **lymphatic vessels** , which resemble small veins in structure but have thinner walls and more valves.
- **Lymphatic Vessels are structurally identical to the veins** – their wall are composed of 3 thin layers of tissues , and **contain valves to prevent backflow**.
- When a lymphatic vessel is compressed, the valves prevent backward movement of lymph.
- **Superficial lymphatic vessels** accompany veins eventually drain into **deep lymphatic vessels** that accompany the arteries.

Larger Lymphatic Vessels Trunks, and Ducts,,,

- As the lymphatic vessels converge along their path, they become larger and larger vessels with changing names.
- ❖ **The route from the tissue fluid back to the bloodstream is: lymphatic capillaries → collecting vessels → six lymphatic trunks → two collecting ducts → subclavian veins.**
- The lymphatic capillaries converge to form **collecting vessels**.
- At irregular intervals, **along the lymphatic vessels**, lymph flows through lymph nodes, consisting of masses of B cells and T cells before it finally returns to the bloodstream.

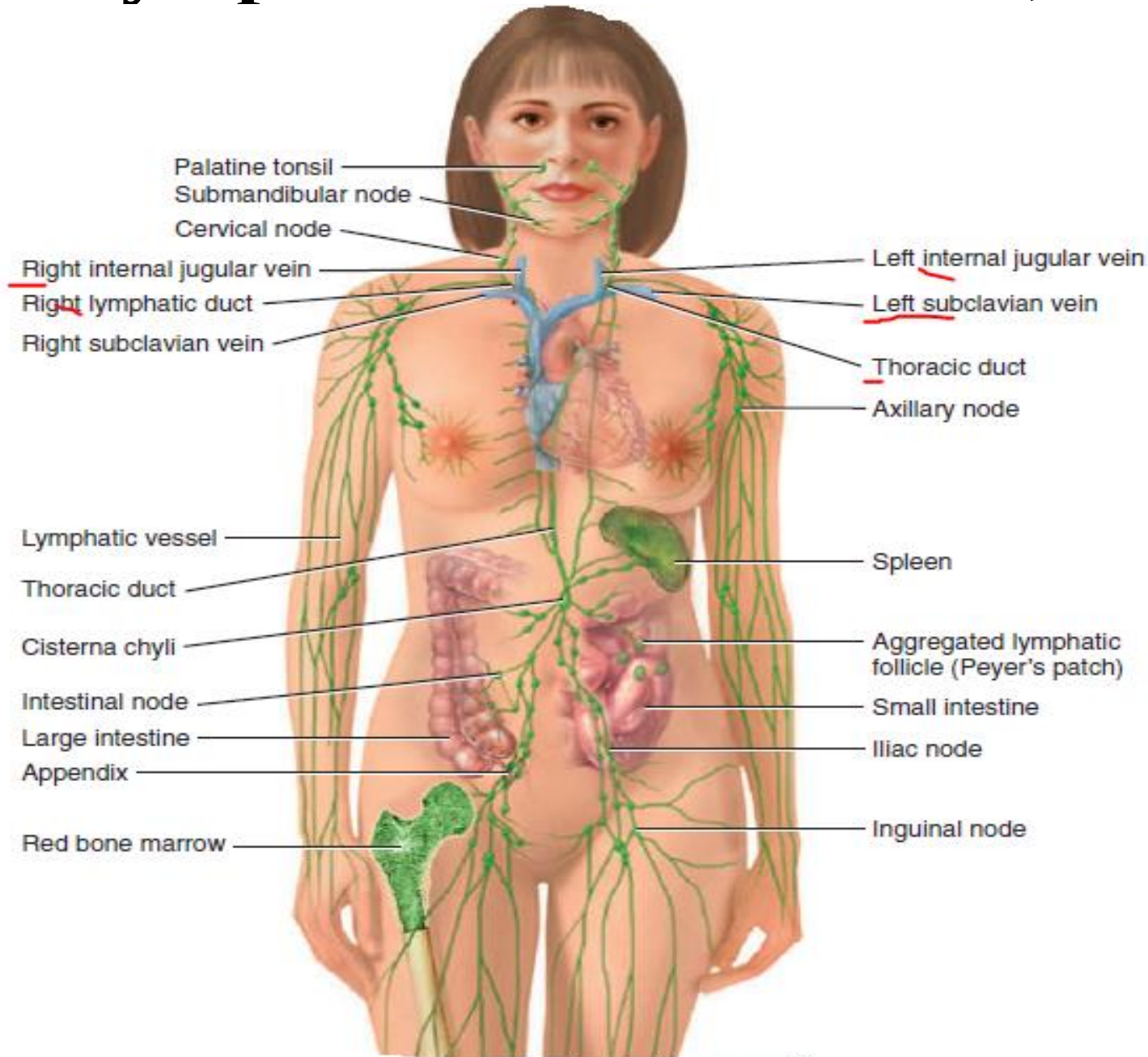
Larger Lymphatic Vessels Trunks, and Ducts,,



Larger Lymphatic Vessels Trunks, and Ducts,,

- Eventually, the collecting vessels converge to form larger **lymphatic trunks**, each of which drains a major portion of the body.
- **There are six of these**, whose names indicate their locations and parts of the body they drain: *the jugular, subclavian, bronchomediastinal, intercostal, intestinal, and lumbar trunks*.
- The lumbar trunk drains not only the lumbar region but also the lower limbs.
- ❖ **The lymphatic trunks** converge to form two **collecting ducts**, the largest of the lymphatic vessels; **the thoracic duct and the right lymphatic duct**.

Larger Lymphatic Vessels Trunks, and Ducts,,



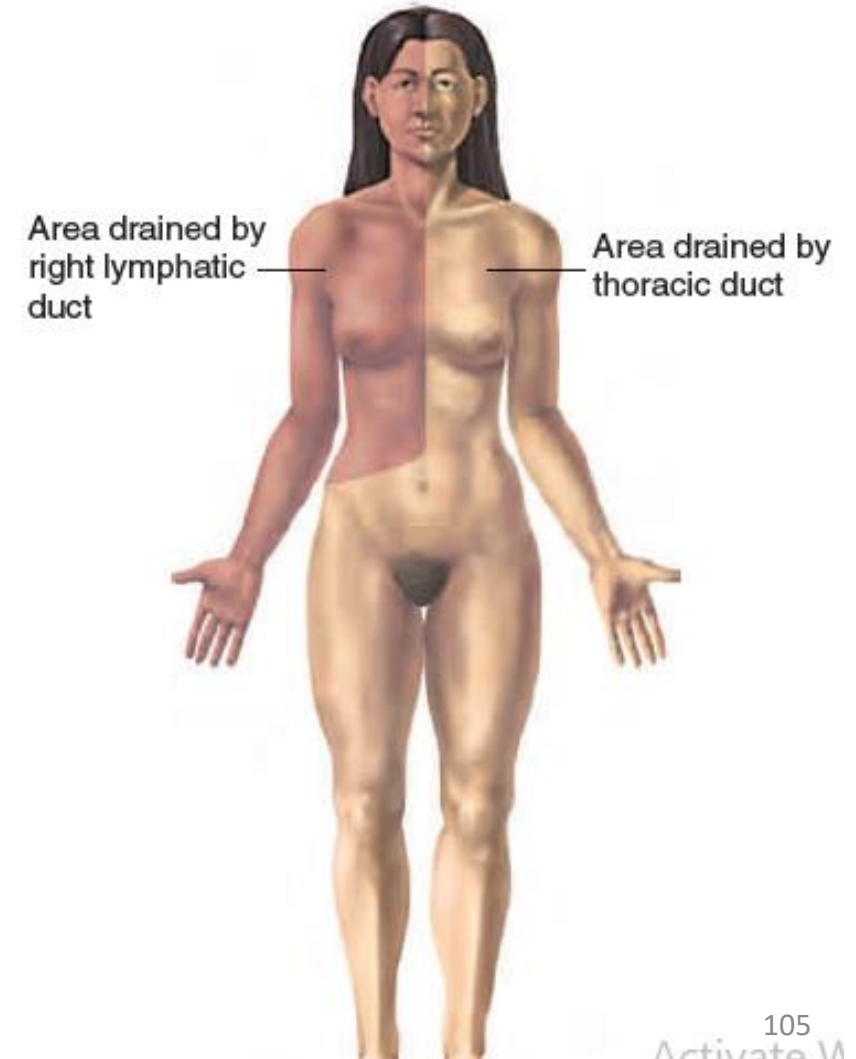
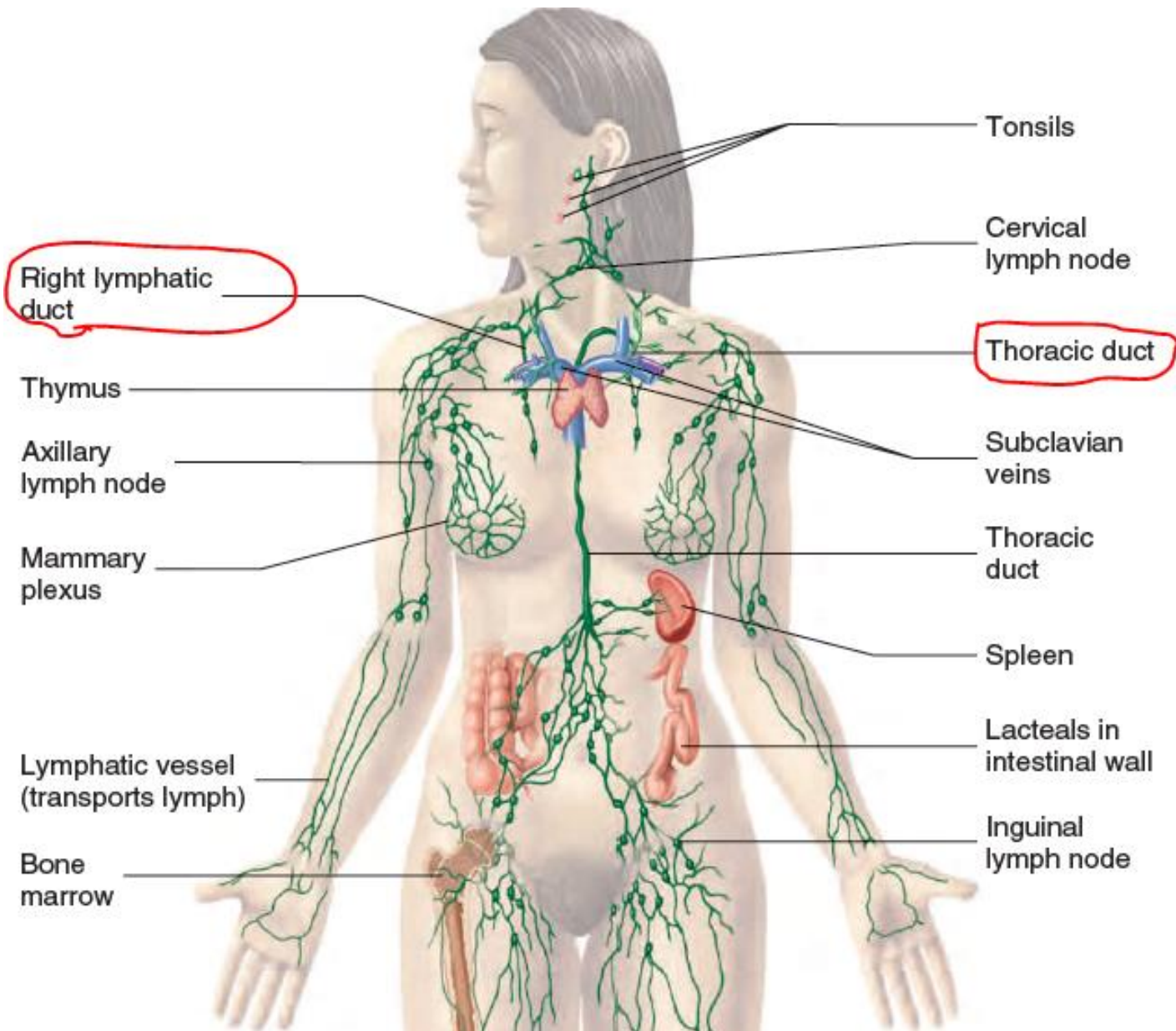
(b) Areas drained by right lymphatic and thoracic ducts

- Area drained by right lymphatic duct
- Area drained by thoracic duct

The thoracic duct

- The **thoracic duct**, on the left, is larger and longer, it is about 38–45 cm long.
- It begins below the diaphragm anterior to the vertebral column at L2.
- Here the two lumbar trunks and the intestinal trunk join and form a prominent sac called the **cisterna chyli**, named for the large amount of *chyle*.
- As it passes through the thorax, it receives additional lymph from the left bronchomediastinal, left subclavian, and left jugular trunks, then empties into the left subclavian vein.
- The **thoracic duct** drains all of the body below the diaphragm, and the left upper limb and left side of the head, neck, and thorax

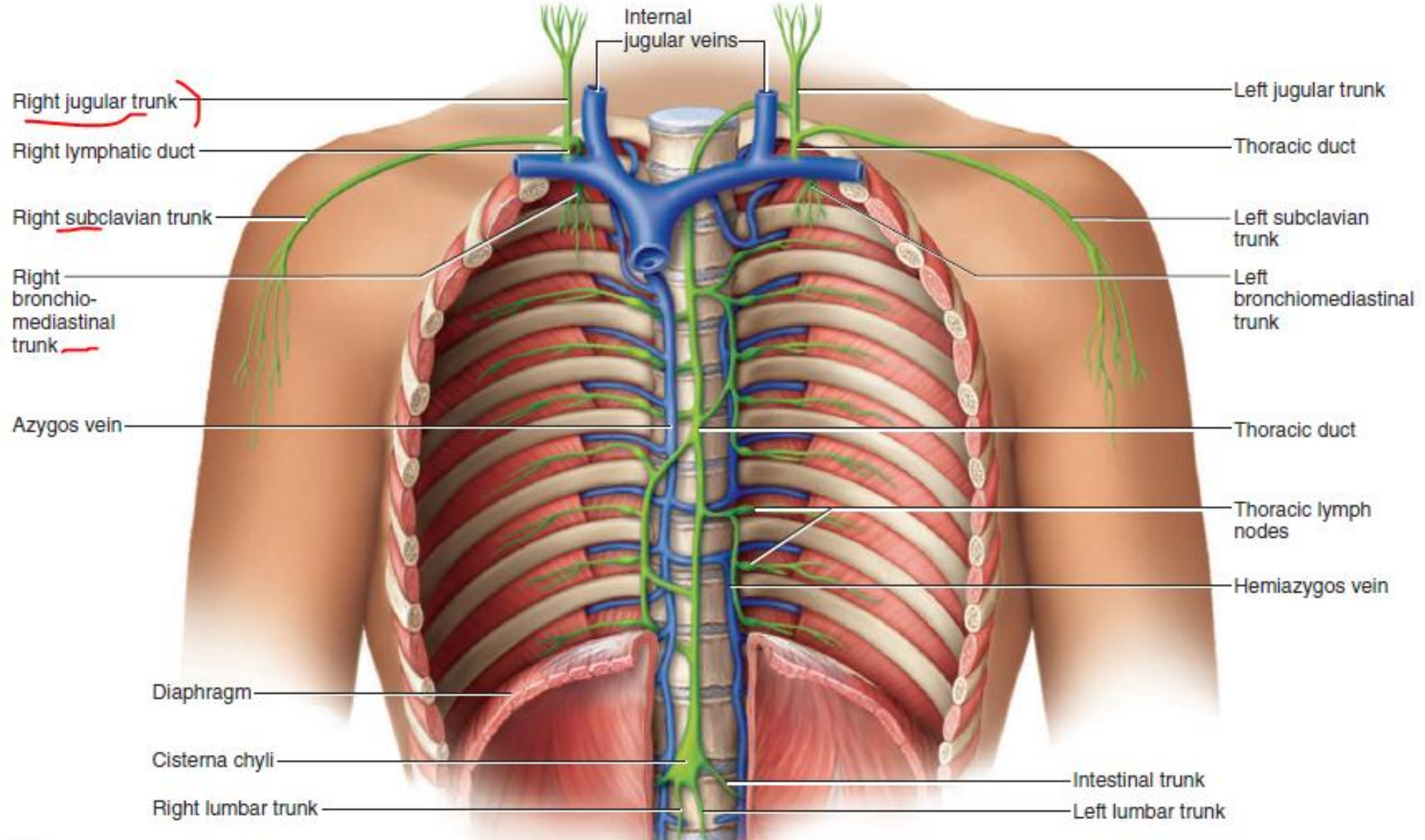
The thoracic duct,,



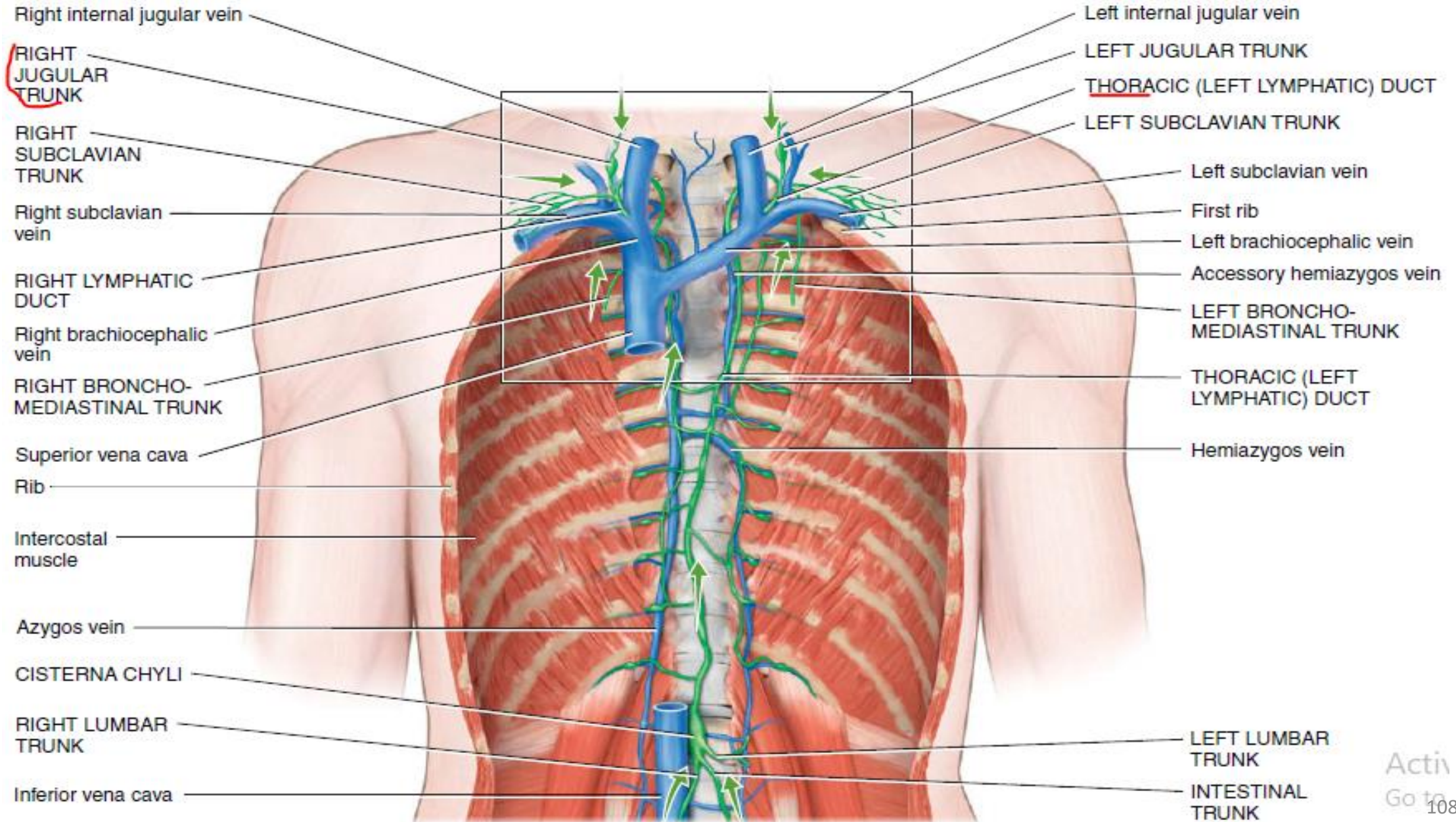
The right lymphatic duct

- The **right lymphatic duct** receives lymph from only the **upper right side of the body**. **The overall drainage system of the body is asymmetrical.**
- It receives lymph from the right jugular, subclavian, and bronchomediastinal trunks in the right thoracic cavity.
- The lymph from the rest of the body enters the bloodstream through the **thoracic duct** via all the **remaining lymphatic trunks**.
- In general, lymphatic vessels of the subcutaneous tissues of the skin, that is, the superficial lymphatics, follow the same routes as veins, **whereas the deep lymphatic vessels of the viscera generally follow the paths of arteries.**

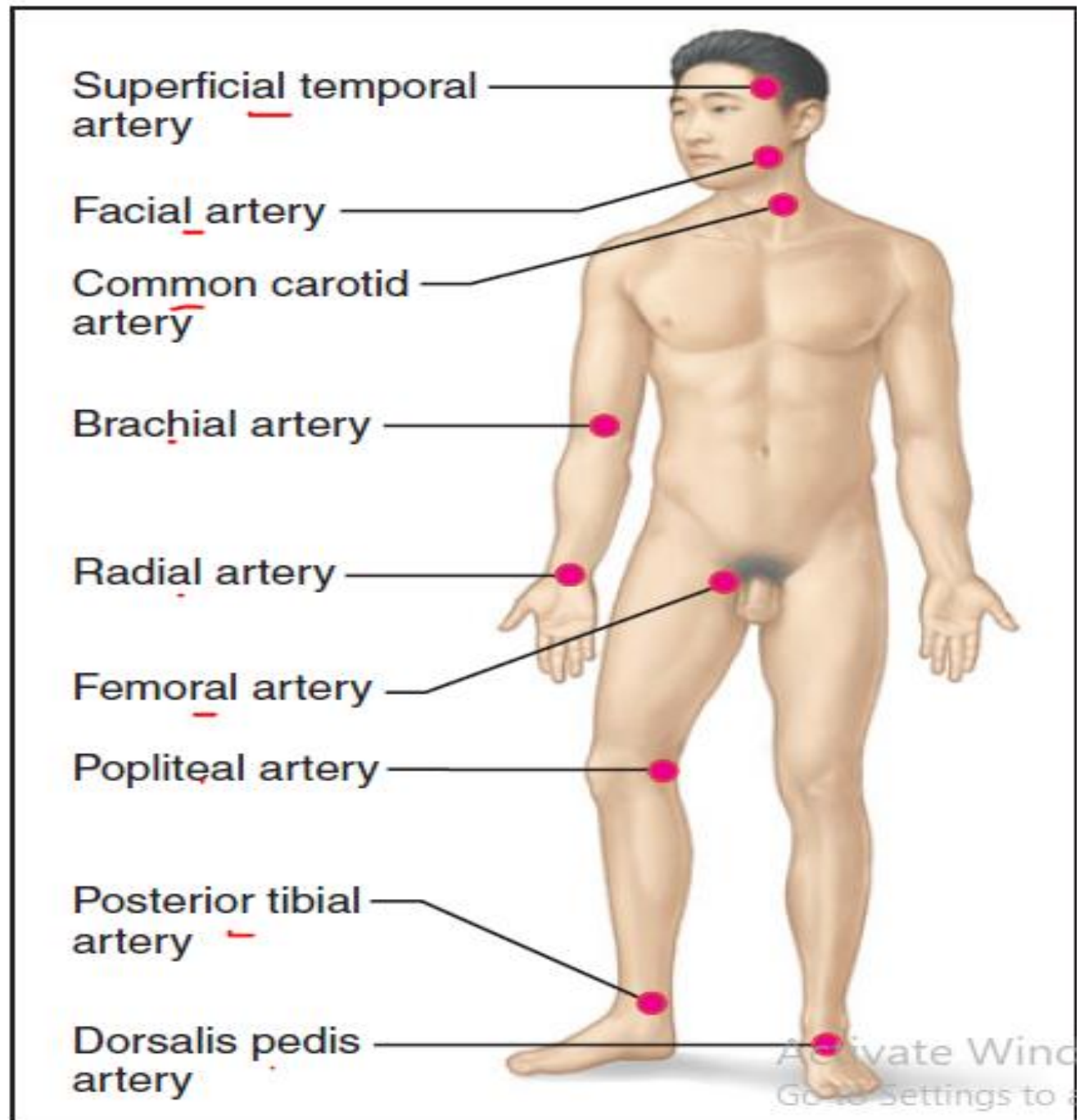
The right lymphatic duct,,



The right lymphatic duct,,



➤ Pulse points

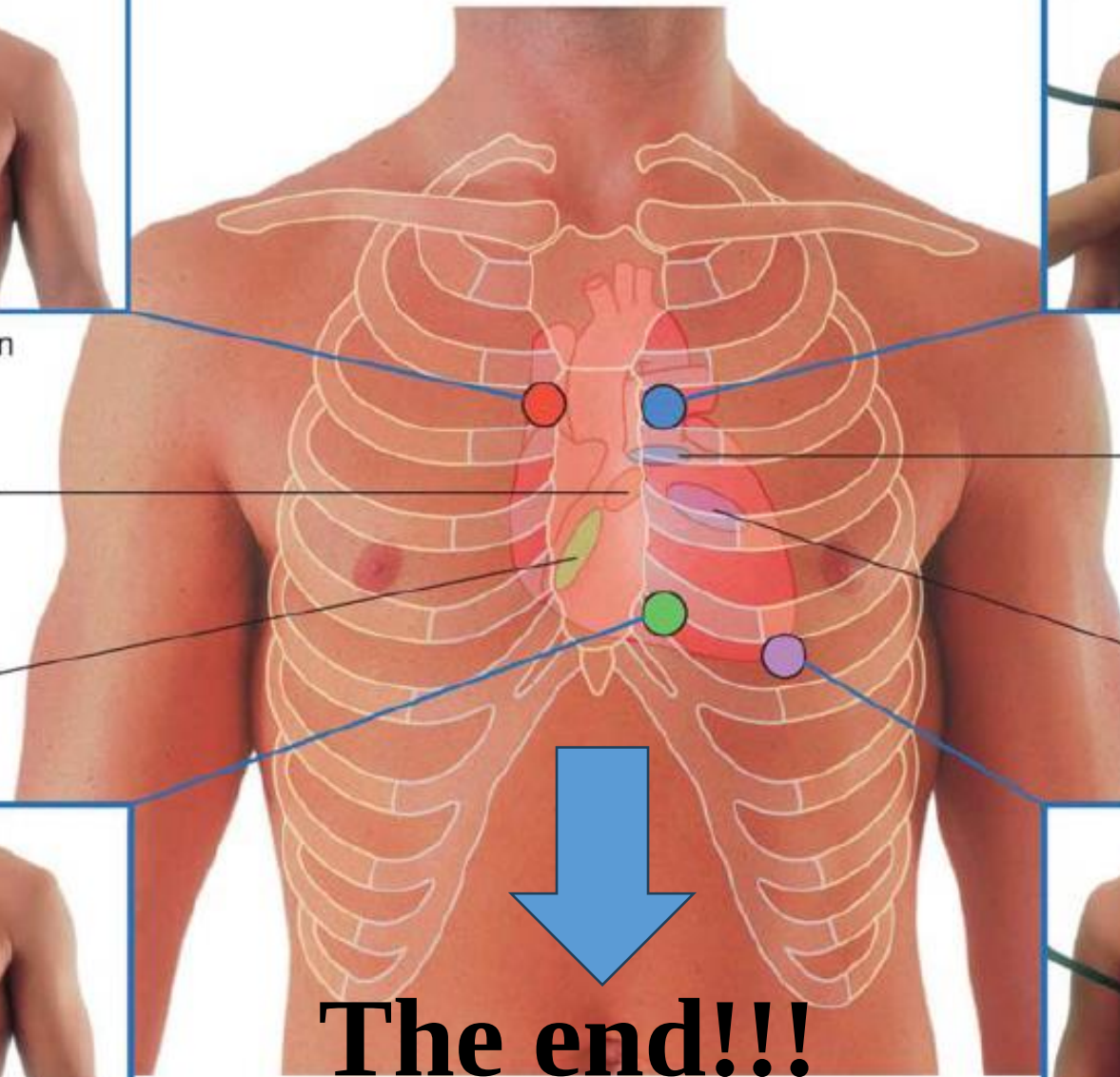




Auscultation position
for aortic valve



Auscultation position
for pulmonary valve



Aortic valve

Pulmonary valve

Tricuspid valve

Mitral valve



Auscultation position
for tricuspid valve

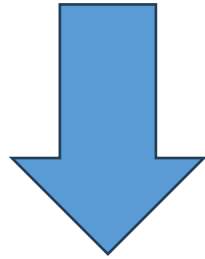


Auscultation position
for mitral valve

The end!!!

Thank You!!!

The end!!!



Thank You!!!